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A Star Called the Sun

George Gamow



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A STAR CALLED THE SUN

George Gamow, Professor of Physics at the University of Colorado, was awarded the 1956 Kalinga Prize by UNESCO for his outstanding contributions in interpreting science for the general public. His previous books on science include *A Planet Called Earth*, *The Creation of the Universe*, *Matter, Earth and Sky*, and *One Two Three . . . Infinity*. His latest book, *Thirty Years that Shook Physics*, was published in 1966.

GEORGE GAMOW

A STAR CALLED THE SUN



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Dedicated to the memory of my book

THE BIRTH AND DEATH OF THE SUN

Published by the Viking Press in 1940

What if the Sun
Be Centre to the World, and other Starrs
By his attractive vertue and thir own
Incited, dance about him various rounds?
Thir wandring course now high, now low, then hid,
Progressive, retrograde, or standing still,
In six thou seest, and what if sev'nth to these
The Planet Earth, so stedfast though she seem,
Insensibly three different Motions move?
— John Milton, *Paradise Lost*, VIII, 122–30

TWENTY-FOUR years ago I published the book *The Birth and Death of the Sun*, devoted to the problems of large-scale liberation of nuclear (subatomic) energy, and to the role of nuclear-energy sources in the life and evolution of the Sun and other stars. The year 1939, during which this book was written, was very significant in the development of nuclear physics and its astronomical applications. That year saw the discovery of the fission of the uranium nucleus, opening the way for the use of nuclear energy for the benefit or peril of humanity, and also brought important progress in the understanding of nuclear reactions which supply energy to our Sun and other stars. Thus *The Birth and Death of the Sun* was full of brand-new, exciting information.

In the following two decades the practical uses of uranium fission became a reality in the form of 'atomic bombs', 'atomic submarines', 'atomic power plants', etc., and numerous studies were carried out concerning the possibility of controlled thermonuclear reaction for the extraction of energy from the heavy hydrogen present in large amounts in the water of the oceans. Great progress also was made in theoretical understanding of thermonuclear reactions in the Sun and other stars, and of the

influence of nuclear transformations on their evolutionary development.

During that period *The Birth and Death of the Sun* was reprinted six times, appeared in several paperback editions, and was translated into the Czech, Danish, Dutch, French, German, Hindi, Italian, Japanese, Norwegian, Persian, Spanish, Swedish, and Yugoslavian languages.

But twenty-four years is much too long a period of time for any book on a rapidly developing subject to stand its ground, and today *The Birth and Death of the Sun* is hopelessly obsolete. Thus I promised my publishers to write an entirely new book on the same subject, on the condition that they subject to auto-da-fé all the remaining copies of the old book except for those needed for bibliographical purposes. It is only natural that I dedicate my new book to the memory of the old one.

G. GAMOW

University of Colorado
December 1963

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The glorious Lamp of Heaven, the Sun,
The higher he's a getting,
The sooner will his race be run,
And nearer he's to setting.

— ROBERT HERRICK

HOW FAR, HOW BIG, HOW HOT?

The chanting Sun, as ever, rivals
 The chanting of his brother spheres
 And marches round his destined circuit –
 A march that thunders in our ears.

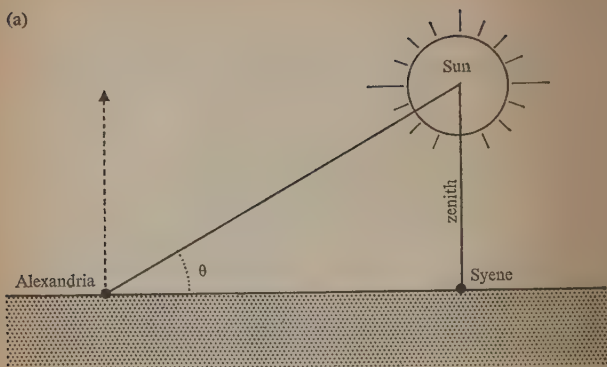
– *Johann Wolfgang von Goethe*
Translated by Louis MacNeice

EARLY EXPLORATIONS

FOR primitive people worshipping the phenomena of nature, the Sun was their most powerful and benevolent god, giving them light and warmth during the daytime and abandoning them only for the night during his journey back from the western to the eastern horizon through the mysterious subterranean passages. The Sun god was given many different names by many different peoples. He was Amon-Ra to the Egyptians, Phoebus-Apollo to the Greeks, Bog-Yarilo (god of brightness) to the Slavs. But thousands of years ago there began to speak voices advocating a more rationalistic and less theological explanation of the phenomena of nature.

About the year 434 B.C. a Greek philosopher named Anaxagoras suggested that the Sun was just a ball of fire about as large as the Peloponnesus, floating in the air about 4,000 miles above the Earth's surface. His arguments which led to that conclusion were quite logical, though factually incorrect. He had heard from travellers to the Upper Nile that in the city of Syene, near the present location of the Aswan Dam, on the day of the summer solstice the noon Sun was directly overhead (in zenith) so that vertical objects threw no shadows. On the other hand, he also knew that at the Nile delta (later the site of the city of Alexandria), about 5,000 Egyptian stadia (500 miles) north of Syene, the noon Sun passes on the same day about 7 degrees away from

(a)



(b)

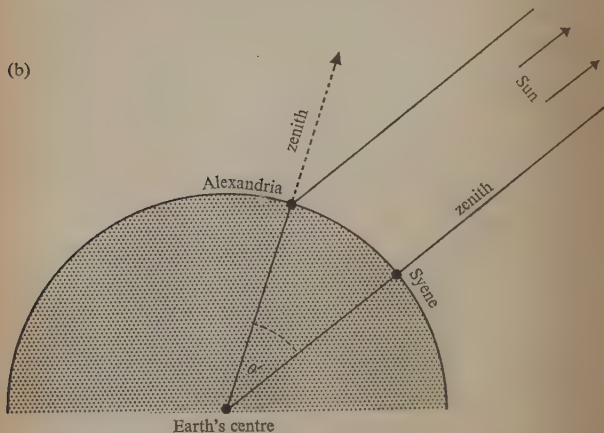


FIGURE 1. Dual interpretation of the Sun's position at noon in Alexandria and Syene: (a) flat-Earth hypothesis leading to an estimate of the distance of the Sun from the Earth; (b) spherical-Earth hypothesis leading to an estimate of the radius of the Earth.

the zenith. Believing that the Earth was flat, Anaxagoras drew a diagram, shown in Figure 1a, which permitted him to obtain an estimated figure of 4,000 miles for the height of the Sun above the Earth's surface. Using that distance and the apparent angular diameter of the Sun (about 0.5 degree), Anaxagoras estimated that the diameter of the Sun is about 35 miles, that is, close to the size of the southern peninsula of Greece. He was arrested as a heretic undermining the established religious dogmas and was banished forever from his native Athens.

Although the mathematical argument of Anaxagoras was correct, his assumption of the flatness of the Earth's surface was wrong, and his argument was turned around about two centuries later by another Greek philosopher named Eratosthenes, who ascribed the difference between the positions of the noon Sun at Alexandria and at Syene at summer solstice not to the Sun's proximity to us but rather to the curvature of the Earth's surface. Assuming that the Sun is so far away that its rays falling on the surface of the Earth are practically parallel to one another Eratosthenes drew another geometrical scheme (Figure 1b) which led to a conclusion that the Earth is a sphere with a radius of about 4,000 miles, an estimate remarkably close to the modern value of 3,963 miles.

The fact that the figure obtained by Eratosthenes for the radius of the Earth is exactly the same as that obtained by Anaxagoras for the height of the Sun above the flat-earth level can easily be understood if we notice that the angle θ in Figure 1a is equal to the angle θ' in Figure 1b. This demonstrates how identical mathematical formulas may lead to either completely wrong or perfectly correct conclusions, depending on physical assumptions which are put in their foundations.

After it turned out that what Anaxagoras believed to be the distance to the Sun was actually the radius of the Earth, new efforts had to be made to find how far away the Sun actually is. A brave effort in this direction was made by Hipparchus (160–125 B.C.), who is often called the father of modern astronomy. His ambitious project involved simultaneous determination of the distances to both the Moon and the Sun, and is shown schematically in Figure 2, in which, for graphic purposes, all angles are

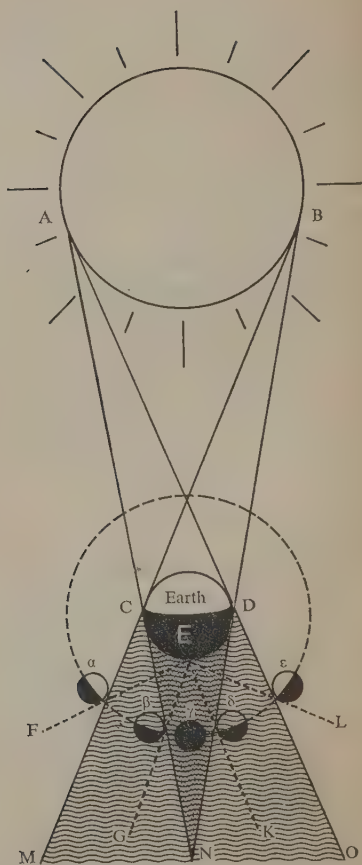


FIGURE 2. Hipparchus's method for estimating the distance from Earth to the Moon and the Sun.

strongly exaggerated and all distances greatly reduced. This figure represents the successive stages of a lunar eclipse caused by the Moon's entering the cones of shadow thrown into space by the Earth. The dark converging cone CDN is the shadow (or *umbra*) cast by the Earth. From any point within that cone no sunlight can be seen at all, and astronauts visiting the Moon's surface at that time would observe a total eclipse of the Sun by the Earth. The lighter diverging cone $MCDO$ is a semi-shadow (or *penumbra*), and from any point within it, astronauts on the Moon would observe only a partial solar eclipse. Hipparchus knew that during a central lunar eclipse (that is, when the Moon passes right through the middle of the shadow) about ten hours elapse between the moment when the Moon first enters the semi-shadow (at F) and the moment when its surface is again brightly illuminated by the Sun's rays (at L). Since the Moon is moving through the sky at the rate of its apparent angular diameter (about 0.5 degree) per hour, the angle FEL is equal to 5 degrees. On the other hand, complete obscuration of the Moon (G to K) lasts for only $5\frac{1}{2}$ hours, indicating that the angle GEK is $2\frac{1}{2}$ degrees. Angles ACB and ADB are also known, since they represent the angular diameter of the Sun as seen from the Earth, being thus equal to 0.5 degree each. Knowing these angles and also the distance CD (diameter of the Earth), one can, by simple geometrical argument, find the distance to the Moon as well as the distance to the Sun.

To do this one assumes some reasonable value for the distance of the Moon, just as a guess, and draws its circular orbit. From the intersections α , β , γ , δ , and ϵ of that orbit with the lines EF , EG , EK , and EL , one draws, according to the numerical values of the angles given above, tangents MC , NC , ND , and OD to the surface of the Earth's globe. Their intersections and A and B will determine the position and the size of the Sun. However, for an arbitrarily assumed value of the Moon's distance, angles ACB and ADB will not necessarily be equal to 0.5 degree, as they should. Thus, to get a correct fit, one will have to change the assumed distance of the Moon until agreement is reached.

Using this method, Hipparchus found that, if r is the radius of the Earth, then the distance of the Moon is $67r$ – quite close,

indeed, to the exact modern value for the mean distance of $60\cdot267r$. He was, however, considerably less successful in his estimate of the distance to the Sun. This was because, since the Sun is very far away, the line pairs MC and ND , as well as the line pairs NC and OD , are almost parallel, thus making the positions of the intersections A and B very uncertain. The only conclusion Hipparchus could reach on the basis of this analysis was that the Sun is at least 37 times farther away than the Moon, and that its diameter is at least 12 times greater than the diameter of the Earth. Although these figures were far smaller than the true values, they remained unchallenged for seventeen centuries, and the correct results were obtained only about two centuries ago, thanks to the invention of the telescope and the development of precise astronomical instruments.

PARALLAX METHOD

The standard method used in modern astronomy for distance measurements within the solar system is based on the observation of the so-called *parallactic displacement* of celestial bodies, or, more simply, their *parallax*. The easiest way to get acquainted with the notion of parallax is to try to thread a needle. If you try to do this with one eye closed, you will find very quickly that it cannot be done skilfully – you bring the thread either too far behind the needle or too close in front of it – and the operation may succeed only by mere chance. But if you use two eyes the task can be done quite easily, as every tailor or dressmaker will testify. Looking at something close with two eyes, you automatically direct both eyes on to the object, thus becoming temporarily slightly cross-eyed. The closer the object, the more your eyes turn toward each other, and the muscular feeling arising from such an adjustment gives you a pretty good idea about the distance. If, instead of looking with both eyes, you close first one eye and then the other, the position of the object (the needle in this case) relative to a distant background (say, a window across the room) appears to change: the left eye sees the needle more to the right in respect to the window, while the right eye sees it more to the left. This angular displacement is known as *parallax* and depends

on the distance of the object from the tip of the observer's nose. Parallaxic displacement decreases with distance, and the two-eye stereovision hardly extends farther out than the arm's length.

For measuring longer distances by this method the two eyes must be placed much farther apart from each other, a task

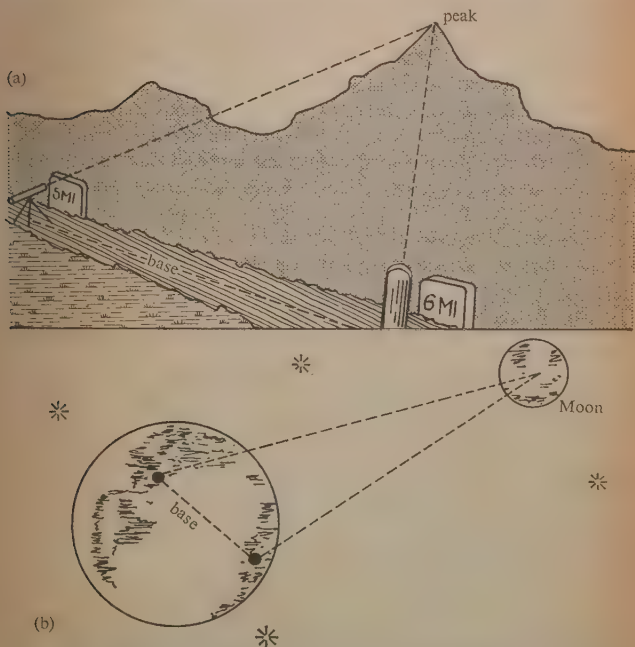


FIGURE 3. Distance measurement by the parallax method in the case of: (a) a mountain peak; (b) the Moon.

which can be achieved in practice by using two theodolites at a considerable distance from each other, and asking two surveyors, using only one eye each, to view through the device a distant object such as a marker placed on the top of some nearby hill (Figure 3a). From the readings on the angle-scale of both

theodolites, the parallactic displacement can be calculated and, hence, the distance of the object in question.

For astronomical parallax measurements, the role of a distant background is of course played by the stars, which are very far away indeed. Thus, for example, two simultaneous photographs of the Moon, taken from two distant locations on the Earth (Figure 3b), show a noticeable displacement of the lunar disk with respect to the stellar background. It was found that when the two observations were made at opposite points of the Earth's equator ('equatorial parallax') the apparent displacement of the Moon with respect to stars is 1 degree, 24 minutes, 5 seconds, which leads to the conclusion that the average distance to our nearest cosmic neighbour is 30.14 times the Earth's diameter at the equator, that is, 384,403 kilometres, or 238,857 miles.

However, the parallactic displacement of the Sun, even in the case when the two observations are carried out from opposite points of the Earth's circumference, is very small and close to the limits of possible observational errors in the measurement of the angles. Thus, resourceful astronomers prefer in this case to use some 'stepping-stone' located between the Sun and the Earth. One of the convenient stepping-stones for that job is the planet Venus, which once in a while comes much closer to the Earth than the distance which separates the Earth from the Sun. The ratio of the distances of the Earth and of Venus from the Sun can easily be compiled from the observed fact that the angular distance of Venus from the Sun on the celestial sphere never exceeds an angle of about 45 degrees, or, to be exact, 43 degrees, 37 minutes, 30 seconds. This maximum angular separation of the beautiful Venus from the brilliant Apollo corresponds apparently to the situation shown in Figure 4, where the line EV is tangent to the orbit of Venus, i.e. perpendicular to the line SV . Trigonometrical tables tell us that in this case the distance SV is 0.7233 of the distance SE . Thus we conclude that, when Venus passes in its orbital motion between the Earth and the Sun, its distance from us is $1 - 0.7233 = 0.2767$ of our distance from the Sun.

If all planets of the solar system were moving in the same plane, we would see Venus passing right across the Sun's disk every 19.2 months. However, because of differences in the tilt of

planetary orbits, these 'transits of Venus' across the Sun's face are rather rare phenomena. The last two transits occurred in 1874 and 1882, and we will have no other transits until 8 June 2004 and 6 June 2012. During the Venus transit in the year

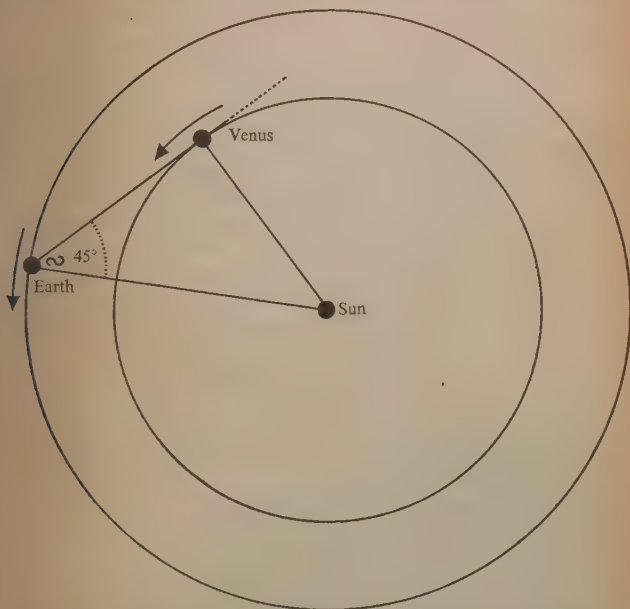


FIGURE 4. A method of measuring the ratio between the radius of Earth's orbit and that of Venus.

1752, the French astronomer Nicolas Louis de Lacaille first used it as a stepping-stone by observing the projection of Venus on the disk of the Sun as seen from two different locations on the Earth's surface. The situation is shown schematically in Figure 5, in which all angles are strongly exaggerated. (The actual displacement of Venus, as observed from diametrically opposite points

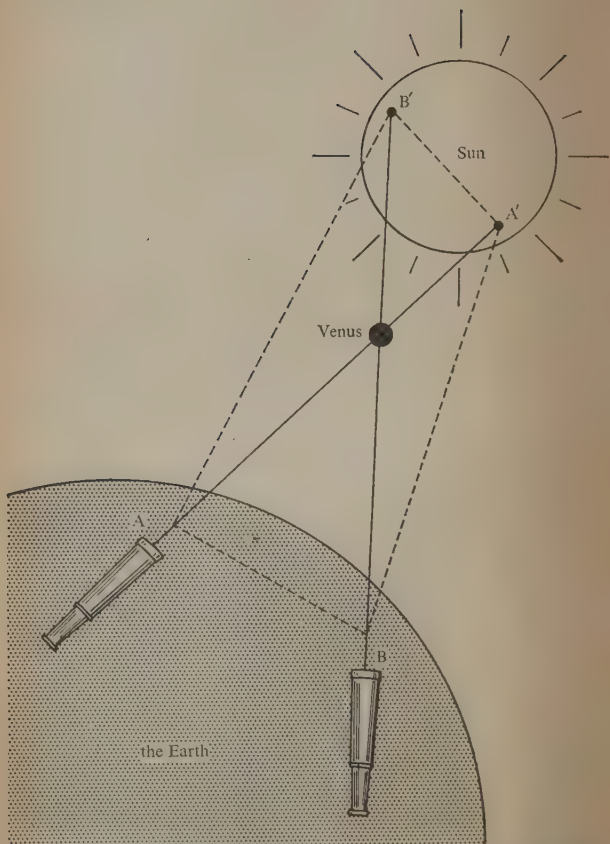


FIGURE 5. Parallaxic displacement of Venus's projection on the surface of the Sun as observed from different locations on the Earth's surface.

on the Earth's circumference, would be only 20 angular seconds, that is, about 1 per cent of the apparent solar diameter.) Knowing from observation the angle $B'AA'$, and noticing that the ratio of sides $B'V$ and AB' in the triangle $AB'V$ is equal to the distance of Venus from the Sun relative to that of the Earth (i.e. 0.7233), one can easily calculate the distance from the Earth to the Sun. Using that method, Lacaille obtained the first reasonably exact estimate of distance to the Sun, exceeding by a factor of about 10 the original value of Hipparchus. Still more exact values of the Sun's distance can be obtained by using as a stepping-stone the asteroid Eros, which sometimes comes as close to us as one sixth of the solar distance. The correct value of the mean solar parallax is now known to be 8.80 angular seconds (8.80'') which corresponds to the mean distance of $149,450,000 \pm 17,000$ kilometres, or $92,870,000 \pm 11,000$ miles. Using this distance and the apparent angular diameter of the solar disk ($31'59.3''$), one finds that the diameter of the Sun is 1,390,600 kilometres (865,400 miles), or 109.1 times the diameter of the Earth.

MASS DETERMINATION

According to Newton's law of universal gravity the attractive force F between two material bodies with the masses M_1 and M_2 placed at the distance R is given by the formula:

$$F = G \frac{M_1 \times M_2}{R^2}$$

where G is the so-called gravitational constant. The first measurement of the gravitational constant was carried out by the British physicist Henry Cavendish (1731–1810), who used an arrangement shown schematically in Figure 6. Two small spheres, both having mass m , are attached to the ends of a light bar, which is suspended on a very thin thread. Close to them are placed two heavy spheres, both with the mass M , which can be moved around a vertical axis. In the original position the line connecting the two massive spheres coincides with the equilibrium position of the suspended bar supporting two light spheres. If the support of the massive spheres is turned through a certain angle, under the action of the gravity forces between the masses m and M the

suspended bar will turn very slowly, twisting the suspension thread and assuming a new position in which the gravity forces will be compensated by the torque of the suspension thread. This apparatus, which nowadays is a standard demonstration gadget in any well-equipped physics lecture room, permitted

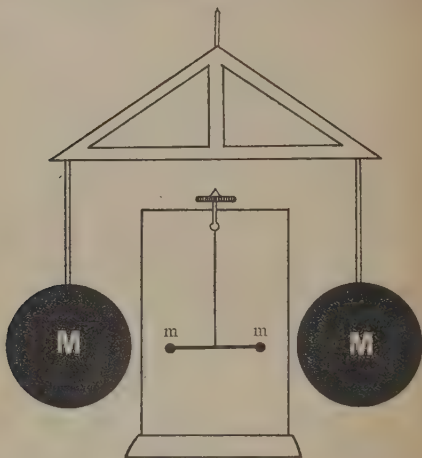


FIGURE 6. Cavendish's arrangement for measurement of Newton's gravitational constant.

Cavendish not only to demonstrate the existence of gravitational attraction between material bodies by a straightforward laboratory experiment, but also to determine the numerical value of the gravitational constant G , which turned out to be equal to 6.664×10^{-8} * in the units based on a centimetre for length, a gram for mass, and a second for time.

* In writing very large or very small numbers it is customary to use exponents to express the powers of ten. Thus $3 \times 10^4 = 3 \times 10,000$ (i.e. four zeros), or 30,000; and $6 \times 10^6 = 6 \times 1,000,000$ (i.e. six zeros), or 6,000,000. On the other hand, $6 \times 10^{-6} = 6 \times 0.000001$ (i.e. six decimal places), or 0.000006. A 'billion' as used in this book is a thousand million, or 1,000,000,000, or 10^9 . (In Great Britain 1 billion means 1 million millions, i.e. 10^{12} .)

Knowing G , one can easily find the mass of the Earth by substituting in the Newton formula the mass of a given object, the force of gravity exerted by the Earth on the object (i.e. its weight), and its distance from the centre of the Earth (i.e. the Earth's radius). This leads to the value $(5.966 \pm 0.001) 10^{27}$ gm. for the Earth's mass, and the value 5.52 for its mean density with respect to water.

In order to find the mass of the Sun it must be remembered that, in the orbital motion of the Earth around the Sun, centrifugal force due to rotation is compensated by the gravitational attraction between the two bodies. Calculating the centrifugal force (which is equal to the Earth's mass times the square of its velocity divided by its distance from the Sun) and substituting it into Newton's formula, one finds that the mass of the Sun is 1.983×10^{33} gm., being 331,950 times larger than the mass of the Earth. Dividing that mass by the Sun's volume, we find that the mean density of the Sun is 1.41 with respect to water. One must remember that this figure represents only the *mean* value of the Sun's density. As we shall see later, the density of the material forming the Sun changes with the depth below its surface, being less than the density of atmospheric air in its outer parts and rising to about a hundredfold water density in its interior.

SURFACE TEMPERATURE OF THE SUN

It is evident that the surface of the Sun must be very hot, but how can one tell exactly what its temperature is? One way of doing this is to compare the light and heat emission of the Sun with that of hot bodies under laboratory conditions, where the temperature can be changed at will. We know that house-heating units whose temperature is below the boiling point of water (100° C.) emit a considerable amount of heat but no light whatsoever. The heating units of electric kitchen ranges at 600° to 700° C. emit much more heat and glow with a dim red light. The filaments of electric bulbs at about $2,000^{\circ}$ C. emit a bright white light which, however, looks yellowish when compared with that of an electric arc operating at a temperature between $3,000^{\circ}$ and $4,000^{\circ}$ C. The electric arc also emits a considerable number of ultra-violet rays.

As was first shown by Isaac Newton, white light represents a mixture of various rainbow colours, which can be separated by sending the light through a glass prism, and we know now that different colours correspond to different wave lengths of the emitted radiation. Our eye is sensitive only to a narrow range of wave lengths between 0.4 and 0.7 microns,* corresponding to the *violet* and *red* ends of the rainbow. Beyond the violet edge of the visible spectrum extends the region of *ultra-violet* radiation, shorter wave lengths invisible to the eye but very active in producing various photochemical reactions such as the blackening of photographic plates and the tanning of the human skin. Proceeding further in the direction of decreasing wave lengths, we find X-rays and gamma-rays emitted by radioactive substances. On the other side of the visible spectrum lies the region of *infra-red* radiation, longer wave lengths mostly observed through their heating effects, and therefore often called 'heat radiation'. Beyond the infra-red band are the regions of short and long radio waves.

It must be emphasized that the apparent qualitative differences between ultra-violet, visible, and infra-red radiation are due to the purely quantitative difference between their wave lengths, which leads to different interactions between the radiation and the material bodies on which it falls. Thus, for example, the fact that our eyes are sensitive to red light, while the photographic plate is not (hence red lanterns in a photographer's darkroom), or the fact that the photographic plate is sensitive to the invisible ultra-violet radiation, is entirely due to the differences between the chemical structures of the eye's retina and of the photographic emulsion.

The only intrinsic characteristic of the radiation of any given wave length is the amount of energy it carries, an amount which can be measured by turning all this energy into heat and measuring the resulting increase of temperature. This is done by means of special gadgets known as *bolometers*, in which the incident radiation falls on a small metal plate painted black to secure nearly 100-per-cent absorption. The heating of the plate is measured by sensitive thermocouples which produce an electric current pro-

* One micron, designated by the symbol μ , is one thousandth of a millimetre, or 10^{-4} centimetre.

portional to the increase of their temperature. Figure 7a gives the scheme of a bolometer used for the measurement of the total amount of incident radiation. Here S is the radiation source, P is the blackened absorbing plate, T the thermoelement, and G the galvanometer measuring the electric current.

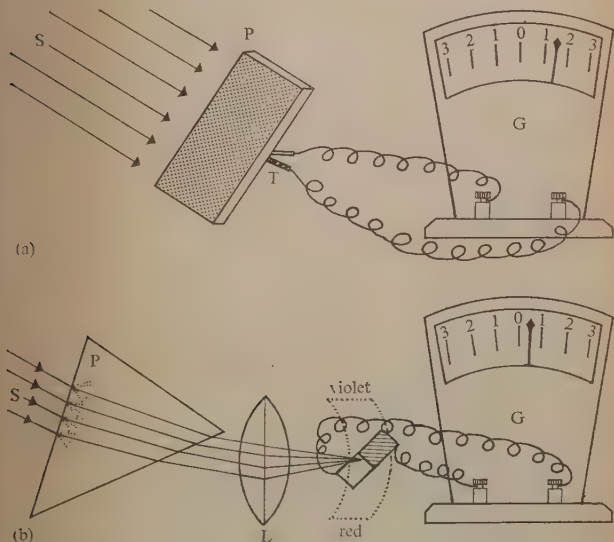


FIGURE 7. Bolometer arrangements: (a) for total spectrum; (b) for spectral distribution.

Figure 7b shows the modification of a bolometer used for measurements of the energy distribution in a spectrum produced by a prism. The difference here is that the plate P is replaced by a blackened thermofilament, which can be placed in different parts of the spectrum, thus measuring the amount of energy carried by different wave lengths.

The curves in Figure 8 show the energy distribution in the spectra emitted by hot bodies at three different temperatures.

Studies of this kind led to the formulation of two important laws concerning the dependence of the emitted radiation on the temperature of the emitting material body:

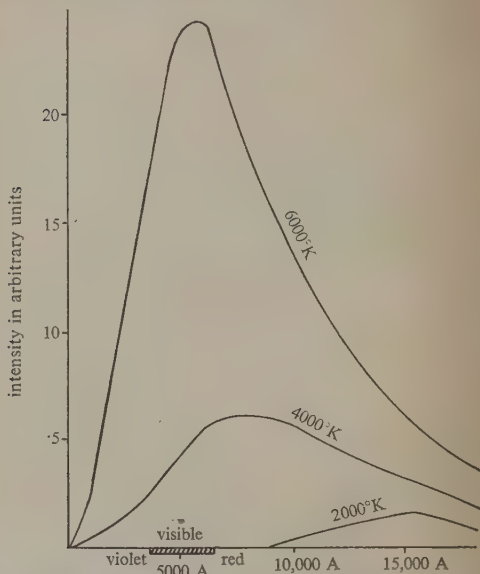


FIGURE 8. Intensity distribution in the spectra emitted by hot bodies at three different temperatures. Wave lengths are expressed in angstrom units, one of which is equal to 10^{-8} cm.

1. *The Stefan-Boltzmann Law*,* which states that *the total amount of energy emitted by a hot body per unit surface per unit time is proportional to the fourth power of its absolute temperature.*†

* Discovered experimentally by the German physicist Joseph Stefan (1835-93), and later derived theoretically by another German physicist, Ludwig Boltzmann (1844-1906).

† Absolute temperature is the temperature counted from the so-called absolute zero point (-273°C.), at which all the thermal motion of the molecules comes to a standstill.

If we express the total amount of energy in calories (the amount of energy needed to raise the temperature of one gram of water by one degree centigrade), and the temperature in degrees Kelvin (centigrade counted from -273°C.), this law can be written as:

$$E = \sigma T^4$$

where σ is known as the Stefan-Boltzmann constant and is numerically equal to 1.34×10^{-12} if E is expressed in calories per second.

2. *Wien's Law*,* which states that *the wave length carrying the maximum intensity in a spectrum emitted by a hot body at a certain (absolute) temperature T is inversely proportional to that temperature*, and one writes:

$$\lambda_{\max} \times T = W$$

where the constant W is equal to 0.289 centimetre degrees Kelvin.

Wien's law just reiterates the earlier statement that, as the temperature increases, material bodies first emit the long-wave infra-red radiation, then red light, then white light, and, at very high temperature, ultra-violet radiation.

Either of these two laws can be used for obtaining the surface temperature of the Sun. If we use the Stefan-Boltzmann law we must know the rate of the total energy emission per unit of its surface, which can easily be obtained on the basis of the observed amount of solar energy which falls per second on the area of one square centimetre placed perpendicular to Sun rays beyond the limits of the terrestrial atmosphere. Until recently, before we learned to send rockets into outer space, estimations of this so-called 'solar constant' required a knowledge of the percentage of solar radiation which is reflected from or absorbed by the terrestrial atmosphere and never reaches the Earth's surface, where the measurements had to be carried out. Nowadays direct measurements can be made by sending sensitive bolometers, which are designed to measure the amount of heat falling on them, beyond the outer fringes of the atmosphere, where solar radiation arrives with its original intensity. Careful experiments show that this amount is equal to 1.94 calories per square centimetre per minute, or 0.032 calories per second. Since the intensity of any

* Formulated by the German physicist Wilhelm Wien (1864-1928).

radiation, whether it is light or heat or any other kind of ray, decreases as the square of the distance, we conclude that one square centimetre of the solar surface emits 0.032 calories per second multiplied by the square of the ratio of the radius of the Earth's orbit to the radius of the Sun. As we have seen earlier, the distance from the Earth to the Sun is 149,450,000 kilometres, while the radius of the Sun is 695,300 kilometres. Thus we find that one square centimetre of the solar surface emits

$$3.2 \times 10^{-2} \left(\frac{149,450,000}{695,300} \right)^2 = 1.5 \times 10^8 \text{ calories per second.}$$

Substituting this value in the Stefan-Boltzmann formula, we obtain for the surface temperature of the Sun a value of about 5,800 degrees Kelvin.

Using Wien's formula, we do not need to know the value of the total amount of energy emitted by the Sun but require instead knowledge of the energy distribution in the spectrum of the Sun. Careful bolometric measurements along the solar spectrum formed by a prism show that maximum intensity falls in the green region of the spectrum at the wave length of about $0.48 \mu = 4.8 \times 10^{-5}$ cm. Using Wien's law, we obtain a value of about 6,000 degrees Kelvin for the Sun's surface temperature, in fairly good agreement with the value obtained by the previously used method.

The method based on Wien's law is especially practical for the surface-temperature measurements of stars, the distances and radii of which may be unknown. Instead of obtaining the entire curve for the energy distribution in the stellar spectrum, it is sufficient for all practical purposes to measure only the intensity ratio for two different wave lengths, since this ratio depends in a unique way on the surface temperature of the star. A customary way of doing it is to compare the relative brightness of a star obtained by *visual* observation with that obtained by the *photographic* method. Since the eye has a maximum sensitivity for yellow light, whereas a photographic plate is sensitive mostly to blue and violet, the ratio of the brightnesses obtained by these two methods is a good indicator of the surface temperature of the star. This ratio has large values for comparatively cool, reddish stars, which emit mostly red and yellow light, and small values

for hot, bluish stars, the light of which is rich in blue and violet rays. In practice one uses the logarithm of this ratio, which is known as the *colour index*. Thus, for example, the colour index of a red star Alpha Orionis, commonly known as Betelgeuse, is 2.3, indicating that its surface temperature is only $3,190^{\circ}$ K. On the other hand, the colour index of the bright blue star Alpha Eridani, known as Achernar, is -0.18 ,* indicating a temperature of 15,000 degrees. The colour index of the Sun is 0.65, which leads to approximately the same value for the Sun's temperature as do the methods described earlier.

Although the figures for the temperature of the Sun's surface obtained by the three methods described are in rather close agreement with one another, they are not close enough to satisfy astronomers or physicists, who are accustomed to agreements within a few decimal points. The reason for the discrepancies in this case is that both the Stefan-Boltzmann and the Wien formulas apply to ideal emitters, known technically as 'black bodies', whereas the surface of the Sun and the heat emitters used in laboratories are not quite ideal. Thus the best method of measuring the temperature of the surface of the Sun and other stars is the much more complicated method of spectral classes, which is described in some detail in Chapter 3.

ENERGY PRODUCTION IN THE SUN

Suppose a person receives for his birthday or for Christmas a present of an electric coffee pot which is advertised as producing heat at the same rate at which heat is produced in the Sun's interior. How long will it take to start the water boiling? Most readers would probably say that it will take a very short time, perhaps a negligible fraction of a second. But this is not correct. To get a cup of hot coffee from that pot one would have to wait for many months or even a year, provided, of course, that the pot is so perfectly insulated thermally that no heat

* The negative value here is due to the fact that astronomers adjust the scale of colour indices in such a way that the colour index of Sirius (Alpha Canis Majoris), with a surface temperature of $11,000^{\circ}$ K., would be exactly 0.00.

produced by the heating unit is permitted to escape. Indeed, using the figures given earlier, we can easily calculate the mean rate of heat liberation in the Sun's body. Since its radius is 6.95×10^{10} cm., its total surface area is 6.07×10^{23} cm.² As was mentioned before, one square centimetre of the Sun's surface emits 1.5×10^3 calories per second, so that the total radiation of the Sun amounts to 8.8×10^{25} calories per second. Thus, since the mass of the Sun is 2×10^{33} gm., the heat should be produced inside of the Sun at the rate of 4.4×10^{-8} calories per second per gram of its material. This is only a fraction of a per cent of the rate of heat liberation in the human body due to the metabolic processes!

But why, then, is the Sun so hot? The explanation is very simple. The rate of energy production within a given volume is proportional to that volume, that is, proportional to the *cube* of its size. On the other hand, the heat losses are proportional to the surface, that is, only to the *square* of the size. Therefore, the larger the body is, the less is the rate of heat production necessary to maintain its temperature; hence the body of an elephant metabolizes about thirty times more slowly than that of a sparrow. If this were not so, we would have roasted elephant, because the accumulated heat would bring the temperature of the elephant's body to a value comparable with that of a kitchen oven. The Sun is much, much bigger than even an elephant, so that its surface is very small in respect to its volume. Therefore, even though the rate of heat production inside the Sun is very small, its surface must be maintained at 6,000 degrees in order to radiate all that heat.

Although the rate of energy production in the Sun is comparatively small, the cumulative amount in the course of the Sun's existence is quite large. Historical geology and palaeontology tell us that the Sun must have been shining with its present intensity for at least 500,000,000 years, the time separating us from the geological eras which left recognizable fossils of living organisms. It is very likely that the Sun has remained unchanged since the time when the solar system was formed, about 5,000,000,000 years ago.

If one assumed that the heat of the Sun is developed by some

chemical reaction, such as oxidation (or burning, in plain language), one would come much too short of the fact. Actually, a complete burning of 1 gm. of gasoline-oxygen mixture (rocket fuel) liberates a total of 200 calories. If the Sun were 'burning', liberating the energy at the rate of 4.4×10^{-8} calories per gram per second, it could have lasted only for $4 \cdot 10^9$ seconds, or for about 100 years, which is certainly not long enough!

In the middle of the last century two famous physicists, Lord Kelvin (of England, of course!) and von Helmholtz (who else but a German?), proposed a new theory of the origin of solar energy. They considered the Sun as a giant gaseous sphere slowly contracting under the forces of its own gravity. Their idea was that the gravitational energy liberated in that process was turned into heat, keeping the Sun hot and luminous. The total amount of gravitational energy which could have been liberated during the past history of the Sun is given by the formula:

$$E = \frac{G \times M^2}{R}$$

where G is the gravitational constant discussed earlier, while M and R are the mass and the present radius of the Sun. Substituting the numerical values, we obtain for E the value of:

$$\frac{(6.6 \times 10^{-8}) (4 \times 10^{66})}{7 \times 10^{10}} \cong 4 \times 10^{48}$$

in the mechanical units of energy known as *ergs*. To turn this into heat units (calories), we must divide that figure by the so-called 'mechanical equivalent of heat': 1 calorie is equivalent to 4.18×10^7 ergs. As a result, we obtain for the entire amount of heat which could be liberated by contraction of the Sun to its present size the value

$$\frac{4 \times 10^{48}}{4.18 \times 10^7} \cong 10^{41} \text{ calories.}$$

Dividing it by the rate of solar energy liberation, we get

$$\frac{10^{41} \text{ cal.}}{10^{26} \text{ cal./sec.}} \cong 10^{15} \text{ sec.} \cong 3 \times 10^7 \text{ years.}$$

While at the time Kelvin and Helmholtz made these calculations the figure obtained seemed to be quite satisfactory, it is

much too small to fit more recent estimates of the age of the planetary system and of the duration of life on the Earth.

The correct answer to the problem of the Sun's energy supply was given only after the discovery of radioactivity, when it was found that tremendous amounts of energy are stored within atomic nuclei. The discussion of nuclear energy liberation inside the Sun and other stars, as well as modern theories of stellar evolution resulting from these nuclear transformations, is the main topic of this book.

SOLAR ECLIPSES

Properly speaking, the discussion of the Sun's eclipses caused by the Moon's passage does not belong to a book about the Sun any more than does that of the Sun's eclipses by the Earth, which occur every twenty-four hours and are called nights. But, while we are all accustomed to daily sunrises and sunsets, the obscurations of the Sun by the Moon are rare events which in olden times led to many superstitions and mass panics and today (especially before the invention of the coronagraph) represent the heaven-given opportunity for astronomers to study the Sun's profile from the bottom of the thick terrestrial atmosphere. Because of a cosmogonical coincidence, the apparent diameters of the Sun and the Moon are just about the same, being equal approximately to the apparent diameter of a dime placed 4 feet away. Due to the ellipticity of the Earth's and the Moon's orbits, the lunar disk appears sometimes a little larger and other times a little smaller than that of the Sun. In the first case, we can observe the total obscuration of the Sun's disk (*total eclipse*), while in the second there is always a bright ring of solar limb protruding beyond the Moon's dark body (*annular eclipse*).

If the plane of the Moon's orbit around the Earth were coincident with the plane of the Earth's orbit around the Sun (the *ecliptics*), solar eclipses would occur monthly, with intermittent eclipses of the Moon. But the two planes are tilted in respect to each other by more than 5 degrees, so that, during the monthly encounters on the celestial plane, the Sun and the Moon most often pass one above the other at a distance larger than their

apparent diameters. Thus an eclipse can take place only when they both pass simultaneously through a *nodal point* which represents the intersection of their circular tracks on the celestial sphere.* The period of complete revolution of the Sun from node to node is 346·62 days,† whereas for the Moon it is 29·5306 days. Now it happens that:

$$346\cdot62 \times 19 = 6585\cdot78 \text{ days}$$

while

$$29\cdot5306 \times 223 = 6585\cdot32 \text{ days}$$

the products being remarkably close to each other. Thus, if the Sun and the Moon met once at the nodal point they would meet there again about 6585·5 days, or 18 years and 11·3 days, later. This periodicity in the eclipses, known as *saros*, was first noticed late in the seventeenth century by the British astronomer Edmund Halley, and one may wonder why the ancient Egyptians and Assyro-Babylonians did not beat him to the discovery. Just as an example, let us take the total eclipse of the Sun which occurred on 30 August 1905, and which was visible in Spain and Egypt. On the basis of saros the next related eclipse was expected to occur on 10 September 1923. And indeed it did occur, but neither the Spaniards nor the Egyptians could see it. The shadow of the Moon ran instead through California and Mexico. The explanation is simple. The saros period has an integer number of days plus one third of a day. Thus, by the time that the Moon's shadow fell on the Earth, the Earth had rotated by $\frac{1}{3} \times 360^\circ = 120^\circ$ westward from its corresponding position in 1905.

Calculation of eclipses in the past and future is one of the most precise as well as one of the most boring branches of astronomy. The bible of eclipses is a multi-volume book entitled *Canon der*

* It is hardly necessary to remind the reader that the 'ecliptics', officially defined as the path of the Sun on the celestial sphere, is equivalent to the plane of the Earth's motion around the Sun.

† The actual time of the apparent revolution of the Sun from equinox to equinox is 365·24 days but, because of the slow rotation of the Moon's orbit caused by gravitational action of the Sun, the time taken by the Sun to move between two intersections of the Moon's orbit with the ecliptics is somewhat reduced.

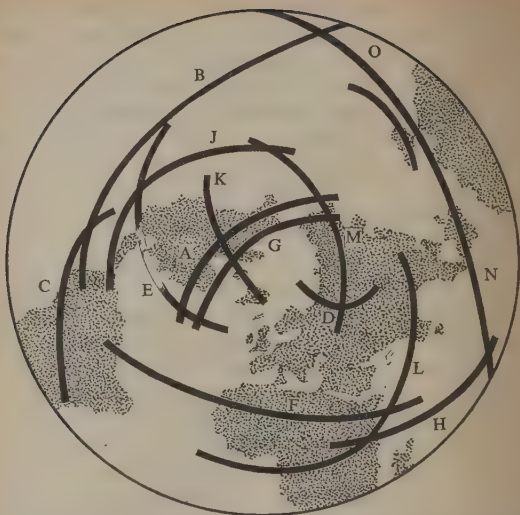


FIGURE 9. Eclipse tracks for 1963 and the next two decades, according to Oppolzer's calculations.

A	1963 20 July	H	1976 23 Oct.
B	1965 30 Apr.	J	1977 12 Oct.
C	1966 12 Nov.	K	1979 26 Feb.
D	1968 22 Sept.	L	1980 16 Feb.
E	1970 7 March	M	1981 31 July
F	1973 30 June	N	1983 11 June
G	1973 10 July	O	1984 22 Nov.

Finsternisse, published in 1887 by the Austrian astronomer Theodor Ritter von Oppolzer, which gives the tracks of all eclipses from 1207 B.C. to A.D. 2162. The earliest recorded eclipse took place on 22 October 2137 B.C., long before the beginning of Oppolzer's tables. According to *Shu King*, the book of historical documents of ancient China, the eclipse brought bad luck to the two royal astronomers Hsi and Ho, who were supposed to be

prepared well in advance to perform the rituals aimed at scaring away the dragon who was going to swallow the Sun in broad daylight. But somehow Hsi and Ho drank too much rice wine and failed in their duties. Thus they were executed in accordance with the law which stated that astronomers who predicted that an eclipse would happen before the time it actually occurred were to be killed outright without respite, whereas those whose predictions set it for after the actual time were to be slain without reprieve.

A historically registered eclipse of the Sun which coincides with the date given in Oppolzer's tables took place in May 585 B.C. during the savage war between the Lydians and Medians. Both sides were so scared by this heavenly demonstration of displeasure in the behaviour of humans that military action was discontinued and a peace was signed. Similarly the eclipse on 28 June 1451 allegedly prevented the outbreak of war between the Mohawk and the Seneca Indians; the eclipse on 29 February 1504 helped Christopher Columbus subdue his rebellious Indian carriers; and the eclipse (not found in Oppolzer's tables) which occurred at three minutes after noon on 21 June, A.D. 528, helped Mark Twain's Connecticut Yankee to discredit the famous magician Merlin and to re-establish his reputation at King Arthur's court. Probably the most memorable eclipse occurred on 29 May 1919 when two British expeditions, one to Africa and another to Brazil, photographed the stars surrounding the Sun and found that their apparent positions were slightly shifted in the radial direction, in complete agreement with the prediction of Einstein's General Theory of Relativity.

For those readers who may wish to observe an eclipse personally we give a map of solar eclipses for 1963 and the next twenty years (Figure 9).

ATOMS AND THEIR SPECTRA

Those lines of rainbow light
Are like the moonbeams when they fall
Through some cathedral window, but the tints
Are such as may not find comparison on earth.

— *Percy Bysshe Shelley*

FRAUNHOFER LINES

As was mentioned earlier, Isaac Newton was the first to find that sunlight passing through a glass prism produces a brilliantly coloured rainbow ribbon known technically as the spectrum of white light. But it was only about a century later that the German physicist Joseph von Fraunhofer (1787–1826), using better optical techniques, noticed that Newton's rainbow ribbon is marred by a large number of thin dark crosslines (Plate 1). Fraunhofer published a list of almost six hundred of these lines, which are now called by his name. He noticed that exactly the same lines appear in the spectra of the Moon and planets, which are luminous because of sunlight reflected from their surfaces. On the other hand, it turned out that the spectrum of light emitted by white-hot laboratory sources is seen in a spectroscope as a continuous rainbow ribbon without any dark lines whatsoever. This proved that the Fraunhofer lines are of solar origin.

The explanation of the Fraunhofer lines came about fifty years later, when another German physicist, Gustav Kirchhoff (1824–87), noticed that the Fraunhofer line usually denoted by the letter *D* became considerably darker if, before entering the spectroscope, the sunlight had passed through a gas-burner flame that was coloured yellow by the introduction into it of some sodium compound (for example, ordinary table salt). When white light from a laboratory source unmarred by Fraunhofer lines was passed through the yellow-coloured flame, Kirchhoff observed a

strong dark line exactly at the position of the *D*-line in the solar spectrum. But, looking through a spectroscope at the sodium-coloured flame in a dark room, Kirchhoff found a bright yellow line at the same position on the dark background. These findings are shown graphically in Plate 1.

The conclusion from these experiments was inevitable. Apparently the white light emitted by the hot surface of the Sun (the photosphere) passes on its way out through the rarefied gaseous layer (the chromosphere) containing the vapours of sodium, as well as other chemical elements, which absorb specific wave lengths from the originally continuous spectrum. The dark *D*-line is intensified if the sunlight passes later through the sodium-coloured gas flame, and appears as the bright emission line in the spectrum of that flame if observed in a dark room. What is true for the *D*-line of sodium must also be true for various lines emitted and absorbed by the other elements.

Kirchhoff's law stating that *each element in the form of hot gas (or vapour) absorbs the same spectral lines as it is able to emit itself* became one of the basic laws of optics and led to the development of *spectrum analysis*, which permits astronomers to learn the chemical constitution of the Sun and distant stars without getting a sample of their material and giving it to a chemist for analysis with his retorts and test tubes.

RESONANCE PHENOMENA

Kirchhoff's law of the relation between the emission and the absorption of light represents a special case of an important physical phenomenon known as *resonance*. It can most easily be understood in the case of simple acoustical examples. Suppose we have two identical tuning forks having exactly the same pitch, that is, the same number of vibrations per second, mounted on a tuning board (Figure 10). If we strike one with a hammer it will begin to vibrate, emitting the sound of a given pitch. But if before the vibrations die out we touch that tuning fork with a hand, thus stopping its vibrations, we will still hear the sound coming from the other fork, which became 'excited' by the sound waves coming from the first fork and began vibrating on its own.

On the other hand, if the two forks have different vibration frequencies, the second fork remains absolutely undisturbed by the vibration of the first one. The same situation exists in the case of radio waves, and everybody knows that in order to get the reception of a given radio broadcasting station one has to 'tune'

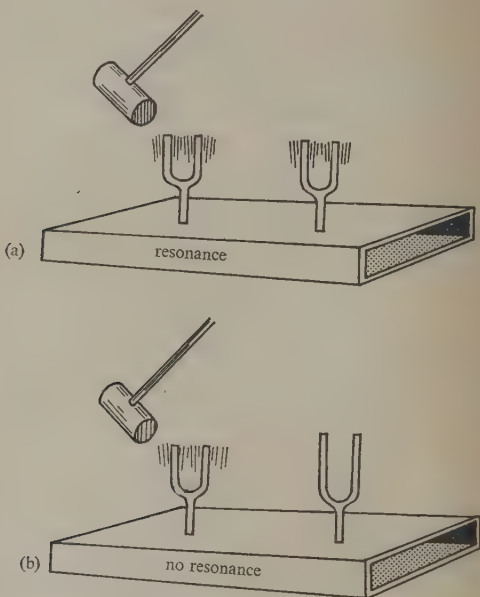


FIGURE 10. Energy exchange between two tuning forks: (a) resonance, if frequencies are equal; (b) no resonance, if frequencies are different.

one's reception set, making the frequency of its own electric oscillations equal to the oscillation frequency of the broadcasting station. In the same way the atoms of gas in the flame, oscillating with the same frequency as the radiation passing through, will resonate to the incoming waves which have the same frequency as that of the atoms in the flame.

Since the energy must be conserved, the oscillators (whether they are atoms, radio sets, or tuning forks) will absorb selectively the energy from the waves passing by, thus reducing the intensity of particular narrow-wave-length regions of the spectrum. In our example of a beam of white light containing all frequencies, passing through a flame containing sodium atoms, these sodium atoms will suck out, from the light waves passing through, the energy corresponding to their own vibration frequency, that is, the frequency corresponding to the *D*-line. However, the light emitted by atoms of sodium 'excited' by the high temperature of the flame will have the characteristic frequency of sodium atoms and will produce in a spectroscope a bright yellow *D*-line. Thus Fraunhofer's lines carry the information concerning the elements which are present in the atmosphere of the Sun, and from their relative intensity we should be able to draw conclusions about the relative amount of different chemical elements present in the Sun's material.

It is proper here to report an incident which, according to a story (probably untrue), took place at a physics examination at a university. When asked what Fraunhofer lines are, one student answered, 'Those are the dark lines crossing the spectrum of the Sun and corresponding to the chemical elements *which are absent* on the Sun.' When the bewildered professor asked him to clarify this statement, the student said that, as far as he understood it, a continuous spectrum of white light is formed by an assortment of all the spectral lines of all the ninety-two chemical elements. Thus, if some elements are absent in the Sun, there will appear dark gaps at the places where these lines should have been. If the reader of this book laughs at the student's explanation, he can rest assured that he has grasped the true significance of Fraunhofer lines.

Since the dark Fraunhofer lines are the result of the absorption within the outer reversing layer of the continuous white light emitted by the solar photosphere, we should be able to observe these same wave lengths as bright emission lines on a black background if, by some miracle, we could stop only for a second the blinding white light of the solar photosphere. Such 'miracles' occur once in a while on the occasions of total solar eclipses. It

happens that, by a sheer cosmogonical coincidence, the apparent angular diameter of the Sun is closely equal to the apparent angular diameter of the Moon, which means, of course, that the diameters of these two celestial bodies are proportional to their distances from the Earth. Thus whenever the Moon comes exactly between us and the Sun it obscures the Sun's brilliant face, permitting us to observe its delicate profile. This situation comes and goes in a flash, and the emission spectrum photographed during this short flash is known therefore as the *flash-spectrum* of the Sun. Instead of dark Fraunhofer lines on the bright background, it shows bright emission lines of the chromosphere standing out against the darkened background. The flash spectra of the Sun stand in about the same relation to its everyday bright-face spectrum as negatives do to positives in photographic work.

ATOMS ARE NOT INDIVISIBLE

Since the notion of an atom was introduced by Democritus some twenty-three centuries ago, and until the end of the last century, atoms were considered as ultimate indivisible building units of matter, and the differences between the properties of the atoms of various chemical elements were ascribed to the differences in their shape. This point of view reached its apogee in the publications of the Swedish physicist J. R. Rydberg, who tried to explain the rapidly accumulating material concerning the physical and chemical properties of atoms entirely on the basis of their geometrical shapes. Rydberg speculated that atoms of hydrogen, which are the simplest of all atoms, must have a perfect spherical shape, whereas the atoms of alkali metals (lithium, sodium, potassium, etc.), which are similar to hydrogen in their chemical and optical properties, must have ellipsoidal shapes with various degrees of elongation. In order to explain the stability of a water molecule, H_2O , Rydberg assumed that the oxygen atom has a toroidal shape (like a doughnut) so that each of the two spherical hydrogen atoms can fit into one side of the cavity. Reactions such as $\text{H}_2\text{O} + \text{Na} = \text{NaOH} + \text{H}$ were interpreted by the assumption that the elongated atoms of alkali

metals can fit into the oxygen atom's cavity better than spherical atoms of hydrogen. Assuming that atoms are elastic bodies of various shapes, Rydberg tried to interpret the observed line spectra of various chemical elements as different modes of the elastic vibrations of the atoms. But the calculated elastic vibration frequencies of spheres, ellipsoids, toroids, etc. did not resemble in any way the observed frequencies in the line spectra.

An entirely different way of studying the atoms was taken at the end of the last century by the famous British physicist Joseph John Thomson (later Sir Joseph), known to his friends and students simply as J. J. He wanted to find out what happens inside the so-called Crookes' tube (invented somewhat earlier by Sir William Crookes), which, being filled by different gases at low pressure and subjected to high electric potentials, emits the bright spectral lines characteristic for the gas used. While the practical development of Crookes' tubes resulted in the modern advertising technique of showing the names of night clubs, cinemas, and other cultural establishments shining in various colours on city streets after dark, the scientific studies of the passage of electricity through gases gave to physicists a deeper insight into the structure and the hidden properties of atoms. Thomson found that when the pressure of the gas in a Crookes' tube is reduced below a certain limit the continuous glow of the gas disappears, but that the glass wall opposite the negative pole (cathode) supplying the electric tension begins to glow with a faint greenish light.

Thomson built another tube in which the positive pole (anode) was moved to the side and the end wall opposite to the cathode was considerably enlarged. Now the entire back wall became fluorescent, and if some metallic object, such as a Maltese cross, was placed in the way, a sharp shadow could be observed. Apparently some mysterious rays emanated from the cathode and rushed along the tube to its opposite end; J. J. called them *cathode rays*. But the cathode rays did not carry any similarity to the X-rays discovered about the same time by the German physicist Wilhelm Röntgen. While the propagation of X-rays was not influenced by a magnetic field, indicating that they were similar to light rays, cathode rays were very strongly deflected when a magnet was brought close to the tube, and the direction

of the deflection indicated that they were swarms of negatively charged particles. But what kind of particles were they? What was the amount of their charge, what was their mass, what were their velocities? Thomson started systematic investigations of the deflection of the mysterious cathode rays by electric and magnetic fields, using the arrangements shown schematically in Figure 11. To cause an electric deflection of the beam, he placed

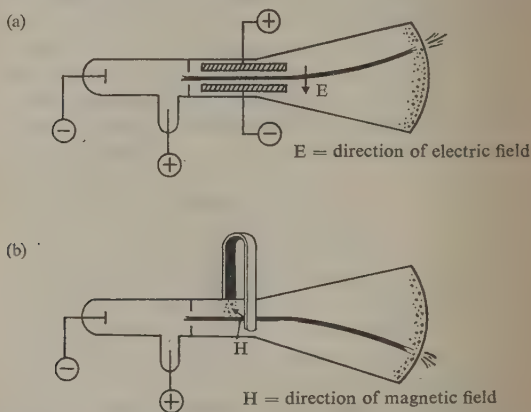


FIGURE 11. J. J. Thomson's method for measuring the velocity v and the specific charge e/m of electrons by sending them through an electric (a) and magnetic (b) field.

inside the tube two metal plates carrying opposite electric charges. The negatively charged cathode-ray particles were attracted to the positively charged plate and repelled from the negatively charged plate with a force proportional to their electric charge, which we shall call e . According to a basic law of mechanics (the second law of Newton), the deflection from the rectilinear motion of a material particle subjected to a certain force is proportional to that force and inversely proportional to the mass, m , of the particle. Thus the deflection of the cathode-ray particles in this experiment must be proportional to the ratio

e/m . In the case of magnetic deflection, a fast-flying electrically charged particle must be considered as an electric current, the strength of which is proportional to the product of the particle's charge, e , and its velocity, v . Since the force exerted by a given magnetic field on an electric current is known to be proportional to the strength of that current, the deflection of cathode rays in this case must be proportional to the quantity ev/m .

Combining these two experiments, Thomson was able to find separately the velocity v and the ratio e/m . While v depends on the electric potential applied to the tube, the ratio e/m , known as the *specific charge* of the particle, is, of course, always the same, and Thomson found that it is numerically equal to 5.28×10^{17} if the charge is measured in so-called electrostatic units* and the mass in grams. This figure was 1,840 times larger than the specific charge of a hydrogen ion (i.e. a charged hydrogen atom) and, since it was expected that in both cases one deals with the same elementary electric charge,† it followed that the particles forming cathode rays are 1,840 times lighter than the lightest of all the atoms. Thomson called them *electrons* and formulated a hypothesis that they form an important constituent part of the atom. He visualized the atom as being a combination of positive electric charge, distributed uniformly throughout the entire atomic body, and a number of negatively charged electrons submerged in it. Electric repulsion between the electrons was supposed to be compensated for by their attraction towards the centre of the positive charge, so that the total system is normally in a state of equilibrium. Under the action of strong electric forces the electric structure of the atom is disrupted; the electrons fly from the cathode to the anode, while the atom, having lost one or more electrons and thus having acquired a positive charge (positive ions), moves in the opposite direction.

Using that model, Thomson tried to explain the origin of line spectra as the result of the vibrations of atomic electrons around their equilibrium positions within the positively charged main body of the atom. However, that picture turned out to be no

* An electrostatic unit is defined as the amount of electricity which acts with a unit force on an equal charge placed at a unit distance from it.

† This assumption was later confirmed by direct experiment.

more fruitful than the earlier attempts to explain the observed spectra by the vibrations of the elastic bodies of various shapes. True enough, one could calculate various frequencies of electric vibration within J. J. Thomson's atomic model, but these frequencies have shown no correlation with the frequencies actually observed.

RUTHERFORD'S MODEL

While Sir Joseph Thomson was trying unsuccessfully to use his model for the explanation of various atomic properties, another great British scientist, Ernest Rutherford, who later received the title of Lord Rutherford of Nelson for his scientific discoveries, was following quite a different approach. In order to obtain information concerning the distribution of positive and negative electric charges inside the atom, he bombarded atoms with alpha particles, the very high-energy projectiles emitted by heavy radioactive elements.

Since alpha particles are positively charged (carrying two elementary charges each), they should be attracted by the negatively charged parts of the atom and repelled by its positively charged parts. But the negative components of atomic structure are the electrons, which are 1,840 times *lighter* than the hydrogen atom, while alpha particles (which are helium atoms stripped of their electrons) are 4 times *heavier* than hydrogen. If electrons were represented by ping-pong balls, an alpha particle would be equivalent to a fast-flying cannon ball 5 inches in diameter. Thus, when an alpha particle passes through or close to an atom, atomic electrons are scattered in all directions without influencing at all the motion of the alpha particle. This disruption of the electron system of an atom hit by a fast-moving heavy projectile is known as the *ionization* of the atom, and is very useful for tracking the path of the bombarding particles. In fact, as was shown by still another British scientist, C. T. R. Wilson, water vapour in the air has a tendency to condense on electrically charged atoms (ions), thus forming thin streaks of fog along the paths of the fast-moving atomic projectiles. The situation is similar to the formation by a jet plane of a white trail in the sky

that can be seen quite clearly, although the plane itself may be almost invisible.

But whereas the motion of alpha particles is practically uninfluenced by their interaction with atomic electrons, it could be greatly influenced by their interaction with the positively charged heavy parts of the atom. According to J. J. Thomson's model, the positive charge and the mass of the atom were supposedly distributed more or less uniformly throughout its entire atomic volume. What would be the electric force exerted by positive-charge distribution on an alpha particle passing through an atom? A comparable problem was first attacked by Newton in relation to gravitational forces, which are similar to electric forces in the sense that both obey the inverse-square law, that is, the force involved is inversely proportional to the square of the distance between the gravitating masses or between the electrically charged particles. Newton showed that the total force with which the Earth acts on a material body placed *outside* its surface (be it the Moon, an artificial satellite, or an apple hanging on a tree) is exactly the same as it would have been if the entire mass of the Earth were concentrated in its centre. If, however, a gravitating particle is placed inside the Earth, for example, a stone dropped into a deep well, the situation changes. The forces exerted on the stone by the masses of Earth above the level of the bottom of the well will cancel one another out and have a total effect of zero. On the other hand, the combined action of the Earth masses located deeper than the well will add up, resulting in the same force as we would have if all these masses were concentrated at the centre of the Earth. Thus, if we make the well deeper and deeper, the gravity force at its bottom will become smaller and smaller as we come closer to the centre of the Earth.

Applying the same argument to an alpha particle approaching and entering an atom having a uniform distribution of positive electric charges, as in J. J. Thomson's model, we would expect the force (in this case, of repulsion) to be represented by the curve shown in Figure 12a; it would have its maximum value when the alpha particle is just entering the atom. Comparing the calculated value of this force with the kinetic energy of the incident alpha particle, one finds that, although the alpha

particle would be deflected from its original trajectory, this deviation would be very small. Rutherford's experiments, in which he bombarded thin foils of different metals with a beam of fast

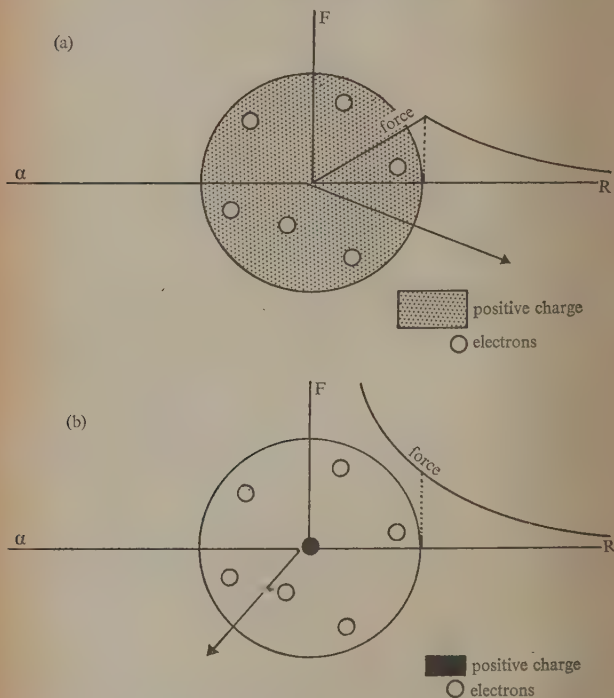


FIGURE 12. Models of the atom: (a) J. J. Thomson's; (b) Rutherford's.

alpha particles, led to an entirely different result. Many alpha particles were deflected by very large angles from the direction of the incident beam, and some of them were thrown practically backward, as if reflected from a solid wall. The only way to explain such a strong force exerted on alpha particles by the

positive charge within the atom was to assume that, instead of being spread thinly through the entire atom volume, this charge (as well as most of the mass) is concentrated in a very small region at the atomic centre (Figure 12b). Rutherford called this concentration of positive charge and mass *the atomic nucleus*. Under the action of the repulsive force changing in inverse proportion to the distance from the centre, alpha particles describe hyperbolic trajectories* and, observing the number of these particles scattered in different directions, one could calculate the value of the positive nuclear charge. It was found in this way that the charges of different nuclei are always integer multiples of the elementary electric charge. Furthermore, these integer values turned out to be equal to the position number of the element in question in the periodic system of chemical elements (atomic number). Thus, for example, in the case of the hydrogen atom, the nucleus (proton) has the same amount of charge as a single electron but, of course, with the opposite sign. In the case of aluminium, which is the thirteenth element in the periodic system, the nuclear charge is $+ 13 e$, and in the case of uranium (the ninety-second element) $+ 92 e$. Since atoms in their normal state are electrically neutral, the discovery meant that the atomic number of any given chemical element corresponds to the number of electrons rotating around its nucleus.

Rutherford's model of an atom led immediately to extremely serious theoretical difficulties. In fact, electrons possessing high electric charge and revolving fast around the nucleus were expected to behave like tiny broadcasting stations, emitting electromagnetic waves into the surrounding space. The estimated wave length of this radiation agreed with the wave length of light, which was reassuring. But what could not be explained was the fact that this electromagnetic radiation would drain the energy of orbital atomic electrons, which as a consequence would spiral inward and fall into the nucleus within about 10^{-8} seconds. Thus, according to Rutherford's model obtained by straightforward experiments, atoms could not exist longer than one hundred-millionth of a second, whereas we know very well that

* In the case of an attractive inverse-square force, as in the case of the planets of the solar system, the trajectories are ellipses.

they exist limitlessly! How so? Why do atoms not collapse, and how can Rutherford's model explain their well-known optical and chemical properties? These questions were answered by a young Danish theoretical physicist, Niels Bohr, who came to work with Rutherford just when Rutherford's experiments were leading to seemingly paradoxical effects. In order to understand Bohr's theory, one should first become acquainted with the memorable events which took place at the turn of this century.

THE QUANTUM OF RADIANT ENERGY

At the end of the nineteenth century classical physics, based essentially on Newtonian mechanical concepts, came to the peak of its progress. Celestial mechanics, the mathematical study of the motion of planets and more distant celestial bodies under the action of gravitational forces, became the most exact of all sciences. Application of the Newtonian laws of mechanics to the motion of tiny molecules forming matter led to a complete understanding of the laws of gases and the kinetic interpretation of heat. Light was explained as short electromagnetic waves, and its interaction with matter was understood in many details. The nature of electromagnetic fields seemed to be understandable in terms of stresses and strains in the world ether, which, although stubbornly refusing to be described as either a fluid or a solid, was still considered as some kind of material medium subject to the laws of classical mechanics.

Then, in the late 1890s, heavy clouds began to appear on the horizon of classical physics, which, as we have said, had been doing quite well, thank you. Lord Rayleigh and Sir James Jeans tried to apply the statistical method, which worked so well in the kinetic theory of gases, to the problems of radiant energy enclosed in a box with perfect mirrors for inner walls. Of course there are no perfect mirrors, and even the best mirror absorbs a little bit of light before reflecting it. But for general theoretical considerations it is permissible to go to the limiting ideal case and consider a closed box with 100-per-cent reflective walls without any trace of absorption. If you shine some light into such a box (Jeans's cube) and then close the lid, the light waves

will travel from wall to wall and back again for an indefinite period of time and will come out with full intensity when the lid is opened again, a second, a year, or a century later. In the case of light waves, such an idealized reflective container plays the same role as a tight, leakproof container plays in respect to the gas filling it.

The kinetic theory of gases, which represented one of the triumphs of classical physics, dealt with a large number, N , of molecules enclosed in a leakproof container and given the total energy E . According to a basic theorem of statistics, known as the law of equipartition of energy, the available energy will be distributed equally among all the molecules so that the average energy of each molecule will be E/N . We say the *average* energy because in statistical considerations one can never get exact data but only averages. In fact, it can be shown that, while in this example the majority of molecules will have the energy E/N , there will always be some molecules which have higher energy and some which have lower.

Rayleigh and Jeans tried to apply the same statistical method to the problem of energy distribution among the different modes of electromagnetic waves which can exist within a mirror-walled container. But here they encountered a fundamental difficulty. While the total number of molecules in a container, even though very large, is represented by a finite figure, the number of different types of vibration within the container is truly infinite. Indeed, let us consider for simplicity's sake only those waves which run to and fro between the right and left walls of Jeans's cube. These waves must fulfil the boundary conditions which state that at the walls the amplitude vanishes (just as in the case of a violin string fastened at its two ends). The possible vibration states are shown in Figure 13. The longest wave will have just two nodes (the points of vanishing amplitude) at both ends. The second wave will have one more node in the middle: three nodes altogether. The third wave will have four nodes; the fourth, five nodes, etc. Since there is no lower limit to the length of electromagnetic waves, there will be no limit to possible wave patterns, with thousands, millions, billions, etc. of nodes. If we add the waves running up and down, forward and backward, and at any

other odd angle within the container, we will have only a 'more infinite' number of them.*

If the law of equipartition of energy holds in the case of electromagnetic waves within a mirror box (and according to classical physics there was no reason why it shouldn't) any given total energy E of the waves in the container will be shared by the infinity of different modes of vibrations, so that the share received

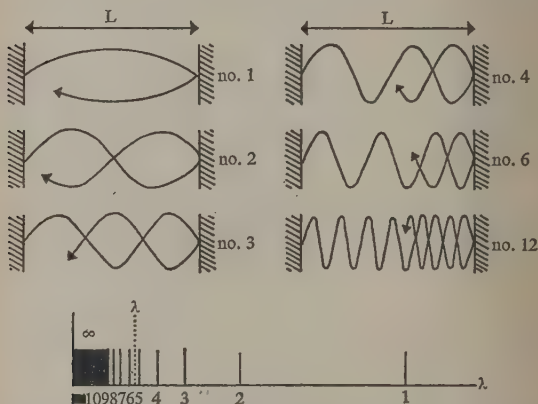


FIGURE 13. Different modes of electromagnetic vibrations in Jeans's cube for dimension only, and the corresponding spectrum of possible wave lengths.

by each particular mode (described by the number of nodes) will be $E/\infty = 0$. The consequences of this situation would be quite catastrophic, as can be seen from the following argument.

Suppose we select a certain mode of vibrations which has, let us say, 200 nodes. There are, in this case, 199 modes with a lesser number of nodes, but there is always an infinite number of modes with a larger number of nodes. Thus, according to the equipartition law, all the available energy will go into the vibra-

* It is known that, if one multiplies infinity by any large number or even by another infinity, one still gets the same infinity. (See the author's book, *One Two Three . . . Infinity*, New York (The Viking Press), 1961.)

tion modes of shorter wave length or, which is the same, of higher frequency (number of vibrations per second). In physical terms, this means that the red glow of the coal burning in a fireplace will turn into yellow, green, blue, violet, ultra-violet, X-rays, gamma-rays, and all the way up to infinitely high frequency and infinitely short wave length. This is apparently a nonsensical conclusion, and we are spared by the laws of nature from being killed by radiation sickness just by putting a match to the wood logs in our fireplace!

When a theory comes into such a sharp conflict with our knowledge concerning the nature of things, the theory and not the nature of things must be wrong. Apparently some additional, hitherto unknown factors must be brought into the picture in order to straighten things out. The way out of the difficulty, which in those days was known as the 'ultra-violet catastrophe', was suggested during the last week of the last month of the last year of the last century* by the German physicist Max Planck (1858-1947). He suggested that the paradox could be eliminated if the law of equipartition of energy is amended by the condition that the amount of energy possessed by any single vibration mode cannot be smaller than a certain value which can be called *a quantum of radiant energy*. Planck proved that, in order to fit the theoretical curves of the energy distribution with the experimental curve of emission of hot bodies, it is necessary to assume that this minimum amount of radiant energy be equal to

$$h\nu$$

where ν is the frequency and h a universal constant known as the quantum, or Planck's, constant. Using Planck's amendment, it is possible to derive theoretically the two basic laws of radiation, the Stefan-Boltzmann and Wien's laws, discussed in the previous chapter.

The numerical value of the quantum constant is exceedingly small (10^{-27} in centimetre-gram-second units), so that the minimum amount of energy in one quantum of light is negligible as compared with the intensity of light we can perceive. Thus, a 60-watt electric bulb emits about 10^{20} light quanta per second.

* At the Christmas meeting of the German Physical Society in 1900.

Therefore, while reading this text under electric light or sunlight, you should not be disturbed by the fact that light is a swarm of very small energy packages, instead of a continuous flow of energy, any more than one should be worried because the water one drinks consists of a very large number of individual molecules.

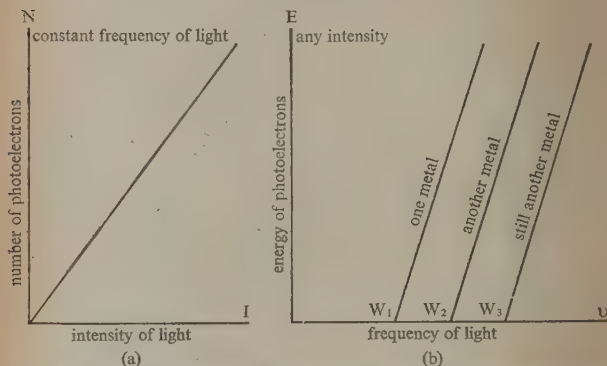


FIGURE 14. Experimental relations found in studies of the photoelectric effect: (a) The number N of photoelectrons depends on the intensity I of the light. (b) The energy E of photoelectrons depends on the frequency ν of the light.

During the first decade of the present century Planck's idea of radiation quanta (which today are called photons) received brilliant confirmation in the study of various phenomena pertaining to the interaction between radiation and matter. In 1905 Albert Einstein showed that the photoelectric effect, that is, the ejection of electrons from metallic surfaces illuminated by violet and ultra-violet light, can be understood only if one makes an assumption that light energy arrives in the form of individual packages as predicted by Planck. In fact, experimental studies of photoelectric effect have shown that the number of ejected electrons (photoelectrons) for the incident radiation of a given frequency increases in direct proportion to the intensity of that

radiation (Figure 14a). This means, of course, that a strictly defined amount of radiant energy is needed to eject one electron from the metal. If, however, one uses radiation of different frequencies, the energy of the photoelectrons increases linearly with the frequency, indicating that the amount of energy communicated to an electron in each case is proportional to the frequency (Figure 14b). The numerical value of the coefficient of proportionality turned out to be exactly equal to the quantum constant h , introduced by Planck for the explanation of a quite different physical phenomenon, that is, distribution of energy in the radiation spectrum.

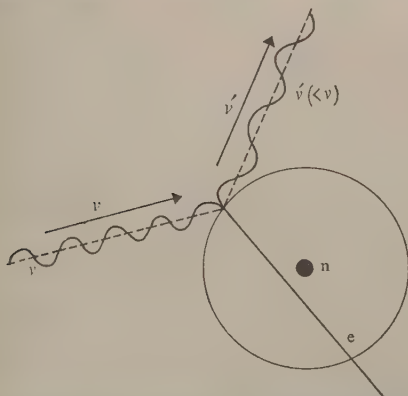


FIGURE 15. The Compton effect, in which a high-energy quantum (ν) ejects an electron from an atom and rebounds with less energy (ν'). The energy of light quanta is so high that the weak binding of the electron to the nucleus can be neglected.

In about 1923 the American physicist Arthur Compton discovered an effect, now bearing his name, that occurs when X-rays interact with free electrons. Compton was able to show that the scattering of X-rays, as well as the recoiling of electrons, takes place in exactly the same way as would be expected for a collision between two elastic balls: a photon and an electron (Figure 15).

These and other experiments left no doubt that radiant energy exists and propagates in the form of individual packages, the energy carried by each package being proportional to the frequency involved.

BOHR'S QUANTUM ORBITS

In order to explain why Rutherford's model of an atom does not collapse in a fraction of a second, the Danish physicist Niels Bohr extended Planck's hypothesis concerning radiant energy to the mechanical energy of electrons circling around the atomic nucleus. If the energy of radiation is quantized, that is, can be had only in packages of well-defined size, argued Bohr, we should also quantize the motion of atomic electrons. Thus the atom, which was previously compared with a miniature planetary system, began to look more like the transmission box in an automobile. Your car can travel in the first (lowest) gear, or in the second, third, or fourth gear, but you cannot put it into first-and-a-half or into second-and-three-quarters gear. Similarly, according to Bohr, an electron in an atom may have only discrete states of motion which can be numbered by the integers 1, 2, 3, etc., which are known as quantum numbers. A peculiarity of this model is that Bohr's atomic gearbox has no neutral gear. The electron within the atom can never stop moving, and the lowest energy corresponds to the first circular orbit (quantum number 1) which is located nearest to the nucleus. Higher energy states correspond to the circular orbits of larger diameters.

The quantum conditions used for selecting the 'permitted' orbits of the atomic electron are somewhat more complicated than those in the case of radiation, and we need not go into that subject in much detail. It is sufficient to say that the quantity which must assume integer values in the case of mechanical systems is the so-called 'action' divided by Planck's constant h . Whereas the notions of the velocity and the energy of a moving body are familiar to any person who has had no special training in theoretical mechanics, the notion of action may sound somewhat odd. Mechanical action has little to do with such common expressions as 'military action' in war or 'Lights, camera, action!'

in a motion-picture studio. It is defined simply as a product of available energy and the period of time during which this energy is available. For any orbital motion of an electron in an atom, the mechanical action is a certain number, and, according to Bohr's postulate, that quantity divided by the quantum constant h must be an integer. This condition selects from all possible orbits of an atomic electron only a discrete set of permitted ones and makes the atom behave somewhat similarly to an automobile's gearbox.

When the hydrogen atom is in a 'normal' state, that is, in the state of lowest energy, its electron moves in the circular orbit closest to the nucleus. When the atom is 'excited', that is, receives some extra energy from the outside (for example, when hit by an electron in a discharge tube), the electron is lifted to one of the higher orbits. There is a classical expression, 'What goes up must come down,' and the excited electron falls back on its original orbit, releasing its excess energy in the form of radiation. But, since the radiant energy can exist only in the form of photons, the transition of an electron from the state of higher mechanical energy E_2 to the state of lower mechanical energy E_1 results in the emission of a light wave with the frequency defined by the condition:

$$h\nu = E_2 - E_1$$

or:

$$\nu = \frac{1}{h}(E_2 - E_1)$$

Thus the transitions of an electron between different quantum states in an atom leads to the emission of a set of different light waves with well-defined frequencies, which will show as sharp lines in a spectroscope.

Bohr was able to show that the energies of the electrons moving along various permitted circular orbits in a hydrogen atom are inversely proportional to the squares of their ordinal numbers. Thus, if the energy of the electron in the 1st (innermost) orbit is E_0 , the energies in the 2nd, 3rd, 4th, 5th orbits should be:

$$\frac{E_0}{4}; \frac{E_0}{9}; \frac{E_0}{16}; \frac{E_0}{25}; \text{etc.}$$

Substituting these values in the previously given formula, we obtain for the frequency of light emitted by an electron in the transition from the n th to the m th quantum orbit the values:

$$\nu_{n \rightarrow m} = \frac{E_0}{h} \left(\frac{1}{m^2} - \frac{1}{n^2} \right).$$

This theoretical conclusion stood in complete agreement with an empirical rule discovered by spectroscopists a few decades earlier that the observed frequencies of different spectral lines emitted by a hydrogen atom are proportional to:

$$\left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

where m and n are integers.

If we take $m = 1$ and ascribe to n the successive values 2, 3, 4, etc., we will get a series of spectral lines with the frequencies proportional to:

$$\begin{aligned} \left(\frac{1}{1^2} - \frac{1}{2^2} \right) &= \left(1 - \frac{1}{4} \right) = \frac{3}{4} \\ \left(\frac{1}{1^2} - \frac{1}{3^2} \right) &= \left(1 - \frac{1}{9} \right) = \frac{8}{9} \\ \left(\frac{1}{1^2} - \frac{1}{4^2} \right) &= \left(1 - \frac{1}{16} \right) = \frac{15}{16}. \end{aligned}$$

These lines, which were actually found by the British spectroscopist Theodore Lyman, are located in the ultra-violet part of the hydrogen spectrum and are known as the Lyman series.

If we take $m = 2$ and ascribe to n the values 3, 4, 5, etc., we obtain the familiar lines in the visible part of the hydrogen spectrum shown in Plate 1. It was a German schoolteacher named J. J. Balmer who published in 1885 the famous paper showing that the visible hydrogen lines can be represented by the series:

$$\begin{aligned} \left(\frac{1}{2^2} - \frac{1}{3^2} \right) &= \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5}{36} \\ \left(\frac{1}{2^2} - \frac{1}{4^2} \right) &= \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{3}{16} \\ \left(\frac{1}{2^2} - \frac{1}{5^2} \right) &= \left(\frac{1}{4} - \frac{1}{25} \right) = \frac{21}{100} \end{aligned}$$

For $m = 3$ and $n = 4, 5, 6$, etc., we get a series of lines in the infra-red part of the spectrum. This series was discovered by Friedrich Paschen and hence carries his name.

In Figure 16 the energy levels of the electron in a hydrogen atom for different values of m and n are shown. The vertical arrows indicate the various possible transitions with the emission

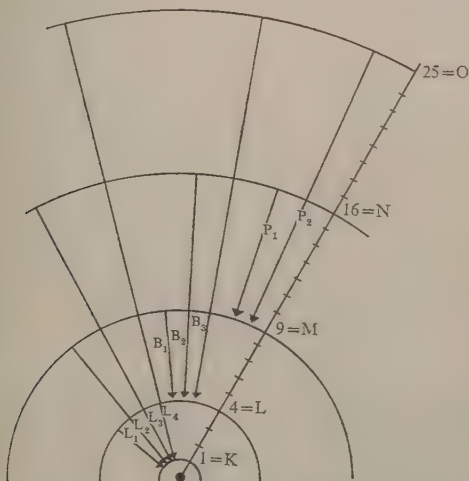


FIGURE 16. The first five circular orbits (K, L, M, N, O) in Bohr's atomic model. The arrows indicate the quantum transitions which lead to the Lyman (L_1, L_2, L_3, L_4), Balmer (B_1, B_2, B_3), and Paschen (P_1, P_2) series.

of light quanta of the corresponding frequencies. The energy levels for the first five circular orbits are denoted by the letters K, L, M, N, and O, according to the traditional spectroscopic designation. The transitions ending at K, the innermost or first orbit, correspond to the Lyman series, those ending at L, the second orbit, are the Balmer series, and those ending at M, the Paschen series.

Thus with a single stroke of intuition Bohr was able to disentangle the complicated results of experimental spectrometry, and to explain all the observed emission frequencies through a simple assumption that the mechanical momentum of atomic electrons must be an integral multiple of the fundamental quantum constant h .

The ideas of Bohr were further extended by the German physicist Arnold Sommerfeld, who conceived the idea that apart from circular quantum orbits there may also be elliptical ones. While a circle can be described by a single quantum number, an ellipse requires two. Sommerfeld's set of quantum orbits is shown in Figure 17. It turned out, however, that the energy of an electron moving along a Sommerfeld elliptical orbit depends only on the sum of two integers, one of them describing the angular motion of the atomic electron and the other its radial motion. Thus, although theoretically more elaborate than Bohr's system of circular orbits, Sommerfeld's system led to practically the same empirical results as the original Bohr theory. But small differences between the two theories resulted in the explanation of the experimentally observed splitting of spectral lines.

THE PERIODIC SYSTEM OF ELEMENTS

Bohr's quantum model of an atom not only led to the explanation of atomic spectra but also resulted in the clarification of the chemical properties of atoms. During the second half of the last century the Russian chemist Dimitri Mendeleev (1834–1907) and the German chemist Lothar Meyer (1830–95) had discovered independently of each other that, if the chemical elements are arranged in a sequence according to their atomic weights, various chemical properties repeat periodically along the sequence. In the modernized version of this sequence its tempo is indicated by the periodic appearance of the *inert* (or *noble*) *gases* helium, neon, argon, krypton, xenon, etc., which stubbornly refuse to enter into ordinary chemical reaction either among themselves (thus forming diatomic molecules) or with other chemical elements. Each noble gas is followed in the sequence by a representative of the so-called *alkali metals* (lithium, sodium, potassium,

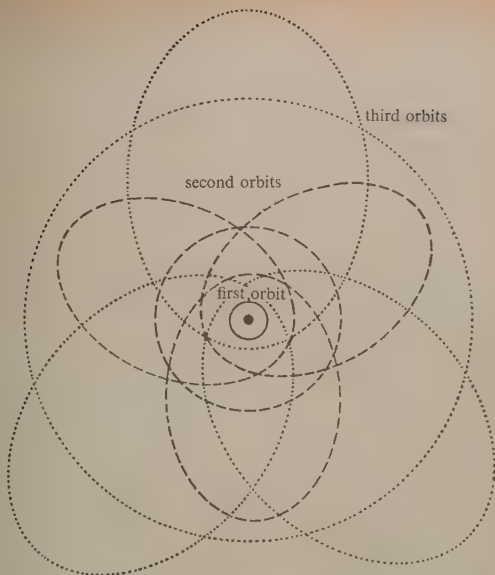


FIGURE 17. Sommerfeld's orbits in a hydrogen atom. The first circular orbit (solid line) corresponds to the lowest energy state. The next four orbits, one circular and three elliptical (dashed lines), correspond to a higher state; the electrons moving in these orbits will each have an equal amount of energy. The next nine orbits (dotted lines), only four of which are shown in the figure, correspond to still higher energy (the same for all nine).

etc.), all of which have very similar chemical properties. At the end of each period stand the so-called *halogens* (fluorine, chlorine, bromine, iodine, etc.) which also show great similarities to one another. The periodic system of elements, which is most conveniently represented on the surface of a cylinder, is shown in Figure 18.

Before Bohr the periodicity of chemical and physical properties of elements presented a deep mystery both for chemists

and for physicists. But with the introduction of the discrete energy levels of atomic electrons, the entire picture suddenly became as clear as crystal. One of Bohr's students named Wolfgang Pauli had introduced what could be called a 'house rule' for atomic electrons, which is now commonly known as the Pauli exclusion principle. According to this rule, each of the quantum orbits can hold no more than two electrons, and, once it is occupied, other electrons must be quartered at orbits with higher quantum numbers. Later, when it was discovered that electrons should be considered as tiny spinning tops, the Pauli principle was amended by the requirement that two electrons occupying the same orbit must spin in opposite (either clockwise or counter-clockwise) directions.

Using the quantization rules and the exclusion principle, one can predict the distribution of atoms between the quantum states, as one proceeds from the simplest hydrogen atom, which has only one electron in the innermost orbit, or K-shell, to the heavier elements, which have more and more atomic electrons. In the case of helium we have two electrons which can move along the same orbit, provided they spin in opposite directions. This completes the K-shell, and the third electron, which is added in lithium, has to move in one of the orbits belonging to the L-level with a considerably higher energy. As we go through the atoms of beryllium, boron, carbon, nitrogen, oxygen, and fluorine, we find more and more electrons accumulated in L-quantum orbits. Considering that the set of L-orbits comprises the circular and the elliptical ones, and that the later ones can be differently oriented in space (also in quantized fashion), one finds that the L-shell contains four different states, which can accommodate a total of eight electrons. Thus, when the eighth electron is added to the atom in neon, the L-shell becomes completely occupied. Therefore in the element sodium, which follows neon, the extra electron is accommodated on the third-level group M and, as we go farther and farther in the sequence of elements, more electrons are added to that level before it becomes in its turn filled up. This process continues, becoming, however, somewhat more complicated, all the way to the end of the natural system of chemical elements.

We notice that one obtains a noble gas when the outer shell is completely occupied. Evidently under these conditions the atom is completely 'satisfied' to be alone and does not 'want' to unite with others. When we have uncompleted shells the situation is different. The third electron in a lithium atom sits all alone in the L-shell, whereas the seven outer electrons in the L-shell of fluorine lack one electron to form a comfortably saturated configuration. Thus the outer electron in lithium leaves its lonely shell and joins the company of seven outer electrons in fluorine. As a result, the lithium atom acquires a positive electric charge (becomes a positive ion), while fluorine acquires a negative charge (becomes a negative ion). Under the action of the electric force between the two ions they stick together, forming a molecule of lithium hydrate. As can be seen from the periodic table, the oxygen atom has only six electrons on the outer shell and in order to saturate it must combine with two hydrogen atoms, thus forming a molecule of water. Similar considerations explain the valence properties of other chemical elements. In Figure 16, pertaining to a single electron in a hydrogen atom, the energy levels belonging to the same shell are shown by a single line, because various circles and ellipses belonging to that shell have the same energy. The preceding statement is not quite true, however; precise optical measurements show that all shells (except the K-shell) consist of several very closely located levels. This splitting is due to the relativistic change of the electron mass with its velocity, and to the effect of electron spin, and is extremely small in the case of hydrogen.

Let us consider the atom of lithium. Here two inner electrons move very close to the nucleus, while the third electron is much farther out. Thus this electron will 'feel' the attraction of the triple positive charge of the nucleus and the double repulsion of the two K-shell electrons. The attraction is partially compensated for by repulsion, and, as a result, the outer electron finds itself moving in a field of forces only slightly different from that produced by a single positive charge. Thus its motion and energy states will not differ very much from those of a single electron in hydrogen. The same is true for other alkaline metals in which the outer shell contains all but one electron necessary for satura-

tion. Since in the optical excitation of an atom it is the outer electron which is lifted to higher energy levels and emits light coming back, one should expect that the optical spectra of alkali metals are similar to that of hydrogen. And so indeed they are. In the case of the elements whose atoms contain two or more electrons on the outer shell, the situation becomes more complicated, but using general theoretical considerations and direct optical observations, spectroscopists are able to disentangle the complicated system of lines in their spectra.

WAVE MECHANICS

During the first decade of its existence Bohr's theory of atomic structure resulted in a saga of glorious successes in the understanding of matter. But toward the end of this eventful period serious difficulties began to accumulate in its further development, and it became more and more evident that Bohr's model represented only a skeleton of the atom and that some flesh should be added to it. In the year 1924 a French theoretical physicist, Prince Louis de Broglie, conceived a seemingly fantastic idea that different quantum orbits in Bohr's atomic model correspond to different modes of vibrations in some kind of 'out-of-this-world' fluid surrounding the atomic nuclei. Thus the atom began to look rather like a pipe-organ or a saxophone, rather than an automobile's transmission box, and instead of different gears the discrete energy states became analogous to different tones and overtones of a musical instrument. Probably the nearest acoustical analogy to modern views on the structure of the atom is presented by the so-called spherical resonators studied a century ago by the German physicist Hermann von Helmholtz. They are simply metal spheres of different sizes with a little opening drilled in the wall. When a tuning fork is placed near the opening the air in the sphere begins to vibrate and, if the frequency of the tuning fork coincides with one of the possible vibration modes of the air inside of the sphere, an increase in sound (resonance) is observed. The layer of air alongside the metal surface remains motionless, being held by the rigid wall, just as are the two ends

of grand-piano strings. We say that we have here a nodal surface of acoustical waves. Because of the spherical symmetry the air in the centre also remains motionless and we say one has here a nodal point. In the case of the first overtone there exists a nodal sphere between the centre and the surface; in the case of the second, there are two. In addition to radial waves, there are also waves running around the centre either in an equatorial or meridional direction. These waves also have discrete frequencies corresponding to different numbers of nodal planes passing through the centre. Mathematically the motion of air in a Helmholtz resonator is represented by the product of two functions: spherical harmonics, which depend on the angular position; and so-called Bessel functions, which depend on the distance from the centre.* Various combined tones calculated mathematically coincide with experimentally measured resonance frequencies.

The vibrations of the mysterious medium surrounding the atomic nucleus can be calculated in a very similar way. Of course, the atom is not surrounded by a rigid spherical shell, but the electric attraction of atomic electrons by the nucleus (which can be compared with a steep slope) plays effectively the same role, preventing electrons from getting away. The angular solutions in this case remain the same as in a rigid sphere (i.e. spherical harmonics), while the radial dependence is given by other functions known as Lagrangian polynomials. Generalizing the original de Broglie idea and writing down what is now known as a wave-mechanical equation, the Austrian physicist Erwin Schrödinger proved in 1926 that the proper vibration frequencies of the electron waves surrounding the proton in a hydrogen atom coincide exactly with the energy levels as calculated on the basis of Bohr's theory which, in turn, coincided with the results of observation.

What is the physical meaning of de Broglie waves and what is their relation to Bohr's orbits? The accepted interpretation is

* 'Spherical harmonics' and 'Bessel functions' are similar to familiar trigonometric functions, such as sines and cosines, being, however, somewhat more complicated, since they pertain to vibrational motion in three dimensions.

that the highest intensity of these waves at any point of the space to which they are confined gives us the probability of finding the electron at that particular point if we carry out a special experiment designed for that purpose. They are, so to speak, 'waves of probability' and do not correspond to any physical reality, as, for example, sound waves. The nearest analogy is probably the case of shooting bullets into a target. Maybe only one or two bullets will hit the bull's-eye, while the others will be scattered among different concentric rings. The number of bullets expected to hit various rings is given by the famous 'error formula' derived a long time ago by the German mathematician Karl Friedrich Gauss. Gauss's formula can be written on paper but cannot be shot from a rifle. Nevertheless, in all processes where one tries to attain certain goals, such as shooting at a target or winning a lottery, or playing a card game, the results are always described by that formula. The radii of Bohr's quantum orbits turned out to be the *average distances* of the atomic electrons from the nucleus, in the same sense as one can determine the average distance of bullet holes from the bull's-eye after many shots. Thus de Broglie waves put flesh on the skeleton of Bohr's orbits.

If the de Broglie waves of probability exist for atomic electrons, they must also exist for electrons flying free through space. And, if so, the beams of electrons in vacuum tubes must show characteristic optical phenomena of diffraction, as in the case of light waves. Soon after de Broglie's proposal, experiments conducted by two American physicists, C. J. Davisson and L. H. Germer, and by a Briton, George Thomson (later Sir George), showed, indeed, that electron beams are subject to characteristic diffraction phenomena. The new theory, known as wave mechanics, confirmed all the predictions of Bohr's earlier theory but also explained a large number of phenomena which Bohr's theory was unable to tackle. One of the advances in our understanding of matter brought about by the theory of wave mechanics pertains to the theory of nuclear transformations and is discussed in one of the following chapters.

The gnosiological meaning of the waves of probability was given by the German physicist Werner Heisenberg in terms of

his famous 'principle of uncertainty'. A discussion of this topic would lead us too far away from the subject of this book, and the interested reader is referred to other books dedicated entirely to physics.*

* For example, the author's book *Biography of Physics*, New York (Harper), 1961.

THE SUN'S TURBULENT SURFACE

There agitate the waves of fire
 That seek but never find a shore;
 There incandescent winds expire
 As in the eons gone before;
 There stones like water boil; on all
 The scorching rains forever fall.

— *Mikhail Lomonosov**

Translated by Barbara P. Gamow

WHAT IS THE SUN MADE OF?

As was stated in the first chapter, the surface temperature of the Sun is about 6,000 degrees Kelvin, and it is thought to increase rapidly as one goes deeper into the body of the Sun. This temperature is well above the evaporation point of all chemical elements, so that we must conclude that the material of the Sun is entirely in a gaseous state. Since, however, the mean density of the Sun is 1.4 in respect to water, these gases must be highly compressed and their atoms tightly packed in the available space. The tightly packed atoms lose their ability to emit harmonious optical tones and their high thermal agitation results in an optical noise, that is, in the continuous spectrum of white light. But the outermost layers of solar material, on the boundary between the dense solar body and the surrounding empty space, are more rarefied and resemble gas flames which we study in our laboratories. When the continuous spectrum of the inner layers (photosphere) passes through the thinner outer layers (chromosphere), selective absorption of certain characteristic wave lengths takes place in much the same way as when light emitted by the hot filament of an electric bulb passes through the flame of a gas burner. This results in the Fraunhofer lines which give us the information on the chemical constitution of the solar atmosphere. All the spectral

* Russian scientist and poet of the eighteenth century.

lines of the elements known on Earth in reasonable abundance are also found in the absorption spectrum of the chromosphere, proving that the Sun is made essentially of the same chemical elements as our Earth.

But it would be a mistake to conclude that the abundance of various elements on the Sun is given by the relative intensity of the corresponding Fraunhofer lines. In fact, the intensity of absorption of various atoms depends in a high degree on the physical conditions to which they are subjected. Take, for example, neon atoms. If we shine white light through a glass tube containing neon and look at it through a spectroscope, we will see a continuous spectrum without any dark absorption lines whatsoever. On the other hand, neon gas becomes luminous and emits bright red light if an electric discharge is passed through it, as is done in neon advertising signs. Does it contradict Kirchhoff's law? Not at all. In the first case all neon atoms remain in the normal state of lowest energy, and the energy of the white-light quanta does not suffice to raise the electrons from this state to the higher state. Thus no quanta of the light passing through are absorbed, and no dark lines appear in the spectrum. However, when a high-tension discharge is sent through a neon-filled tube, the electrons rushing through the tube possess enough energy in collisions with the neon atoms to excite them and to lift atomic electrons to the higher energy levels. In returning to the ground state, the excited electrons in the neon atoms emit the light quanta, producing a bright red line characteristic of neon in the spectroscope. And if we shine, through a luminous neon tube, white light of high intensity, we will see in the spectroscope a bright continuous background with a dark line at the place where the red emission line of neon was, just exactly as Kirchhoff's law says. Instead of exciting neon atoms by a high-tension electric current, one can excite them by bringing the gas to a temperature of several thousand degrees, at which thermal collisions between the atoms are sufficiently violent to cause the excitation. That is exactly what happens in the Sun's chromosphere, causing the formation of Fraunhofer lines.

At this state of excitation neon atoms begin to emit the characteristic red line, and the intensive white light from the photo-

sphere passing through will show a dark Fraunhofer line, since the amount of light of that wave length absorbed from the continuous spectrum exceeds the comparatively weak emission of gaseous neon.

What is true for neon is, of course, true for all other chemical elements, but since different atoms have different *excitation potentials* (i.e. the work necessary to bring them from the normal to the excited state), their spectra will be differently affected by the temperature.

By still stronger impacts the electron can be thrown clear out of the atom and the material becomes 'thermally ionized'. Here again we encounter large differences between the atoms of various chemical elements. Table 1 gives the values of *ionization potentials* (i.e. the work necessary for the extraction of an electron) for some elements of astrophysical significance. Columns I, II, III, etc. give the work necessary for the extraction of the first, second, third, etc. electrons from the atom in question, thus leaving a singly, doubly, triply, etc. positively charged ion. The values are given in so-called 'electron volts', one of which is the energy obtained by one electron from the electric tension of one volt. At the same density and pressure, atoms with lower ionization potential become ionized at lower temperatures, where they are subjected to more moderate collisions, than atoms having a higher ionization potential.

We notice first of all that the ionization potential of noble gases (helium, neon, and, in a way, argon) is very high, because their atoms possess completed electron shells which are not too easy to break. On the other hand, all metals have exceptionally low ionization potentials, indicating that their outer electrons are bound very loosely. This stands in good agreement with the fact that all metals are good conductors of electricity. Indeed, the electrical conductivity of metals is due to the presence of the so-called 'free electrons' which are detached from their atoms (not by high temperature but by close packing of atoms in the metal crystal lattice) and flow through a metallic wire if electric potential is applied at its two ends. We also notice that the more electrons one tries to detach from the atom the more difficult it becomes.

The difference in ionization potentials of different elements is of

paramount importance in the interpretation of Fraunhofer lines, as can be demonstrated by the simple example of hydrogen. At the Sun's surface temperature of 6,000 degrees Kelvin, hydrogen atoms are almost completely ionized, that is, they lose the single electron they possess. While other atoms, having more electrons,

TABLE 1 *The Ionization Potentials of Some Abundant Elements (in ev)*

Element	Degree of Ionization						
	I	II	III	IV	V	VI	VII
Hydrogen, H	13.530						
Helium, He	24.46	54.14					
Carbon, C	11.212	24.26	47.64	64.17	390.12	487.55	
Nitrogen, N	14.46	29.44	47.20	77.04	97.40	549.08	663.73
Oxygen, O	13.55	34.94	54.63	77.03	113.30	137.42	735.20
Neon, Ne	21.47	40.91	63.30	97.00			
Sodium, Na	5.11	47.06	71.30				
Magnesium, Mg	7.61	14.96	79.74				
Aluminium, Al	5.96	18.73	28.31				
Silicon, Si	8.11	16.26	33.33	44.92	165.66		
Sulphur, S	10.31	23.3	34.9	47.08	63.00		
Argon, A	15.69	27.49	40.48	61.00	78.00		
Potassium, K	4.32	31.67	45.5				
Calcium, Ca	6.09	11.82	51.0				
Titanium, Ti	6.81	13.6	27.6				
Vanadium, V	6.72	14.1	29.6				
Chromium, Cr	6.74	16.6	31				
Manganese, Mn	7.41	15.56	34.4				
Iron, Fe*	7.83	16.16	30.48				
Cobalt, Co	7.84	16.9					
Nickel, Ni	7.61	18.4					
Strontium, Sr	5.67	10.98					

* The high ionization potentials of iron, which are very important in the interpretation of Solar corona spectrum (see later in this chapter), are: Fe VIII, 150.4; Fe IX, 233.5; Fe X, 261.0; Fe XI, 288.9; Fe XV, 454.0.

may lose one, two, or more of them and still have some left, the singly ionized hydrogen becomes just a bare nucleus (a proton) which does not interact with light waves. Thus the hydrogen absorption lines must be all but absent in the Fraunhofer spectrum. The fact that these lines are still observed, even though as extremely faint ones, indicates that hydrogen must be very abundant on the Sun. Indeed, more exact calculations indicate that

hydrogen forms about 50 per cent (by weight) of the solar atmosphere.

Another interesting example is presented by helium, whose atoms have normally two electrons. At the Sun's surface temperature one of these two electrons is usually stripped away, and helium is present in the form of positive ions He^+ , consisting of a nucleus carrying a double elementary charge ($Z = 2$), and a single electron rotating around it. According to Bohr's theory, the energy levels of He^+ ions will be given by the formula

$$E_n = Z^2 R \frac{1}{n^2} = 4R \frac{1}{n^2}$$

where R is a constant and n is the quantum number of the orbit. Using the formula, we obtain the same spectral lines as in the case of hydrogen atoms, in which we double the value of n .*

Thus in the Fraunhofer spectra the lines due to ionized helium may be (and indeed have been) mistaken for the lines of the non-ionized hydrogen. The correct assignment of these lines was made only after it was noticed that they have a slightly different wave length, because of the difference between the masses of hydrogen and helium nuclei. The lines of the non-ionized helium in the Sun's spectra are very faint indeed, which originally led to the conclusion that the abundance of helium in the Sun's atmosphere is very low. This is amusing, since the element helium was first discovered in the Sun's atmosphere and its name was derived from the word *helios*, which is the Greek word for the Sun. We know now that this conclusion was all wrong. In the Sun's atmosphere helium is just about as abundant as hydrogen, but, because of the high degree of its thermal ionization, normal helium lines in the Fraunhofer spectrum are very weak.

As a third example, let us take the case of calcium, which normally has 20 electrons per atom. At a temperature up to 3,000 degrees Kelvin, the atoms in the calcium vapour maintain all their 20 electrons. As the temperature increases, thermal collisions between calcium atoms strip them of one of their electrons, forming the positive calcium ions Ca^+ . At the temperature of about 4,000 degrees one half of the calcium atoms have become calcium ions,

* Indeed: $4R \frac{1}{(2n)^2} = 4R \frac{1}{4n^2} = R \frac{1}{n^2}$.

and at 5,000 degrees the number of non-ionized calcium atoms drops practically to zero. At a temperature of 6,000 degrees Kelvin, which is the temperature of the Sun's surface, calcium exists exclusively in the form of singly ionized atoms. If the temperature rises still higher (as is the case in some hotter stars), the second calcium electrons begin to be stripped off, and the process is practically completed at 9,000 degrees, at which most of the calcium atoms exist in the double ionized state, i.e. as Ca^{++} ions. At further increases of the temperature, Ca^{+++} ,

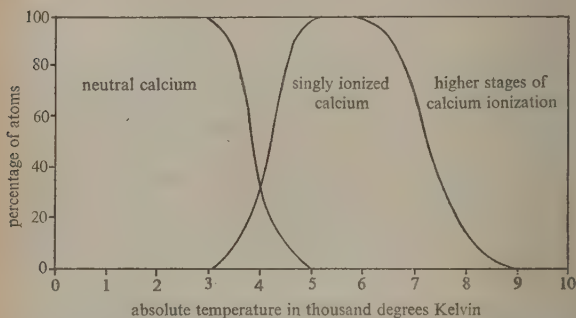


FIGURE 19. The percentage of neutral, singly ionized, and more highly ionized calcium atoms as a function of the absolute temperature. These curves pertain to the condition when the pressure of electron gas is 0.00001 atmospheres.

Ca^{++++} , etc. ions are formed. In Figure 19 we show the changes of relative amounts of different calcium ions with the increasing temperature of the gas.

Since the non-ionized calcium atoms and the calcium ions with different numbers of electrons missing have different absorption properties, the increase of the temperature of the gas results in great changes of the Fraunhofer spectra for the hot gas containing the same percentage of calcium. The same is, of course, true for all other members of the periodic system of elements.

Apart from the temperature, the important factor in the ionization of atoms is the density of the gas, which is, of course,

connected with the pressure to which it is subjected. At high density there is less place for the electrons stripped from the atoms to move around, which naturally suppresses the degree of thermal ionization. Thus, at the same temperature, the degree of ionization of any kind of atom will drop if the density is increased as a result of increased pressure. For example, the curves of calcium ionization given in Figure 19, which are calculated for the pressure of about one one-hundred-thousandth of atmospheric pressure (on the surface of the Earth) will be shifted to the right if the gas pressure is increased. The entire picture of the effect of changing temperature and pressure on the state of ionization of a gas formed by a mixture of all different chemical elements is extremely complicated, but not beyond the understanding of a person who knows the laws of atoms and is willing to spend some time on it.

TABLE 2 *Abundances of Elements in the Solar Atmosphere*

	Percentage Volume	Mass of Vapour in Milligrams per Sq. Cm. of the Sun's Surface
Hydrogen	81.760	1,200
Helium	18.170	1,000
Carbon	0.003000	0.5
Nitrogen	0.010000	2.0
Oxygen	0.030000	10.0
Sodium	0.000300	0.1
Magnesium	0.020000	10.0
Aluminium	0.000200	0.1
Silicon	0.006000	3.0
Sulphur	0.003000	1.0
Potassium	0.000010	0.003
Calcium	0.000300	0.2
Titanium	0.000003	0.003
Vanadium	0.000001	0.001
Chromium	0.000006	0.005
Manganese	0.000010	0.01
Iron	0.000800	0.6
Cobalt	0.000004	0.004
Nickel	0.000200	0.02
Copper	0.000002	0.002
Zinc	0.000030	0.03

This important task was performed in the early 1920s by the Indian astronomer Meghnad Saha, who was able to solve the complicated jigsaw puzzle of the Fraunhofer lines and to arrive at the correct proportion of different elements in the atmosphere of the Sun. His results, corrected by more recent studies, are shown in Table 2.

SUNSPOTS

In former times the Sun was considered to be a divine fire and therefore perfect and infallible. It was known that the bright face of the Sun is sometimes marred by darkish spots, but these were thought to be objects passing in space between the Sun and the Earth. That is why Galileo, who constructed the first astronomical telescope and thus opened a new era in the study of celestial bodies, hesitated a long time before he made the following announcement: 'Repeated observations have finally convinced me that these spots are substances on the surface of the solar body, where they are continuously produced and where they are also dissolved, some in shorter and some in longer periods. And by the rotation of the Sun, which completes its period in about one lunar month, they are carried around the Sun, an important occurrence in itself and still more so for its significance.'

What Galileo must have seen through his telescope is demonstrated in Plate 2. Sunspots vary largely in their sizes. The smallest ones, called 'pores', are just a few hundred miles in diameter, and it is very likely that there are even smaller ones, invisible to the telescope. The larger ones may measure tens of thousands of miles, and a large group of spots may extend more than 100,000 miles. The largest spot group yet recorded occurred in April 1947; its total area was 6,000,000,000 square miles, that is, more than 1 per cent of the total apparent area of the solar disk. The tendency of sunspots to appear in pairs or larger groups is a very important factor in solving the problem of their origin, as we may see later in this chapter.

The great majority of spots occur between the latitudes 5 degrees and 30 degrees both north and south of the solar equator, in what could be called the 'tropical' regions of the Sun. Only

very rarely can spots be observed above 35 degrees and almost never above 40 degrees. However, during the period of high solar activity in August 1956 a group of spots appeared at the unprecedented latitude of 50 degrees from the solar equator.

Observing the motion of spots located at different distances from the equator, one finds that the Sun does not rotate as a solid sphere. The spots located very close to the equator go around the Sun in 25.14 days; those at the latitude of 15 degrees in 25.50 days; and those at 30 degrees in 25.53 days. Plate 3 shows photo-

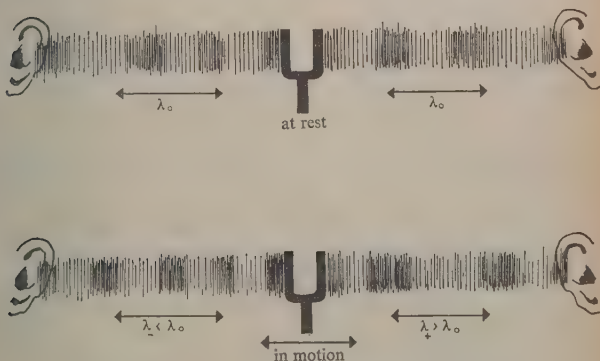


FIGURE 20. The Doppler effect in the case of sound waves.

graphs taken on successive days of the great sunspot group of 1947 as it was carried twice across the disk by the Sun's rotation.

In order to study the rotation of the Sun's surface farther away from the equator, we have to use a method based on the so-called 'Doppler effect', which describes the influence of the motion of a source on the wave length of the light or the sound it emits. For example, sound waves emitted by a tuning fork moving toward you will be shortened in the direction of its motion, as indicated in Figure 20. Shorter waves mean higher frequency of sound vibrations, and thus a higher pitch of sound. Conversely, if the tuning fork is moving away from you, the sound waves are lengthened, the frequency decreases, and you will hear a lower pitch. The

same phenomenon occurs in the case of light waves, except that, since light propagates much faster than sound, the effect can be observed only at much higher velocities.

Consider a moving source emitting light of a certain frequency, and two spectroscopes, one of them located in the direction of the motion and another in the opposite direction. If the light source were at rest, one would observe the Fraunhofer lines at exactly the same positions in both spectroscopes. However, because of the motion of the source, light waves emitted in the direction of its motion will be 'squeezed' and will thus appear to an observer located in that direction to be shorter, that is, shifted toward the violet end of the visible spectrum. On the other hand, the light waves emitted in the direction opposite to the motion of the source will be 'stretched', so that the spectroscope located in that direction will register a shift toward the red end of the spectrum. It can be shown that the relative change of the wave length $\Delta\lambda/\lambda$ is equal to the ratio v/c of the velocity of the source and the velocity of light.

Thus, measuring the Doppler shift of spectral lines, we can tell not only whether the source of light is approaching us or receding from us, but we can also get the exact value for its velocity in radial direction. The Doppler-shift method has wide application in all fields of astronomy, such as the study of the motion of stars and of gaseous nebulae, as well as that of stellar galaxies located billions of light-years away.

Studying the Doppler shift in light coming from different points of the solar disk makes it apparent that the eastern (left) rim of the Sun is approaching us, while the western (right) rim is receding. It also confirms the conclusion based on the observation of spots that the rotation period increases with the distance from the equator, as indicated in Table 3.

The situation resembles that of a broad river where the velocity of flow is larger in the middle and smaller near the shores, and it is difficult to escape the impression that sunspots may be remotely analogous to the whirlpools formed by water streams moving with different velocities. But such an analogy would not explain anything unless we understand why the angular rotation of the Sun varies with the distance from the equator. In fact, the

formation of whirlpools will dissipate the energy and tend to equalize the angular velocities at different solar latitudes. Thus, what is the agent supporting the energy for the continuous formation of sunspots?

TABLE 3 *Rotation of the Sun**

Distance from the Equator	Rotation Period (Days)
0°	24.64
15°	25.41
30°	26.45
45°	28.54
60°	30.99
75°	33.07

* As determined by Doppler-effect method.

Although, because of the contrast with the rest of the solar surface, the spots appear as dark regions, they are actually quite luminous. Their temperature is only a couple of thousand degrees less than that of the rest of the solar surface, and, if we could obscure the entire surface, leaving only a good-sized group of sunspots, they would illuminate the Earth like a hundred Moons, which it would be painful to view with the naked eye.

The detailed structure of sunspots is very complicated and grotesque. But they always contain the darker central part (*umbra*) separated from the rest of the solar surface by a somewhat brighter ring (*penumbra*). The boundaries between the umbra, the penumbra, and the outlying surface are rather irregular and rapidly changing. Sometimes the wing-shaped formations can be seen extending toward the centre of the spot; sometimes the luminous bridges cross the entire spot from edge to edge. Plate 4 shows a beautiful photograph of a group of sunspots, obtained by Martin Schwarzschild of Princeton University by sending a telescope and a camera on a large unmanned balloon to an altitude of over 80,000 feet, well above the major part of the terrestrial atmosphere.

One apparent way to study the motion of luminous gases in sunspots is to measure the Doppler effect in the light emitted from their different parts. It was found that, when the spot is near

the centre of the Sun's disk, the Doppler shift is negligible, indicating that there is practically no vertical motion of the luminous gases. However, when the spot is close to the rim of the Sun, one observes very large Doppler shifts of Fraunhofer lines, which indicate the approach of the side of the spot closer to the observer and the recession at its farther side. These proved that the hot gases flow more or less parallel to the Sun's surface, diverging from the centre of the spot. It turned out later, however, that the Fraunhofer lines of some other elements show exactly opposite behaviour, indicating convergence toward the centre of the spot. It seems from observations that the convergent streams move at higher altitudes above the surface of the Sun than do the divergent ones. The whole thing seems to look like a flat smoke ring blown from a cigar and projected on the ceiling above the smoker's head.

SMALL FRY

While the sunspots give us the most impressive proof of the unquietness of the Sun's surface, astronomical observations show that, even away from the spots, solar material is in a state of violent, turbulent motion, swirling, spurting, and boiling as lava does in the crater of an active volcano. This phenomenon is known as *granulation* and appears in the telescope as a rapidly changing network of brighter and darker areas looking, as one astronomer put it, like grains of rice in a violently boiling chicken-rice soup. The individual granulae are only a few hundred miles across, and their pattern changes completely within only a few minutes, indicating gas velocities up to 100 miles per second. Until recently the granulation of the Sun's surface was at the limit of astronomical observation, mostly because of terrestrial atmosphere through which the sunlight must pass before entering the telescope. The photographs taken recently by Schwarzschild with the airborne telescope show the granulation of the Sun's surface in much greater detail than it could be shown before (Plate 5). The bright rice-grains are apparently streams of hot gases rising to the surface from the deeper and hotter interior. Plate 4 also shows the granulation in the neighbourhood of sunspots.

Apart from granulae, which represent a rather common sight

on the surface of the Sun, one observes from time to time larger increases of local surface brightness known as *faculae* (torches), *floculi* (flakes), and *flares* in the case of especially bright ones. Those appear mostly in the vicinity of sunspots and are undoubtedly associated with the increase of turbulent motion in these regions.

To characterize a person (wanted by the police, or otherwise) it is desirable to give two photographs: the full face and the profile. In everyday observation of the Sun we see the motion of hot gases rising above its turbulent surface essentially in the direction of our line of vision, which is equivalent to photographing, let us say, the Old Faithful geyser from a helicopter hovering in the air directly above it. Although such photographs may be quite interesting and informative, it is also very desirable to have pictures taken in profile, that is, in the direction perpendicular to the motion of the material. Photographing the profile of the Sun has also the advantage that the gaseous streams we want to study are projected on the dark background of the sky and not on the brilliant disk of the Sun. At first sight the problem may look easy. Keep a coin in front of your eye at such a distance that it just covers the disk of the Sun, and you will see the edge of the disk protruding beyond it.

The fact is, however, that although this simple method of obscuring the Sun with a coin will probably work if the observer is on board an artificial satellite, on a spaceship, or on the Moon, it is not too satisfactory when used by an observer from the surface of the Earth with a thick layer of terrestrial atmosphere above his head. The sky is blue in the daytime because the sunlight is scattered in all directions by small inhomogeneities of the air (statistical density fluctuations), the same kind of inhomogeneities which cause stars to twinkle. In addition, close to the Earth's surface there is always a certain amount of dust which also contributes to the scattering of sunlight. When we look at the sky close to the rim of the Sun's disk, the glare produced by these factors becomes particularly intense and prevents any observation of the Sun's profile, although the disk itself is obscured by a coin. Even if we climb to the top of a high mountain where the air is rarer and the dust practically absent, the conditions are

still not good enough for observation. The best optical glass, from which the telescope's lenses are made, is still not homogeneous enough to eliminate the scattering of the Sun's rays passing through it, and microscopic flaws and air-bubbles inside the lenses, or minor scratches on their surfaces, produce too much glare to permit observation of the fine details of the Sun's rim.

Until 1930 there was a general opinion that it is quite impossible to eliminate the glare surrounding the Sun when seen through the Earth's atmosphere, and astronomers waited patiently for total solar eclipses and travelled to far-away corners of the Earth in order to observe the Sun's profile when our faithful Moon covered its face. Observations of that kind revealed the presence of long tongues of flame, known as *protuberances* or *prominences*, rising 100,000 miles above the surface of the Sun (Plate 6). Since the average integrated duration of the total solar eclipses which can be seen from somewhere at the surface of the Earth is only about 3 minutes per year, it was a long wait for such observations and called for long trips to be made by astronomers with heavy astronomical equipment. Their troubles were solved, and their travel-lust curtailed, by the French astronomer Bernard Lyot, who constructed an instrument which replaced the Moon by a coin, thus permitting us to observe an artificial solar eclipse any place, any time – with the exception, of course, of cloudy days. Lyot called his instrument a *coronagraph*, since it permits us to see not only the chromosphere but also the much more delicate and less luminous aureole called the *solar corona*, which extends far out from the Sun's surface (see next section). In principle, Lyot's instrument is nothing but a coin obscuring the disk of the Sun. But in practice it is a very ingenious and complicated optical system designed to reduce as much as possible the glare around the Sun's disk by using specially constructed lenses, polaroids, and many other tricky devices which can be provided by a good optical workshop.

Despite all its advantages, the coronagraph is not very effective if used at sea level and has to be installed on top of high mountains. Lyot's original coronagraph sits on the top of Pic du Midi in the French Pyrenees at an elevation of 9,351 feet. The coronagraph of the High Altitude Observatory of the University of

Colorado is at 11,200 feet alt. on Mount Climax, and still another one is on Sacramento Peak, New Mexico, at 9,200 feet. The coronagraphs carry continuous day-by-day observations of the profile of the artificially obscured Sun and give us invaluable information concerning its turbulent surface. Since the 'natural' total eclipses of the Sun have the integrated duration of only 3 minutes per year, and assuming an 8-hour working day for the coronagraph attendants, the accumulation of information by this instrument can be about 60,000 times faster than it was by the old-fashioned eclipse expeditions.

The most common view of the Sun's profile is presented by a large number of small flames shooting beyond the dark rim of the Moon. They resemble a wheat field on a windy day and are called *spiculae*. It is likely that *spiculae* are the same as *granulae*. But once in a while, especially during periods when a large number of spots appear on the Sun's surface, one observes the much larger prominences or protuberances that are apparently associated with the flares surrounding sunspots. One of the most spectacular eruptions of that kind occurred on 4 June 1946, and Plate 7 shows several frames of a long film taken on that day at the High Altitude Observatory on Mount Climax. The hot gases shot up in the form of an arch reaching a height equal to 70 per cent of the Sun's radius before breaking up and raining down back to the surface.

THE SOLAR CORONA

Every king is entitled to wear a crown, and so is the Sun, which, from our point of view at least, is the King of the Sky. The glare of the sunlit sky prevented the astronomers of older generations from observing the solar corona except in the rare cases of the total solar eclipses, but today it is 'open to the public', and everybody who gets permission to visit one of the coronagraph observatories and does not mind driving up the steep mountain roads can see it seven days a week. Plate 8 shows how the solar corona looks when observed during a total solar eclipse. Aesthetically inclined astronomers ascribe the beauty of the solar corona to the delicate fanlike structure of its rays, which sometimes extend

as far as ten solar diameters. To the present author it looks very much like the hair of Albert Einstein in his old age.

The light of the corona shows a faint continuous spectrum, with bright emission lines superimposed on it. The continuous spectrum is very similar to that of the Sun itself, including even some of the strongest Fraunhofer lines. There is no doubt that we deal here with the sunlight scattered by the material of the corona, most likely by the free electrons which are plentiful in the highly ionized rarefied gas which forms it. The bright emission lines most prominent in the coronal spectrum represented for a long time a complete mystery for the solar astronomers. In fact, these lines did not coincide with the spectrum of any chemical element known on the Earth, which led the astronomers to believe that they must be due to some new element, which was named 'coronium' and which apparently was not present on the Earth. But such a hypothesis contradicted anything known about atoms. Indeed, the modernized version of Mendeleev's periodic system accounted for all known chemical elements, and there were no vacant seats in it into which coronium could be put. The mystery was solved in 1940 by the Swedish physicist B. Edlén, who, combining the ingenious experimental studies of the emission spectra of various atoms with the quantum theory of the atomic structure, was able to prove that the mysterious coronium lines are actually due to *highly ionized* atoms of elements quite common on Earth.

We have seen in the beginning of this chapter that, at the very high temperature of the solar chromosphere ($6,000^{\circ}$ K.), many helium atoms lose one of their two electrons and masquerade in their line spectrum as normal atoms of hydrogen, and that, under the same conditions, all calcium atoms lose one and sometimes two of their outer electrons. At still higher temperatures more and more outer electrons are stripped off by violent thermal collisions, and the atomic spectra become quite unrecognizable for the observer accustomed to see them under normal laboratory conditions. Thus Edlén decided to investigate how the spectra of common elements will look under the conditions of extreme ionization (electron-stripping) which can be achieved by subjecting them to intensive high-energy electric discharges. Working

with iron vapour, in which the atoms of iron were stripped of 13 of their 26 electrons, he observed in the spectrum the famous green line (5,303 angstrom units) that is characteristic of coronal emission and was previously ascribed to the unknown element existing only on the Sun. Other coronal lines were also explained as due to iron atoms stripped of 12, 10, or 9 electrons, and to the highly stripped atoms of nickel (normally 28 electrons) and cal-

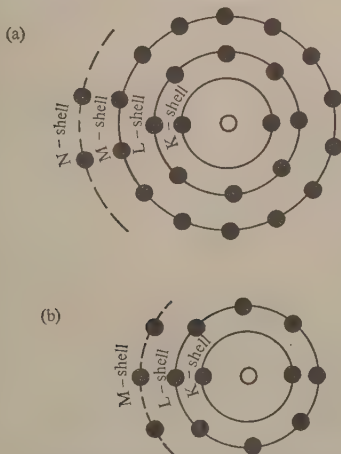


FIGURE 21. An atom of iron with a complete set of 26 electrons (a), and an iron atom (b) which has lost 13 electrons and is masquerading as an atom of a 'new element', coronium.

cium (normally 20 electrons). A highly schematic presentation of how a high degree of ionization affects the atomic-emission spectra is given in Figure 21. Above is the iron atom, with all its 26 electrons distributed among the 4 quantum shells. While 24 atoms occupying the three inner shells (K, L, and M) are sitting tight inside the atom's body, the remaining two are comparatively loosely bound in the fourth (N) shell. When the atom is mildly excited, one of the outer electrons can be thrown to one of the

outer empty quantum orbits. Returning to its normal orbit in the N-shell, this electron will emit one of the familiar lines of the standard iron spectrum. The drawing below represents the same iron atom when it is involved in a violent strip-tease dance at a state of high ionization. The outer electron skirt is completely stripped off, and only 3 electrons are left on the one immediately below. Thus, the excitation of the atom will result in raising an electron from its quantum state in the M-shell to the higher energy level, which was previously occupied. On returning, this electron will emit spectral frequencies which are not present in the atom's normal non-ionized state.

Edlén's solution of the mystery of the coronal emission lines led straight to another, just as puzzling, problem. In order to ionize iron atoms to such a high degree, a temperature of about $1,000,000^{\circ}$ K. is needed. One comes to the same conclusion by studying the width of coronal lines, which is much greater than that of the ordinary Fraunhofer lines originating in the chromosphere. The width of spectral lines is usually ascribed to the Doppler effect, which was discussed earlier. The excited atoms usually emit sharply defined wave lengths, which appear as very thin lines in a spectroscope. But the atoms in a hot gas are not at rest and rush in all directions with velocities which increase with the temperature. Thus, because of the Doppler effect, the light waves arriving from the atoms moving toward the observer will be somewhat shortened, while those coming from the atoms moving in the opposite direction will be correspondingly lengthened. The atoms moving in various directions in respect to the line of vision will show intermediate Doppler shifts so that, instead of the original thin lines, the observer will see them considerably broadened. This broadening increases with the increasing temperature; thus, by measuring the width of the lines, we can tell how hot the gas is. The width of coronal lines indicates that the temperature of the gas is about 1,000,000 degrees, in complete agreement with the figure obtained on the basis of the degree of ionization of iron atoms.

What makes the corona so hot? It clearly can not be heated by the solar photosphere, the temperature of which is only 6,000 degrees Kelvin. For a while astronomers thought that the heating

came from the deep solar interior, which, as we see in the next chapter, has a temperature of 20 million degrees. But nobody could suggest any reasonable mechanism to explain how the heat from the solar interior could be carried out through the much cooler outer layer. The most likely solution of the problem was suggested by Martin Schwarzschild, who proposed that the gases of the corona are heated by 'the noise' produced by the turbulent motion of the solar photosphere. Just as a choppy sea covered with numerous whitecaps growls and roars even at a distance from the shoreline, turbulent eddies associated with granulation send sound waves into the rarefied chromospheric gases above the Sun's surface. It is known in the theory of sound that sound waves propagating from a denser into a more rarefied medium gradually turn into the so-called shock waves characteristic of the crack of a whip, the shot of a pistol, or the explosion of a bomb. Instead of a smoothly alternating sequence of successive compressions and rarefactions, even a pure musical tone may turn into shock waves with strong compression and heating in a narrow advancing front. According to Schwarzschild, the same phenomenon takes place when sound produced by surface turbulence of the photosphere propagates farther and farther out into the very rarefied gases of the solar corona. In fact, he was able to prove by direct calculations that the energy supplied by such waves can suffice to maintain the corona at its high temperature.

SOLAR MAGNETISM

Probably the most important discovery concerning sunspots (apart, of course, from the proof of their existence) was made in 1908 by the American astronomer George Ellery Hale. Studying the Fraunhofer lines in the spectrum of light emitted by sunspots, he noticed that each line was split into several components located close to one another. This phenomenon, called the *Zeeman effect* after the Dutch physicist who discovered it, had been known in physics for a decade before Hale's work and was always observed when a flame emitting a line spectrum was placed between the poles of a strong electromagnet. The pattern of Zeeman splitting

of spectral lines varies from element to element, but in each case the separation of the components is proportional to the strength of the magnetic field. Thus, the presence of the Zeeman effect in the line spectrum indicates that the light source is subject to the action of a magnetic field and, by measuring the separation of the components, one can find the strength of that field.

The origin of the Zeeman effect can be understood in simple terms in the following way. Consider a single electron rotating around the nucleus along a circular orbit. If we place this atom between the poles of an electromagnet, the plane of the electron's orbit will become oriented perpendicularly to the lines of magnetic force, and the electron will move along its orbit in either a clockwise or counterclockwise direction (Figure 22). A moving electron constitutes an elementary electric current; in fact, an electric current flowing through an ordinary wire connecting the two poles of a battery is nothing else but a flux of electrons which are broken off from the atoms of the conductor and are driven along the wire by applied electric tension. Early in the nineteenth century the French physicist André Ampère found that if a wire carrying an electric current is placed in a magnetic field it will experience a force perpendicular both to the direction of the current and to the direction of the magnetic field. The magnitude of this force is proportional to the strength of both the current and the field. To find which way the force is directed, you use a simple three-finger rule of the left hand: if you point the forefinger in the direction of the field and the middle finger in the direction of the current, the thumb will indicate the direction of the mechanical force. Since an ordinary electric current represents the flux of negatively charged electrons and not of the positive charges, as was believed in Ampère's time, the left-hand rule can be applied to the atom by pointing the middle finger in the direction *opposite* to the motion of the electron. Thus the motion of an atomic electron in a magnetic field will be as shown in the drawings. In case (a) the electron will be pulled away from the nucleus, while in case (b) it will be pushed toward it. This will result in two different orbits (indicated by the dotted lines in the drawing) which correspond to two different states of energy of the atomic electron. The stronger the magnetic field, the larger

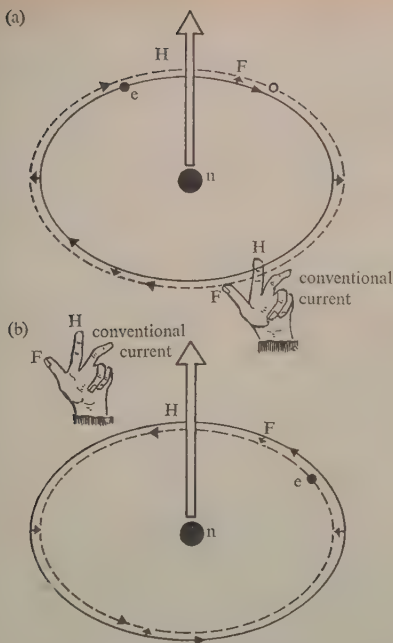


FIGURE 22. The change in orbit of an electron placed in a magnetic field H . Above (a) the force F pushes the electron outward; below (b) it pushes it inward. (To understand the left-hand rule, it may be helpful to think of a rotating disk and to point the forefinger along the axis and the middle finger in the direction of rotation.)

will be the energy difference between the clockwise and counter-clockwise motions of the electron. According to the discussion in Chapter 2, the frequency of light emitted during the various quantum transitions within the atom is determined by the energy difference between the initial and final quantum states of the electron in question. Thus, the splitting of the energy levels because of the clockwise and counterclockwise motions of atomic

electrons will result in the corresponding splitting of the spectral lines.

The Zeeman effect observed in the light emitted by sunspots indicates the presence of extremely strong magnetic fields which exceed by a factor of several thousand the magnetic field of the Earth. If an ordinary compass could be brought to the Sun and placed in the middle of a spot, it would take a strong man to deflect the needle from the direction it assumed. Considering

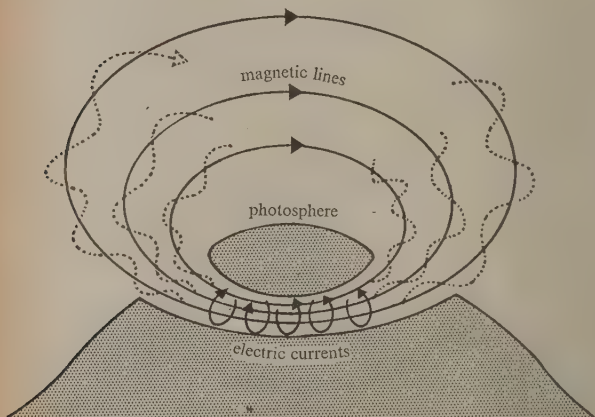


FIGURE 23. The structure of sunspots, according to Hale.

sunspots as giant electromagnets, one may calculate that the electric current feeding them must amount to several thousand billion amperes. The analogy with an electromagnet is amplified by the fact that sunspots usually occur in pairs and that in this case the two members of a pair always have opposite magnetic polarities. It looks as if we have here a giant cylindrical whirl of electrically charged material with its two ends opening through the surface of the photosphere (Figure 23). A remarkable fact is that the polarity of sunspot pairs in the northern hemisphere is always opposite to the polarity of pairs in the southern hemi-

sphere. Thus, for example, if, in a sunspot pair located in the northern hemisphere of the Sun, the one on the right corresponds to the N-magnetic pole and the one to the left to the S-magnetic pole, in a pair located in the southern hemisphere the S-magnetic pole will be on the right and the N-magnetic pole on the left. If the poles are openings of electricity-carrying whirls, this means that these whirls in the two hemispheres rotate in opposite directions. Still more mysterious is the fact that this disparity changes periodically from one epoch of high solar activity to the other. As we will see later in this chapter, the number of sunspots shows a pronounced variation in time, with a period of about eleven years. Although at the years of consecutive sunspot maxima the Sun's face may look the same, spectroscopic studies show that in the northern hemisphere we have now a S-N arrangement (instead of the N-S arrangement of eleven years ago), and the reverse order in the southern hemisphere. The reason for these changes of polarity remains just as mysterious as the origin of the sunspots themselves.

Magnetic fields associated with sunspots must have a pronounced influence on the motion of ionized solar gases ejected high above the surface of the photosphere. Electrically charged particles have the tendency to follow the direction of the magnetic field, spiralling around the lines of forces as is shown by dotted lines in Figure 23. A glance at Plates 6 and 7, showing the shapes of prominences, strongly supports that assumption.

MAGNETOHYDRODYNAMICS

In a way, sunspots can be compared with the cyclones in the terrestrial atmosphere which originate on the boundaries between cold-air whirlwinds in the arctic and antarctic regions and the warmer air masses of the temperate zone. Once in a while the smooth gliding of the cold polar air masses, along the front that separates them from the warmer air masses of the temperate zones, becomes turbulent, resulting in the formation of smaller whirls (cyclones) which detach themselves from the polar front and propagate in an easterly direction into the lower latitudes, bringing cold winds, rain, snow, and other unpleasant things to

the inhabitants of the parts of the Earth's surface over which they move.*

But, whereas in the case of the terrestrial atmosphere we have essentially a two-dimensional problem because its thickness is very small as compared with the diameter of the globe, the circulation in the Sun is a problem of three dimensions, since the Sun is gaseous all the way from the surface to the centre.

The role of polar whirlwinds on the Earth in producing cyclones is played, in the case of the Sun, by strongly convective regions in its deep interior. In fact, as we shall see in Chapter 4, about 10 per cent of the Sun's material surrounding its centre is in a highly convective state, with hot gas streams carrying the nuclear energy produced at the centre toward the surface, and with cooler gases which carry fresh 'nuclear fuel' descending into the thermonuclear-reaction region. Just as cyclones in the terrestrial atmosphere spiral around their centres because of the rotation of the Earth, convective streams inside the Sun deviate from the radial motion because of the rotation of the Sun. This results in a very cumbersome circulation system of convective currents whirling around the Sun's centre. Beyond this central convective zone lie the quieter regions of the solar interior, through which the heat continues its journey to the surface by the more 'conventional' process of heat conduction. According to recent studies of two Swedish astronomers, H. Alfvén and C. Walén, the turbulence at the boundary between the convective and conductive regions of the Sun's interior leads to the formation of doughnut-shaped whirls which rise toward the surface and open up upon reaching the photosphere, thus resulting in sunspot pairs as we see them (Figure 24).

The properties of doughnut-shaped whirls in a frictionless fluid were studied about a century ago by von Helmholtz, who was able to prove that, once such a whirl originates within a fluid, it cannot be dissipated and moves along as if it were a real material object. The properties of the similar whirls which originate in the solar interior and rise to the surface are much more complicated,

* For a more detailed discussion of the circulation of terrestrial atmosphere, see the author's book *A Planet Called Earth*, New York (The Viking Press), 1963.

because the Sun's material is strongly ionized and thus possesses a very high electric conductivity. Electric currents caused by the motion of the ionized gases in the Sun produce strong magnetic fields, which, in their turn, influence the motion of the gases, which, in their turn, change the magnetic fields, which, in their turn . . .

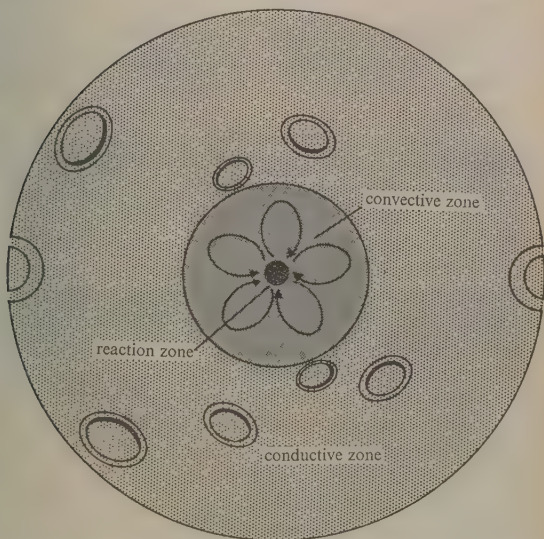


FIGURE 24. The origin of sunspots, according to Alfven and Walen.

The study of this interplay of electromagnetic and hydrodynamical phenomena is known under the not-too-easily-pronounceable name of electromagnetohydrodynamics (magnetohydrodynamics for short). The reader may get some notion of this budding new field of science from the following simplified discussion. As was discovered in 1820 by the Danish physicist Hans Christian Oersted, an electric current flowing through a wire exerts a force on a magnet (compass needle) placed in the vicinity. It follows from a generalization of Newton's law of the equality of

action and reaction that a magnet should exert an equal force on a wire carrying an electric current. In fact, if we bring a magnet close to a loosely hanging wire carrying an electric current, the wire will be displaced from its original position. In 1831 the British physicist Michael Faraday found that a changing magnetic field produces an electric current in a wire, a phenomenon known as *electromagnetic induction*.

An impressive example of induced current is given by a special type of electric motor first constructed by the American inventor Nikola Tesla. It consists essentially of three solenoids (wire coils) placed at a 120-degree angle to one another (Figure 25a). They are fed by a so-called three-phase current, that is alternating currents carried by three wires, with the phases of each component shifted with respect to one another by one third of a period. One can show that such an arrangement results in a magnetic field which is constant in strength but rotates in space with the period of the current. Between the three magnets is a copper disk which can rotate around the axis perpendicular to the plane of the magnets. If the disk is at rest, the rotating magnetic field induces in it Foucault currents, tending to turn it in the direction in which the field rotates. As a result, the disk begins to spin with the period of a three-phase current. This gadget gives a vivid example of how a moving magnetic field can drag along electrically conductive material bodies.

The reverse of the Tesla motor is the so-called self-generating dynamo shown in Figure 25b. It consists of a copper disk rotating around a fixed axis, and a wire coil, the two ends of which are in gliding contact with the axis and with the rim of the wheel. If we spin the disk, the magnetic field of the Earth will induce in the copper a current flowing, let us say, from the rim to the axis. Passing through the coil, the current will produce an additional magnetic field in the same direction as the original field which induced it. This will strengthen the magnetic field in which the disk spins, thus increasing the induced current which, in its turn, will add to the strength of the magnetic field. Thus, within a short time the dynamo will operate, producing a strong electric current and a strong magnetic field.

Both gadgets we have discussed above could be classed as

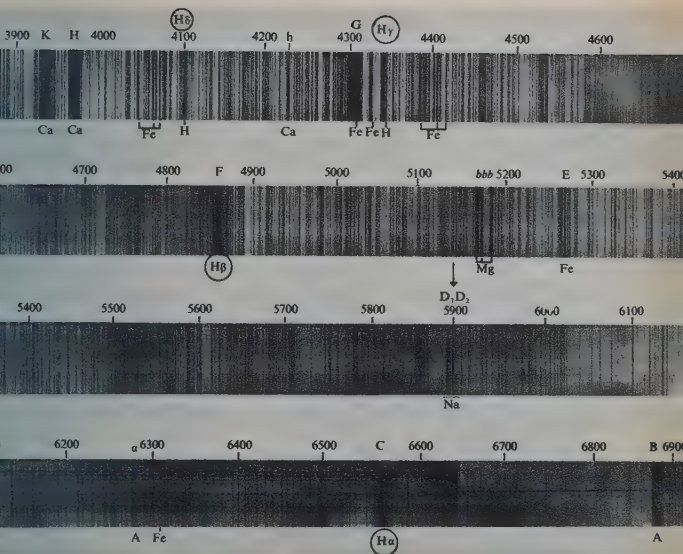


PLATE 1. Fraunhofer lines in the visible part of the solar spectrum, photographed by Mount Wilson's 13-foot spectroheliograph. Notice the famous *D* lines in the yellow region, due to sodium (arrow), and a series of hydrogen-absorption lines, or Balmer series (circled). Figures give the wave lengths in angstrom units

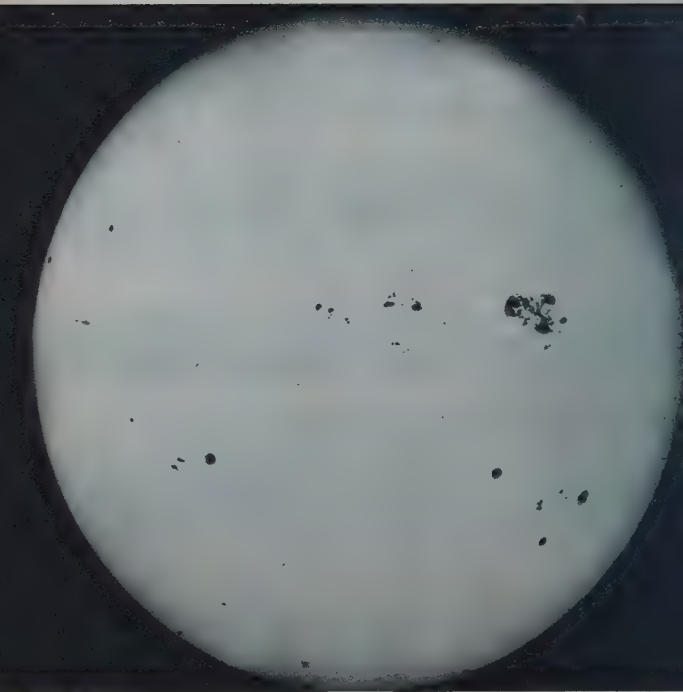


PLATE 2. The Sun's disk and sunspots photographed with the Mount Wilson 60-foot tower telescope on 12 August 1917 (Mount Wilson Observatory)



PLATE 3. Photographs of the great sunspot group of 1947 as it moved twice around the Sun's disk (Mount Wilson and Palomar Observatories)

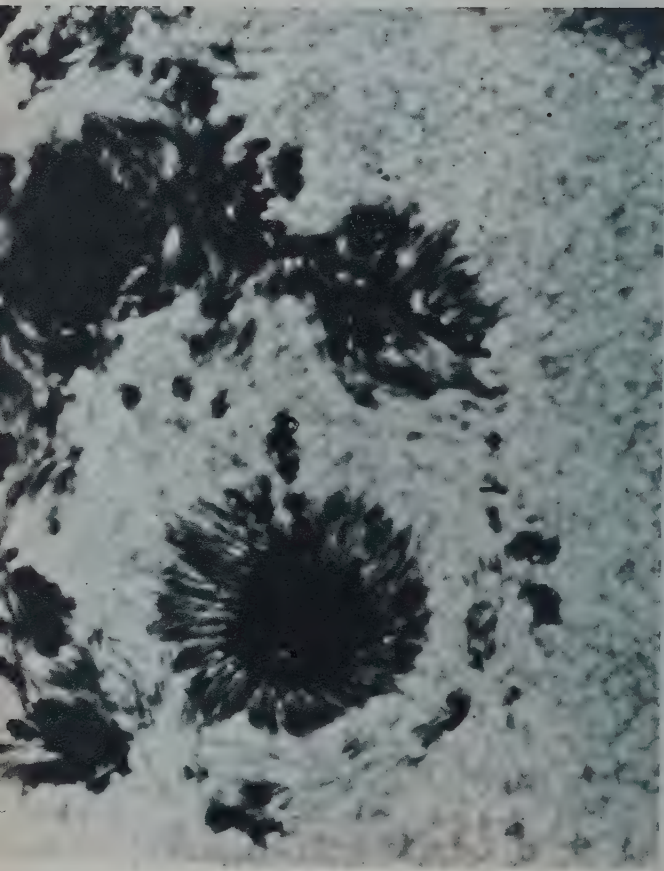


PLATE 4. A group of sunspots, photographed by Martin Schwarzschild of Princeton Observatory (Project Stratoscope)

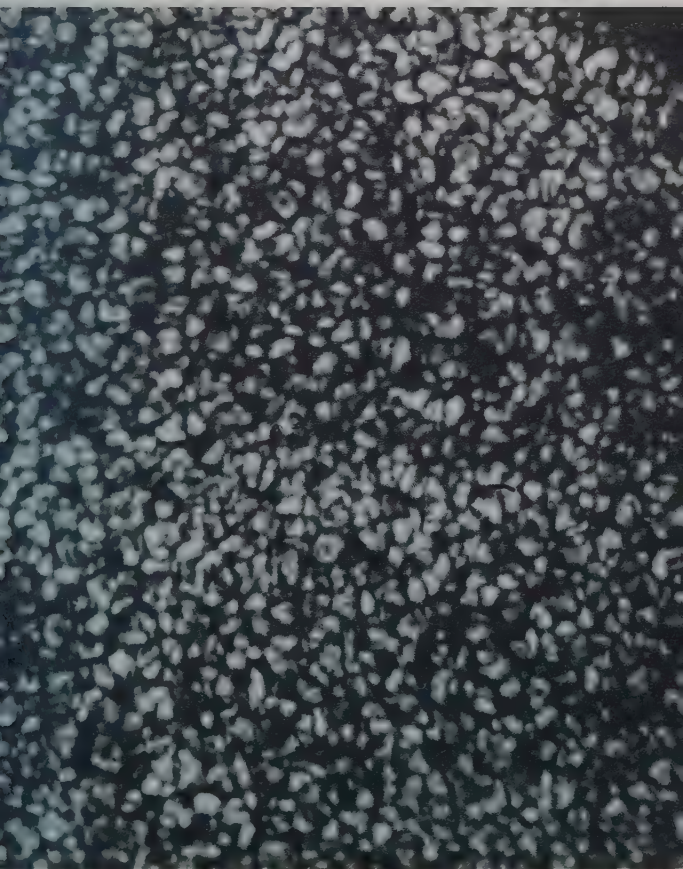
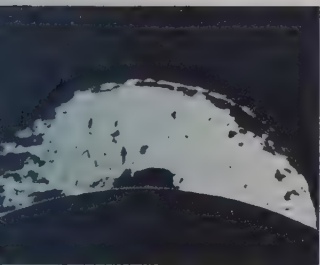


PLATE 5. Granulation on the clear Sun's surface, photographed by Martin Schwarzschild of Princeton Observatory (Project Stratoscope)

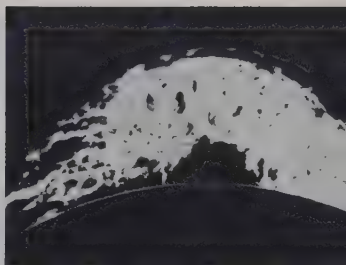


PLATE 6. Prominences on the surface of the Sun, photographed during an eclipse on 9 December 1929 (Mount Wilson and Palomar Observatories)

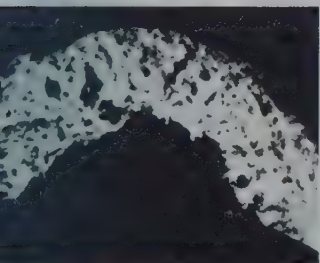
Early stage



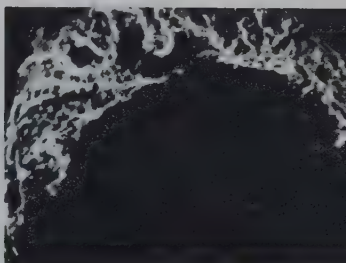
16 minutes later



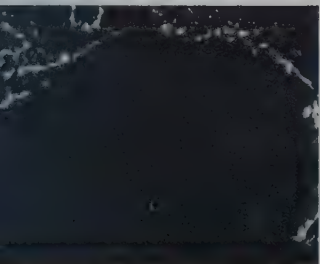
33 minutes later



46 minutes later



52 minutes later



1 hour and 19 minutes later



PLATE 7. A giant eruption on the surface of the Sun which took place on 4 June 1946, as photographed by the coronagraph of the High Altitude Observatory of the University of Colorado (Courtesy of Dr Harold Zirin)

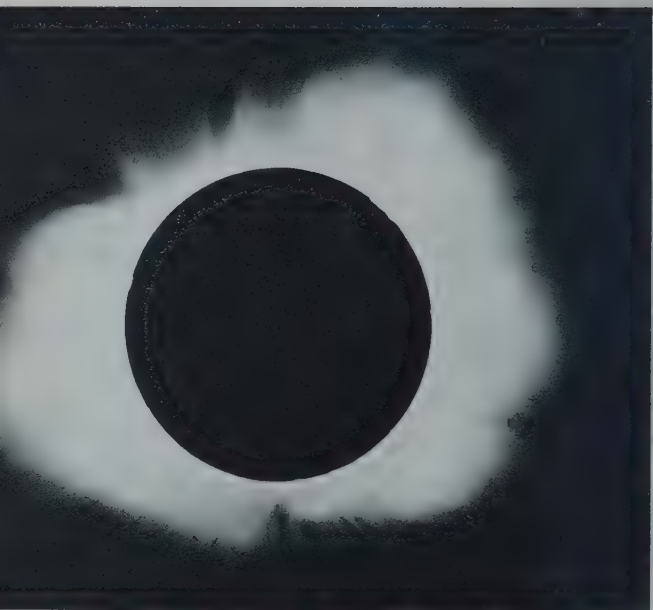


PLATE 8. Solar corona photographed during eclipse of 8 June 1918
(Mount Wilson and Palomar Observatories)

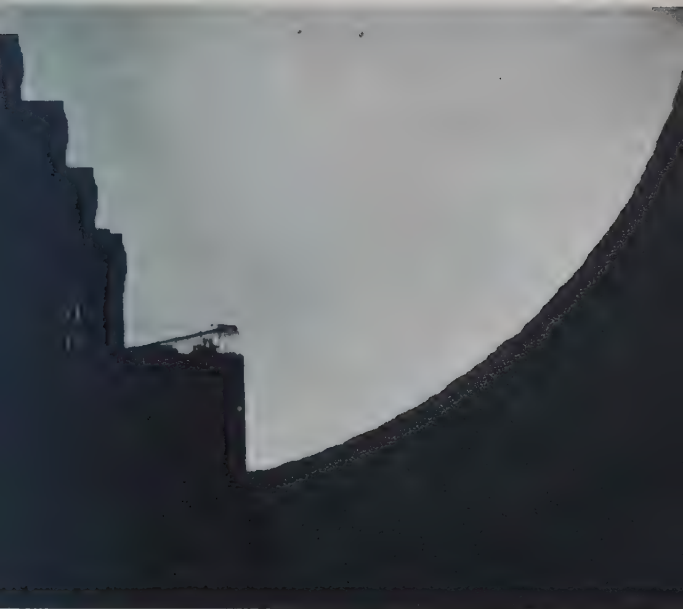


PLATE 9. This is how the Sun will look as the background of the Empire State Building's balcony when it begins to expand about five billion years from now. This photograph was actually taken by an amateur astronomer, Charles Cuevas, with a home-made telescope located on the roof of his house in Long Island City, New York, some three miles from the Empire State Building. At this distance the building looks very small, and the sun seems to be very large by comparison (Courtesy of Charles Cuevas)

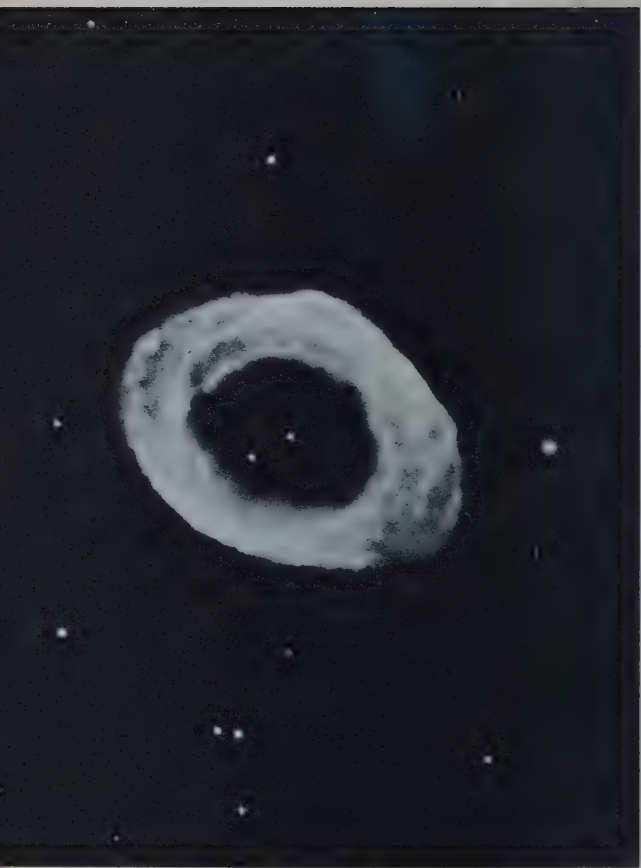
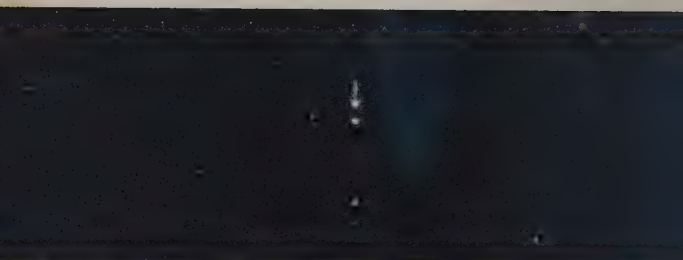


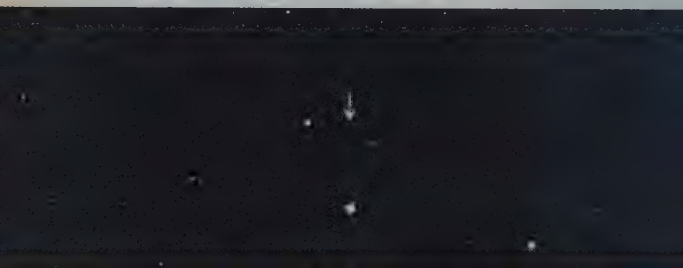
PLATE 10. The 'Ring' Nebula in Lyra. 200-inch photograph (Mount Wilson and Palomar Observatories)



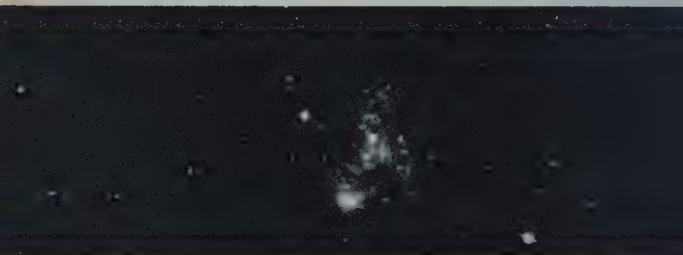
PLATE 11. The Crab Nebula, the expanding mass of luminous gases which resulted from a violent stellar explosion in the constellation of Taurus that was recorded by Chinese astronomers nine centuries ago. 200-inch photograph (Mount Wilson and Palomar Observatories)



(a)



(b)



(c)

PLATE 12. Supernova in IC-4182: (a) 1937, maximum brightness; (b) 1938, faint; (c) 1942, too faint to observe. 100-inch photographs (Mount Wilson and Palomar Observatories)

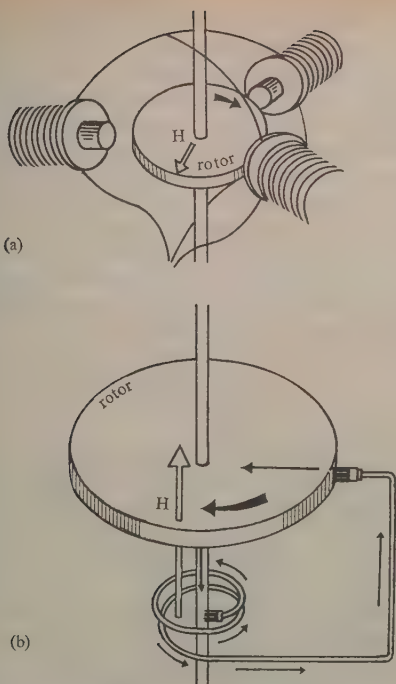


FIGURE 25. Interactions between the motion of conductive material and the magnetic field: (a) Tesla motor, in which a rotating magnetic field causes the rotation of a conductive disk; (b) self-inducing dynamo, in which the rotation of a conductive disk generates a magnetic field. (White arrow represents magnetic field; heavy black arrow represents motion of the conductor; thin arrows, electric current.)

'hardware', since they are built from rigid parts which can be given special shapes and fitted in the desired way into one another. Inside the Sun we have, of course, no wires, no copper disks, etc. But the interior of the Sun *is* made of electrically conductive material and it *does* contain violent gaseous convection streams.

Thus all that we have said about the interaction of electric currents and magnetic fields, and the motion of material bodies, must apply to solar matter in a less rigid and, so to speak, more fluid way. Hence the name electromagnetohydrodynamics, mentioned at the beginning of this section. Combining the laws of electromagnetic fields with the laws of the motion of fluids, one obtains an extremely complicated mathematical system which can crack the brains of most gifted mathematicians and the teeth of the most advanced computing machines.* But considerable progress is nevertheless being made in this difficult field, and we begin to understand more and more such things as sunspots, prominences, etc. in terms of the simpler 'hardware', such as self-generating dynamos and Tesla three-phase motors.

SOLAR ACTIVITY AND ITS INFLUENCE ON THE EARTH

'Solar activity' is one of the misuses of terminology which is encountered in any branch of science; what it actually means is the *periodicity* of phenomena taking place on the surface of the Sun. It was noticed first in the case of sunspots in 1843 when the German astronomer Heinrich Schwabe published the result of his seventeen years' study of the number of spots. This study suggested that their number varies with the period of about eleven and a half years. The Swiss astronomer Rudolf Wolf, excited by Schwabe's findings but unwilling to wait for several decades to confirm them, undertook an investigation of astronomical records concerning sunspots during past decades, and confirmed that the periodicity found by Schwabe for just one period is really true. Since that time, of course, the number of sunspots has been carefully recorded by astronomers up to the present day. The curve in Figure 26 shows the variation of sunspot numbers for the last 213 years. From this curve one concludes that the average period of sunspots is 11.2 years, although the individual time intervals between the two neighbouring maxima may be as short

* The term 'skip a tooth' originated from the old-time cogwheel computers, but can just as well be applied to modern machines, in which electron tubes or transistors represent the teeth.

as 7.5 and as long as 16 years. The number of spots during various maxima of solar activity also varies a great deal.

Since the Sun is the source of energy received by the Earth, it is natural for us to expect, without believing in astrology, that the change in number of sunspots is reflected on various terrestrial phenomena important for the life of man. The honest-to-God

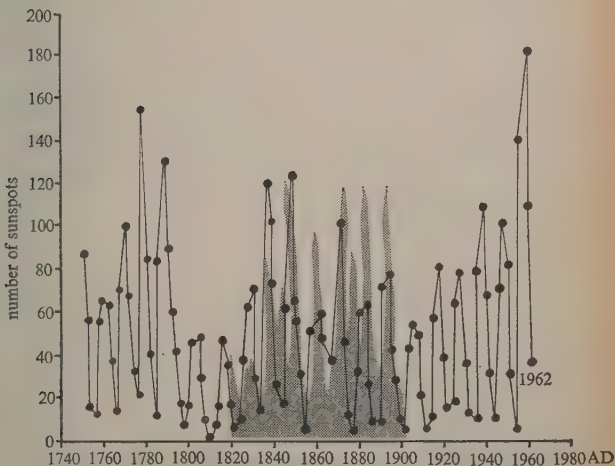


FIGURE 26. Variations in the number of sunspots over more than two hundred years, showing a periodicity of about 11.5 years. The shaded curve shows the variation in the rate of tree growth (in England, Norway, Sweden, Germany, Austria, and Arizona) from the data collected by A. E. Douglass.

correlation between the number of sunspots and the growth of trees, as obtained from the annual rings in felled trees, is shown by the shaded curve in Figure 26. It indicates that trees grow faster during the years of sunspot maxima. A related correlation seems to exist between the number of sunspots and the price of wheat: the smaller the number of sunspots, the higher is the price of wheat. This can probably be explained by saying that during

the lean sunspot years wheat grows per acre so that its price shoots up. Less understandable is the correlation between the number of sunspots and the number of lynx and fox skins bought by the Hudson's Bay Company: the more sunspots, the larger the trade. This unexpected correlation can probably be explained by the likelihood that the arctic hunters and trappers have a better chance of success during polar nights illuminated by bright auroras, and by the scientific fact that polar lights occur more often and are much brighter during the years of sunspot maxima.

There have also been attempts to correlate sunspots with political events. Thus, for example, the French Revolution (1789) and the Russian Revolution (1917) occurred at the sharp maxima of spot number. The fact that the American Revolution (1776) took place when the spot number was at the minimum suggests that, if such a correlation exists at all, it applies only to certain types of revolutions!

What is certainly true is that variations in the number of sunspots, and also in the number of flares, prominences, and other phenomena associated with them, affect very strongly the upper layers of the terrestrial atmosphere. This can be explained by the increased number of charged particles, mostly protons and electrons, ejected from the Sun during its maximum activity. Since sunspots possess strong magnetic fields that may vary rapidly with time, these magnetic fields can be expected to speed up the charged particles to very high velocities. The phenomenon is somewhat similar to the acceleration of charged particles in the strong magnetic fields of the modern cyclotrons used for the study of nuclear transformations.* But whereas the cyclotrons are carefully designed for production of narrow particle beams of the desired energy, and can be considered as 'hardware' in the sense in which this word was used earlier, particle acceleration on the turbulent surface of the Sun is a random magnetohydrodynamical phenomenon which is very difficult to visualize or to describe in any detail. When the streams of these particles, sometimes called 'solar wind', approach the Earth, they enter into the terrestrial

* For a more detailed discussion of cyclotrons and other particle accelerations, see Chapter 5.

magnetic field and follow the lines of force being deflected toward the arctic and antarctic regions, causing brilliant displays of polar light. The faster and less deflectable particles from the Sun may enter the atmosphere in lower latitudes and cause changes in the ionization of its upper levels, called the ionosphere. Of course, since the ionization of these layers is mostly due to the ultra-violet radiation of the Sun, the ionosphere is also affected by changes in that radiation, which, in its turn, is connected with the intensity of the solar activity.

All this takes place high above the Earth's surface and should not affect us any more than do the meteors, which completely burn out in the upper layers of the atmosphere before they can hit us on the head. However, it *does*, since we depend on the upper layers of the atmosphere in round-the-world radio communication. The only reason we can send radio signals between two distant points on the surface of the Earth is that electromagnetic waves curve around the Earth's surface, being alternatively reflected from the ground and from the ionized layers of the upper atmosphere. When the ionization of these layers changes, the regular reflection is distorted and for the radio operators in, say, London, Moscow, or Tokyo, the broadcast from New York effectively goes off the air. These 'radio fade-outs' hamper world radio communication quite a bit.

Even telephone and telegraph communications are affected by solar activity. Electric currents in the ionosphere produce erratic magnetic fields (magnetic storms) which overlap on the regular magnetic field of the Earth caused by the convection currents in its molten iron core.* Compass needles jiggle, and strong currents may be induced in telephone and telegraph wires and cables, blowing fuses and causing other damage.

Last, but not least, we should mention the solar 'radio noise' which was discovered during the Second World War by British early-warning radar installations. One late afternoon all the radar stations along the British coast watching for the approach of German bombers indicated one direction: westward across the Atlantic. It seemed very unlikely that the Luftwaffe would choose that avenue of approach, and indeed no bombers appeared. The

* See the discussion in the author's book *A Planet Called Earth*.

strong radio waves which were recorded were ascribed to the Sun, which was setting at that time beyond the waters of the ocean. It turned out that a giant eruption of hot gases had taken place that day at the surface of the Sun, and it must have been responsible for the strong radio signals coming from the west.

THE HOT SOLAR INTERIOR

I would have cast me into molten glass
To cool me, when I enter'd; so intense
Raged the conflagrantly mass . . .

— *Dante Alighieri*

Translated by H. F. Cary

BROKEN ATOMS

ACCORDING to a Greek legend, Prometheus flew all the way to the Sun in order to bring back to mortals some of the heavenly fire. But even Prometheus would not risk diving into the Sun's photosphere to see what was under it. However, this feat was carried out by the British astronomer Sir Arthur Eddington, who was able to find out everything about the interior of the Sun and other stars without leaving his comfortable study at Cambridge University. 'It should not be too difficult,' Sir Arthur used to say, 'to understand such a simple thing as a star.' And he had very good reasons for that statement. Indeed, while geophysicists are still unable to agree on the exact value of the temperature in the centre of the Earth, which is only about four thousand miles below our feet, astronomers can tell the temperature of the central regions of the Sun and of many other stars within a few percentage points and be quite sure about the figures they quote. How come?

The answer is that, whereas the body of the Earth is made of solid and molten materials, the Sun and all other stars are made of *plasma*, the 'fourth state of matter', which possesses very simple and easily predictable properties. Classical physics recognized three states of matter. In the solid and liquid states the molecules are as tightly packed as herrings in a barrel and are either rigidly cemented together by intermolecular forces (as solids) or are able to slide in respect to each other, without,

however, losing contact (as liquids). In the gaseous state the kinetic energy of thermal motion becomes high enough to break the bonds between the individual molecules, and they fly apart, rushing madly through the available space. The transition temperature between the condensed (solid or liquid) and the gaseous states depends on the strength of the intermolecular attraction. Thus, for example, helium becomes liquid only at the temperature of -269°C. , while iron becomes a gas only at the temperature of $+3,000^{\circ}\text{C.}$ In the condensed state, matter possesses a very small compressibility, since the repulsive forces between the electrically charged parts of the molecules stubbornly resist their mutual interpenetration. In the gaseous state, when the molecules move freely through the available space, matter is easily compressible, since the decrease in volume of a certain amount of gas can be accounted for by the reduction of the free space through which these molecules are permitted to move.

In Figure 27 the drawing at the upper left shows helium gas with the individual helium molecules (in this case, single helium atoms consisting of a nucleus and two electrons each) flying freely through intermolecular space. The drawing at the lower left represents liquid helium, in which the atoms are packed closely together and resist any further compression. However, the situation becomes quite different when the matter is subjected to a temperature which is so high that the kinetic energy of thermal motion becomes comparable to the binding energy of the electrons in the individual atoms. In this case the collisions between the atoms become so violent that all atomic electrons are completely stripped from their normal orbits around the nuclei and, instead of the ordinary gas formed by the atoms of different chemical elements, we get a plasma-gas consisting of bare nuclei and free electrons rushing wildly through space (see Figure 27, upper right). Under such conditions matter regains its compressibility, since, indeed, the particles forming it are no longer atoms or molecules with diameters of about 10^{-8} centimetres but nuclei and free electrons measuring only about 10^{-12} centimetres across. Thus there is enough space for them to move about until the distances between them are reduced to one hundred-thousandth of the distances between the atoms in ordinary condensed

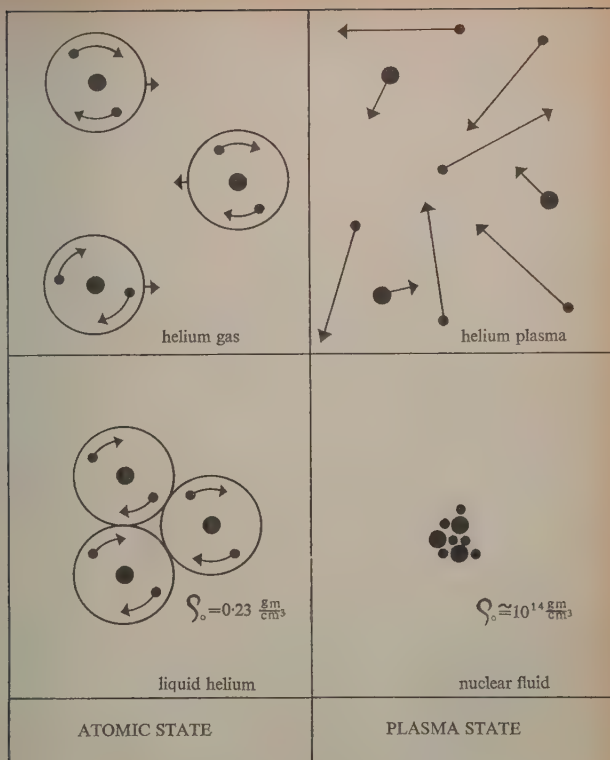


FIGURE 27. Four states of matter.

materials, and we can treat plasma as gas until its density becomes comparable to the density of the atomic nucleus, which is 10^{14} grams per cubic centimetre. One cubic centimetre of this condensed plasma or nuclear fluid would weigh one hundred million tons, but there is no evidence that nuclear fluid exists anywhere in the universe today. Thus, at densities of about 100 grams per

cubic centimetre, which exist in the central regions of the Sun, or even at densities of 10^6 grams per cubic centimetre found in certain stars known as white dwarfs, plasma is still well below the density at which deviations from the perfect gas law can be expected.

It may be noted here that ordinary matter can be turned into plasma not only by extremely high temperatures but also by extremely high pressures. Just as it is inadvisable to use fresh eggs (even if they were spherical) as billiard balls, it is also inadvisable to sit down on a basket filled with them. Even though the temperature of matter may be quite low, very high pressure will crush electronic shells and turn the material into a mixture of free electrons and bare nuclei. This phenomenon is known as pressure ionization, and the pressures necessary to cause it are well above those produced by all high-pressure devices existing in our laboratories. Even in the centre of the Earth the pressure is not high enough to crush the atoms, but in the central regions of a bigger planet, such as Jupiter, the material may be partially ionized by the weight of its corpulent body. We will speak more about this in Chapter 7.

THE LAWS OF PLASMA

The equation of state of a plasma, that is, the relation between its density (ρ), pressure (P), and temperature (T), looks exactly like the equation of state of ordinary gases, such as atmospheric air. The British scientist Robert Boyle, a friend of Newton, showed that the density of a gas increases in direct proportion to the pressure to which it is subjected. Later the French physicist Jacques Charles proved that the pressure of the gas confined to a given volume is directly proportional to the absolute temperature, that is, the Kelvin scale counted from absolute zero at -273°C . If we combine these two laws, the formula for the pressure P may be written:

$$P = \frac{R}{\mu} \rho T$$

where R is the universal constant known as the gas constant, which has the same numerical value for all gases. The Greek

letter μ in this formula stands for the molecular weight of the gas in question. Thus, in the case of hydrogen, the molecules of which contain two atoms each (H_2), we take $\mu = 2$. For oxygen (O_2) we take $\mu = 2 \times 16 = 32$, since the atomic weight of oxygen is 16. At high temperatures thermal collision breaks hydrogen and oxygen molecules into individual atoms (thermal dissociation), in which case we must use for μ the values of 1 and 16 respectively. In the case of a mixture of two different gases we must use for μ a mean molecular weight, $\bar{\mu}$, which is obtained by combining the molecular weights of the components with their percentages in the mixture. The formal definition of the mean molecular weight is: the weight per particle of the gas, regardless of which particular particles they are.

How do we find the mean molecular weight for a plasma state of some chemical element? Very simply indeed. Suppose we take hydrogen, for which $\mu = 2$ at room temperature and $\mu = 1$ at a few thousand degrees, when hydrogen molecules are broken into two atoms each. But, as we have discussed, at a much higher temperature each hydrogen atom breaks into a proton and an electron. We now have two particles sharing the total atomic weight 1, so that the mean molecular weight of hydrogen in a plasma state is $\frac{1}{2}$. Similarly, in helium plasma each atom breaks into three particles: the helium nucleus and two free electrons. Since the atomic weight of helium is 4, the mean molecular weight of helium plasma is $\frac{4}{3}$. In general, the plasma of any chemical element with atomic weight A and atomic number Z is $\frac{A}{Z+1}$

since each atom breaks into one nucleus and Z electrons. Glancing through the periodic system of elements, we notice that atomic weight increases approximately in direct proportion to the atomic number, being roughly twice as large (thus for oxygen, $Z = 8$, $A = 16$, while for silicon $Z = 14$, $A = 28$). Since for large Z s the 1 in the formula can easily be neglected, it follows that the mean molecular weight of the plasma state for all elements heavier than helium is close to 2. A refinement can be made by taking the exact mean molecular weight of the so-called 'Russell Mixture', composed of the most abundant elements heavier than helium in proportion to their actual relative

abundances. The recipe for the Russell Mixture is (by weight): O = 50 per cent; Na and Mg = 25 per cent; Fe = 13 per cent; Si = 6 per cent; K and Ca = 6 per cent, and its mean molecular weight is 1.85, that is, just a little less than 2.

Thus, whereas the molecular weights of, let us say, aluminium vapour, iron vapour, and lead vapour will be quite different (27, 56, and 207), in the plasma state the mean molecular weight of all three will be close to 2. Therefore, in studying the stellar interior we have the advantage not only of being able to use the simple gas formula, but also of not having to care about the details of the relative abundance of all the elements heavier than helium. Thus, see Table 4:

TABLE 4 *Mean Molecular Weight of Completely Ionized Elements*

Element	Mean Molecular Weight	Relative Abundance
Hydrogen	1/2	X
Helium	4/3	Y
Everything else	about 2	$1 - X - Y$

Therefore, if we know the percentages of hydrogen and of helium in a star, we know all we need to write a correct equation of state.

Now let us turn to another problem: the way in which the energy generated in the stellar interior propagates toward the surface to be radiated into the surrounding space. Here again the fact that matter is present in the state of plasma makes the theory infinitely simpler. The absorption of light by molecules of matter, which have a very complicated structure of various electron shells, is an extremely complicated and difficult problem. In principle it is possible, of course, to calculate theoretically the absorption of different wave lengths by atoms and molecules the structure of which is known. One should be able to predict that sulphur crystals will be yellow, copper sulphate bluish, chlorophyll green, and haemoglobin red. But even the most advanced electronic computing methods today do not give more help in solving these problems than we could get by counting on our fingers.

In the plasma state of matter there are no molecules, no atoms,

but just a mixture of free electrons and bare nuclei. Nuclei, practically, do not interact with the radiation of the wave length involved. Thus all we have to do is to consider a simple problem of interaction of electrons and the light quanta. This problem can be treated exactly by using the methods of wave mechanics, and one can construct tables showing how much radiation will be transmitted through the plasmas of different constitutions (i.e. hydrogen, helium, and the rest) at any given temperature. The formula first derived by H. A. Kramers for the opacity of plasma for the radiation passing through it is very simple and can be written as:

$$K = K_0 \rho^2 T^{-3.5}$$

where K is the opacity (or the light-absorption factor); K_0 is the function of chemical constitution of stellar matter (essentially a function of X and Y); ρ is the density; and T is the absolute temperature.

The third equation which we need for studying the interior of a star is the relation between the rate of decrease of gas pressure with the increasing distance from the centre, and the gravity forces pulling the gas masses toward the centre. But this is not much different from the 'barometric formula' for the change of atmospheric pressure above the surface of the Earth, which the meteorologists use every day.

THE CONVECTIVE ZONE

The analogy between the hydrodynamics of the interior of the Sun and the hydrodynamics of the terrestrial atmosphere is augmented by the existence in both cases of the so-called convective currents.

During a hot summer day we observe on the blue background of the sky beautiful white clouds, known to meteorologists as *cumulus clouds*, which sometimes resemble a man's face, sometimes a dragon, and sometimes a mushroom. These clouds are due to the vertical currents of the moist air which is heated at the sunlit surface of the Earth and rises to higher altitudes. Getting into the cooler air layers, the moist air loses its humidity, forming large and often odd-looking bodies of what one can

also call steam or fog. The reason for these ascending currents is a large temperature difference (temperature gradient) between the air near the ground and the air high above it. Physicists call that kind of gas (or liquid) motion convective currents, and they are in principle not different from the water currents in a kettle

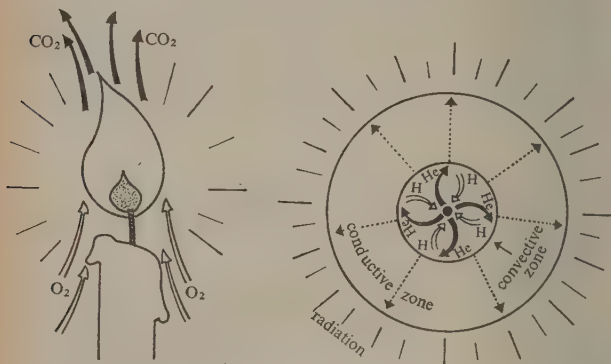


FIGURE 28. Convective processes in a candle and in the Sun.

on a hot stove. The necessary condition for the convective currents is a large temperature gradient (rapid change of temperature with the altitude), which on our planet exists in the summer, when the ground is heated by direct rays from the Sun, but usually does not exist in the winter, when the ground is cold.

Coming back to the interior of the Sun, we find that all the heat is produced by nuclear transformation in a very small region near the centre. To drive this heat through the body of the Sun toward the surface very large temperature gradients are necessary. It was first shown by the British astronomer Thomas George Cowling that the temperature gradients existing near the Sun's centre are sufficiently large to cause convective currents within about 10 per cent of the Sun's material surrounding it. These currents carry a double duty. They remove the product of nuclear transformation (helium) from the central furnace, and they bring in fresh fuel (hydrogen). They operate on the same principle as the air

convection currents near a candle flame, which bring fresh oxygen from below and remove the carbon dioxide produced in the burning (Figure 28). Being brought to the outer border of Cowling's zone by convective currents, the heat diffuses through the outer envelope to the Sun's surface, from which it is radiated into surrounding space and brings life and joy to the people of the Earth.

Using the laws of hot plasma and starting with the known physical conditions on the surface of the Sun, one is able to follow

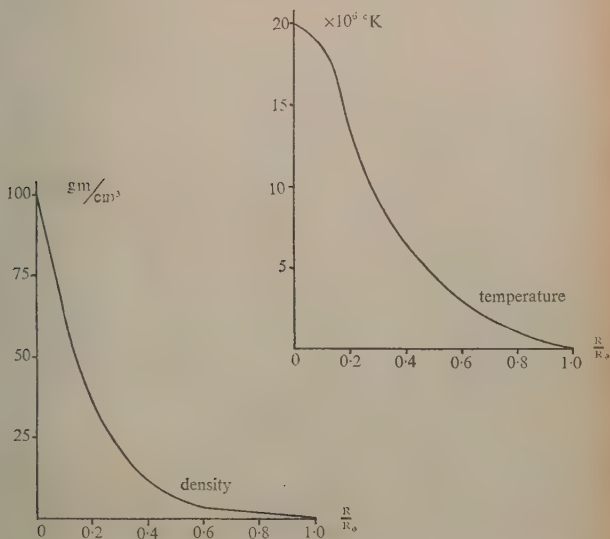


FIGURE 29. Distribution of density and temperature inside the Sun.

mathematically the temperature, pressure, and density changes all the way to its centre. Eddington did this with the help of a large number of people computing by the laborious processes of high-school arithmetic or, at best, with the use of primitive table

computing machines. Today, using modern electronic computers, we can do this much, much faster and more accurately. The curves in Figure 29 represent the results of such computations. They show that the material near the centre of the Sun has a density about 100 times that of water, and a temperature of about 20 million degrees. To give an idea of the tremendous value of that temperature, it is enough to say that if we could maintain the temperature of a pinhead at 20 million degrees its thermal radiation would burn everything within the radius of several miles.

NUCLEI AND THEIR ENERGY

... I saw the flaring atom-streams –
And torrents of her myriad universe
Ruining along the illimitable inane
Fly on to clash together again, and make
Another and another frame of things.

– *Alfred, Lord Tennyson*

EARLY DISCOVERIES

IN the year 1896 the French physicist Antoine Henri Becquerel made a remarkable discovery. He found that minerals containing the element uranium emit a peculiar type of radiation that is invisible to the eye but which darkens photographic plates even when they are wrapped in black paper. In this sense the radiation was similar to X-rays, discovered a year earlier by the German physicist Wilhelm Röntgen. But, whereas to produce X-rays a continuous high-tension electric discharge had to be sent through an evacuated tube, the newly discovered radiation flowed steadily from the uranium minerals without any external excitation and without showing any noticeable tendency to diminish its intensity with time. Furthermore its intensity did not change, no matter to what physical or chemical agents the mineral was subjected, and continued at the same rate when this mineral was placed into molten lead or into liquid air. Becquerel called this new and mysterious phenomenon *radioactivity*. Later it was found that some other elements are also radioactive.

Excited by Becquerel's discovery, Madame Marie Sklodowska Curie, the Polish-born wife of the French physicist Pierre Curie, started intensive chemical studies of radioactive substances. Comparing the radioactivity of uranium ores with that of pure metallic uranium, she noticed that the ores are about five times as radioactive as would be expected from their uranium content. This

indicated that the ores must contain some other element present in such a small amount that ordinary methods of analytical chemistry could not detect it, but so strongly radioactive that one could easily notice its radiation. The attempt to separate such a rare element called for carloads of expensive uranium ores. Fortunately, the financial problem was easily solved. The uranium ore known as pitchblende was mined near Joachimsthal in Bohemia and, after uranium was extracted, the 'worthless' leftovers, which still contained the unknown rare element, were just piled up near the separation plants. Thus the Austro-Hungarian government was only too glad to send Madame Curie any amount of that refuse that she needed. Led by the Theseus thread of penetrating radiation, Madame Curie, working with her husband, succeeded in separating minor quantities of a radioactive element similar in its chemical properties to ordinary bismuth and called it *polonium* in honour of her native country. But polonium was not yet the solution of the problem, since its radioactivity diminished appreciably with time, dropping by a factor of 2 each 140 days. More and more pitchblende went through the Curies' laboratory, and finally they managed to separate the new element, which they called *radium*. They found that it was two million times more active than uranium and did not decay noticeably in the human time scale (decay period 1,840 years). This initial success was followed by the discovery of many more radioactive elements, and both physicists and chemists busied themselves with their study and classification.

A young New Zealander called Ernest Rutherford, working with another British physicist, Frederick Soddy, was the first to suggest that the phenomenon of radioactivity is due to a slow spontaneous transformation of the atoms of one heavy element into the atoms of another – the old dream of medieval alchemists. Rutherford later proved that the atom consists of a small central nucleus surrounded by a swarm of electrons (see discussion in Chapter 2), and the site of radioactivity was transferred to this central, highly condensed body.

ENERGY LIBERATION

The high degree of penetration of the newly discovered radiation through all kinds of material was ascribed to the extremely high energy which it carried. This energy could be measured by experiments in which the radiation was sent through a strong magnetic field. It turned out that there were three kinds of radiation, which were called by the first three letters of the Greek alphabet:

Alpha (α)-particles, which are the nuclei of helium, carrying the double positive charge and moving with velocities from 5 to 7 per cent of the velocity of light.

Beta (β)-particles, ordinary negatively charged electrons, identical to those studied by J. J. Thomson, but moving with velocities unattainable in discharge tubes at that time.

Gamma (γ)-rays, similar to ordinary X-rays, but having much shorter wave length and, consequently, much higher energy per radiation quantum.

In nuclear physics energies of particles and the radiation quanta are measured in units known as million electron volts. According to the laws of classical electrostatics, the energy gained by a particle carrying a certain electric charge and accelerated by an electric field with a certain potential difference between the anode and cathode is equal to the product of that charge and that potential difference. If an electron is accelerated by a potential of 1 volt, the energy it gains is called an electron volt (ev); if the potential is one thousand volts the energy will be a kilo electron volt, or kilovolt (kev); and if it is a million volts we speak of one million electron volts (mev). Energies encountered in ordinary chemical reactions are of the order of several volts per molecule; in radioactive transformations energies are several million electron volts. Thus nuclear transformations are millions of times more energy-producing than chemical transformations, for the same amount of material.

To give a discrete example, let us consider the problem of how much water can be heated from the freezing point to the boiling point by the fission process in one kilogram of uranium.* How

* Today everybody knows what uranium fission is, but a more detailed discussion is given later in this chapter.

many U^{235} nuclei are there in one kilogram? A hydrogen atom weighs 1.66×10^{-24} gm.; the weight of the U^{235} nucleus is 235 times larger, i.e. 3.9×10^{-22} gm. Thus one kilogram of uranium contains $\frac{10^3}{3.9 \times 10^{-22}} = 2.56 \times 10^{24}$ atoms. In the fission process

a uranium nucleus liberates 200 mev. The relation between the nuclear unit of energy (i.e. mev), and the thermal unit of energy (i.e. a calorie), is:

$$1 \text{ mev} = 3.8 \times 10^{-14} \text{ calories.}$$

Thus the total energy liberated in fission of one kilogram of uranium is $200 \times 3.8 \times 10^{-14} \times 2.56 \times 10^{24} = 1.95 \times 10^{13}$ calories; and the amount of water it can heat from 0° to 100° C. is 1.95×10^{11} gm., or about 200,000 tons of water.

LIFETIMES

Radioactivity is the result of spontaneous decay of heavy nuclei which are unstable, each of which ejects either an alpha-particle (a helium nucleus) or a beta-particle (an electron). The gamma-ray is nothing but very short electromagnetic waves radiated by a nucleus as it readjusts itself after alpha or beta emission. Both alpha and beta decay are subject to a very simple law of probability: the number of nuclei which decay at any given short interval of time is proportional to the number of as yet non-decayed nuclei present at that moment. The half-life T of a given radioactive element is defined as the period of time during which its original amount is reduced by 50 per cent (i.e. when one half of it has decayed). Thus, if we start with 1 gm. of a radioactive element, there will be $\frac{1}{2}$ gm. after time T , $\frac{1}{4}$ gm. after time $2T$, $\frac{1}{8}$ gm. after time $3T$, etc.

Various radioactive elements possess widely varying half-life times, from a negligible fraction of a second to many billions of years. Some examples of the energy of emitted particles and the corresponding half-life times are shown in Table 5.

One notices that in the case of both alpha and beta decay the half-life increases with the decreasing energy involved in the transformation. But, whereas in the case of alpha decay com-

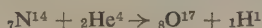
paratively small changes of energy result in the change of a life-time from a few thousandths of a second to several billion years, for beta decay the variation in the lifetime is much more moderate.

TABLE 5 *Half-Life Times*

Alpha-decaying Elements		
Symbols	Half-life	Energy (in mev)
U ²³⁸	4.5×10^9 years	4.05
Ra ²²⁶	1,590 years	4.79
Rn ²²²	3.82 days	5.49
Po ²¹⁸ (RaA)	3.05 minutes	6.11
Po ²¹⁴ (RaC')	0.0002 second	7.68
Beta-decaying Elements		
Symbols	Half-life	Energy (in mev)
Pb ²¹⁰ (RaD)	22.2 years	0.025
Th ²³⁴ (UX ₁)	24 days	0.13
Pb ²¹⁴ (RaB)	26.8 minutes	0.65

THE FIRST SPLITTING OF THE NUCLEUS, AND MODERN ATOM-SMASHERS

When during the First World War Rutherford was asked by the British Admiralty to direct his efforts toward war work, he declined, stating that he had more important tasks to do than crushing Germany: he was crushing the nucleus of an atom. And it was a very good excuse indeed, since in 1919 he announced that he had succeeded in breaking up the nucleus of a nitrogen atom by bombarding it with high-energy alpha-particles from a radioactive source. What he observed was that a fast alpha-particle colliding with a nitrogen nucleus penetrates once in a while into its interior and ejects a proton, remaining itself stuck in the nucleus. This process can be expressed by an 'alchemical' formula:



where the lower indices give atomic numbers, and the upper, atomic weights of the nuclei in question. Thus an atom of nitrogen was transformed artificially into an atom of oxygen – not ordinary oxygen, which constitutes a major part of the Earth's atmosphere (${}_8\text{O}^{16}$) but its heavier isotope (i.e. an atom with the same atomic number but a different atomic weight), which is present in the air in very small but detectable amounts. Ironically enough, in this first experiment on nuclear transformation the energy was not liberated but absorbed, and the ejected proton carried less energy than the incident alpha-particle. In subsequent experiments Rutherford was able to transform a few other light elements, such as aluminium and magnesium, in which case the balance of energy was positive.

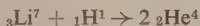
Using alpha-particles of different energy and aiming them at targets of various materials, Rutherford established two basic rules of artificial nuclear transformations:

1. The yield of ejected protons decreases very rapidly with the decreasing energy of alpha-particles and, whereas the high-energy alpha-particles from polonium-214 (RaC') (see Table 5) produce excellent results, the slow particles from uranium-238 had apparently not enough energy to crack the nucleus.

2. When alpha-particles with the same energy are aimed at different targets, the yield of protons rapidly decreases with the increasing atomic number of the element used, and no effect can be noticed except with the elements at the beginning of the periodic system.

In all these pioneering experiments the only projectiles which could be used for nuclear bombardment were alpha-particles ejected from natural radioactive elements. Theoretical considerations indicated, however, that high-energy protons would serve as much better projectiles, since they carry a single electric charge and are therefore less repelled by the nucleus than the double-charged alpha-particles. At the request of Rutherford, the author of the present book carried out the theoretical calculations and was able to prove that, if protons were used as the projectiles, their energy needed to be only half that of the alpha-particles in order to produce the same yield. Encouraged by this prediction, Rutherford asked two of his colleagues, John Cockcroft and

E. T. S. Walton, to construct a machine which would produce proton beams with the energy of a few mevs. By 1931 this first atom-smasher was ready, and Cockcroft and Walton directed a beam of protons at a target of lithium, which was expected to be most fragile under proton bombardment. And crack, crack went the lithium nucleus, breaking into two equal parts, each a helium nucleus. The reaction was:



and it liberated very large amounts of nuclear energy.

Today the original home-made atom-cracker of Cockcroft and Walton has grown into various types of giant atom-smashers which can lick any nucleus in the house. In one type of these ingenious machines, which was named for its inventor, the American physicist Robert J. van de Graaff, particles are accelerated by very strong electric fields. Another type, invented by the American physicist Ernest O. Lawrence, is based on the spiral motion of particles between the poles of giant electromagnets. Modern types of particle-accelerators carry such odd names as cyclotron, synchrotron, bevatron, cosmotron, etc., and accelerate nuclear particles to many billion electron volts.

THE POTENTIAL BARRIER

In the year 1928 Rutherford published a very interesting paper describing his new experiment, in which he had bombarded a uranium target with very fast alpha-particles emitted by polonium-214 (RaC') and having the energy of 7.68 mev (see Table 5). He had found that, even in head-on collisions with uranium nuclei, when alpha-particles bounce almost directly backward, they do not experience any forces other than those of simple electrostatic repulsion. The approach to a nucleus of an incident alpha-particle travelling against the forces of electrostatic repulsion can be compared with the motion of a ball rolling at a certain speed up the slope of a steep hill (Figure 30). If the initial velocity of the ball is comparatively small, it will stop before reaching the top and will then roll back. If, however, the velocity is high enough, it will reach the top and roll down the other side

of the hill. Thus, Rutherford's experiment proved that the height of the 'hill' surrounding the uranium-nucleus and separating the internal region of nuclear attraction from the outer region of electrostatic repulsion was larger than 7.68 mev. This result stood in direct contradiction to the fact that alpha-particles 'roll out' of the uranium nucleus, going 'over the top' even though their energy is only 4.05 mev (see Table 5).

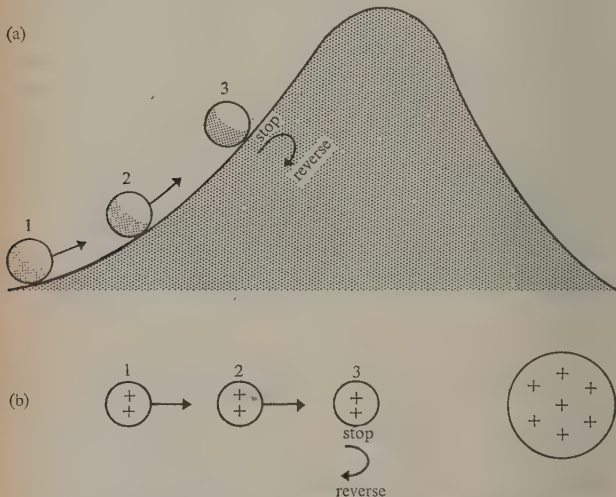


FIGURE 30. The analogy between a ball thrown uphill with not enough energy to reach the top (a), and an alpha-particle moving toward an atomic nucleus with not enough energy to overcome electrostatic repulsion (b).

After reading that article, the author of this book formed the idea that this apparent paradox arising in classical mechanics could be explained on the basis of wave mechanics (see Chapter 2), which had been developed only two years earlier. In fact, as it was put by the British mathematician R. H. Fowler during a discussion of this subject at the meeting of the Royal Society of London in

1930, 'According to wave mechanics one can leave this auditorium without going through the door or being thrown out of the window.' Things which were impossible from the point of view of classical mechanics became possible in the quantum theory, even though their probability was negligibly small. Using wave mechanics, the author was able to explain the 'paradox' of alpha decay of radioactive nuclei, and to give a perfect theoretical interpretation of the dependence between the energies of the emitted alpha-particles and the half-lives of the radioactive elements shown in Table 5. As often happens to people working on the frontiers of science, the same idea occurred simultaneously and independently to the British physicist Ronald Gurney, who, in collaboration with the American physicist Edward Condon, published a paper containing very similar arguments.

If an alpha-particle can get out of the nucleus even though its energy is less than the height of the nuclear potential barrier, an alpha-particle coming from the outside with insufficient energy can also pass through the potential barrier and penetrate into the nucleus, thus disturbing its inner structure and causing the artificial transformation of one chemical element into another. The calculations carried out by the author have shown that this is really true, and Rutherford's two laws of artificial nuclear transformation, stated earlier in this chapter, have been explained in full detail on the basis of the potential-barrier theory.

THERMONUCLEAR REACTIONS

During the year following these advances in nuclear physics the Austrian physicist Fritz Houtermans and the British astronomer Robert Atkinson put their heads together to see whether the energy production inside the Sun and other stars could be explained by nuclear transformations caused by the extremely high temperatures in their interiors. The temperature of about 20 million degrees in the stellar interior corresponds to thermal energy of only about 3.5 kev, which is much lower than the energy of particles used in laboratory experiments for artificial nuclear transformations. But there is another side to the coin. While a

multimillion-electron-volt nuclear projectile produced in a cyclotron or any other particle-accelerator slows down and rapidly comes to rest, thermal motion does not subside as long as the material is kept at a constant high temperature. Thus thermal collision between nuclei continues for thousands, millions, and billions of years until, as a result of a lucky hit, two colliding particles penetrate into each other and liberate the hidden treasure of nuclear energy.

Using the quantum formula for the probability of the interpenetration of the two colliding nuclei, and Eddington's data on physical conditions inside of stars, Atkinson and Houtermans were able to calculate the rate of thermonuclear reaction between different elements in the conditions existing in the stellar interior. They found that the only reactions which could liberate enough energy for maintaining the observed radiation are those between the nuclei of hydrogen and of some other light element, though, because of lack of data at that time, they could not tell what this light element is that reacts with hydrogen. They suggested that the process consists in the consecutive capture of four protons by this unspecified nucleus, and the formation of an alpha-particle carrying a large amount of energy. Thus the unspecified nucleus was playing the role of what chemists call a catalyst, helping the process of the transformation of four hydrogen atoms into one atom of helium. They tentatively called their paper 'How to Cook a Helium Nucleus in the Potential Pot', but when it was published in the magazine *Zeitschrift für Physik*, the editor, who lacked a sense of humour, changed the title to a less imaginative one. Table 6 gives the rates of various thermonuclear reactions which may occur in the stellar interior, based on more exact formulas derived later by George Gamow and Edward Teller, and more recent data concerning nuclear properties.

Inspecting this table, one finds that there is a very narrow choice for the reactions which could be responsible for the stellar energy production. Indeed, the reaction between very light elements, such as $\text{H}^2 + \text{H}^2$ (used in the hydrogen bomb) or $\text{Li}^7 + \text{H}^1$ (the first one obtained experimentally by Cockcroft and Walton), runs too fast and could not possibly last for the long life of stars. On the other hand, the reactions involving heavier elements, such

as $O^{16} + H^1$, or $Si^{30} + H^1$, run too slowly and cannot possibly produce enough energy to support stellar radiation. Thus we are left with only carbon and nitrogen reacting with hydrogen. In the year 1938 a detailed study of the possibilities that presented themselves was carried out simultaneously and independently by Hans Bethe in the United States and Carl von Weizsäcker in Germany.

TABLE 6 *Mean Reaction Times and Rate of Energy Production for Various Thermonuclear Reactions*

(at temperature = 2×10^7 °C., density 80 times that of water, and H-concentration 35 per cent)

Reaction	Rate of Energy Liberation Cal./Gm. Sec.	Time Period During which 1/2 of the Reacting Element is Consumed
$H^2 + H^1 \rightarrow He^3 + \gamma$	1×10^{12}	2 seconds
$H^2 + H^2 \rightarrow He^3 + n$	2.5×10^6	0.00003 seconds
$Li^7 + H^1 \rightarrow 2He$	2.5×10^{10}	1 minute
$B^{11} + H^1 \rightarrow 3He^4$	4×10^6	3 days
$C^{12} + H^1 \rightarrow N^{13} + \gamma$	1.5	2.5×10^6 years
$C^{13} + H^1 \rightarrow N^{14} + \gamma$	2.5×10^{-3}	5×10^4 years
$N^{14} + H^1 \rightarrow O^{15} + \gamma$	2.5×10^{-2}	4×10^6 years
$N^{15} + H^1 \rightarrow C^{12} + He^4$	5×10^3	20 years
$O^{16} + H^1 \rightarrow F^{17} + \gamma$	1.5×10^{-12}	10^{12} years
$Mg^{26} + H^1 \rightarrow Al^{27} + \gamma$	7.5×10^{-14}	10^{17} years
$Si^{30} + H^1 \rightarrow P^{31} + \gamma$	1.5×10^{-17}	3×10^2 years

They found the remarkable fact that the group of reactions involving carbon and nitrogen form a *cycle*, which repeats again and again as long as the hydrogen lasts. The nuclei of carbon and nitrogen pass through a series of transformations and reappear at the end of each cycle, as a phoenix from the ashes. This cycle, known as the carbon cycle (although it could just as well be called the nitrogen cycle), is shown in Figure 31. The total duration of the cycle is of course equal to the sum of the durations of individual steps and amounts to 6.5×10^6 years. If the carbon and nitrogen nuclei were not regenerated at the end of each cycle, both elements would vanish on the Sun within ten or fifteen million years and the thermonuclear reactions would stop. However, as it is, only hydrogen is consumed, being turned into

helium, and there is enough hydrogen on the Sun and other stars to last at least for many billion years.

Thus Houtermans and Atkinson were quite justified in calling the process 'cooking helium in a potential pot'.

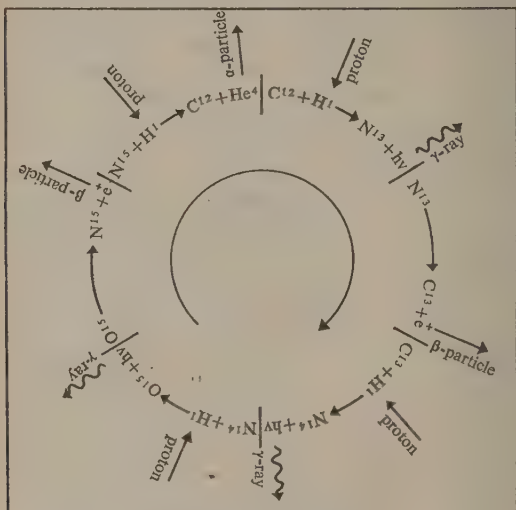


FIGURE 31. The carbon cycle.

BETA DECAY AND THE NEUTRINO

We must now discuss another important feature of radioactive nuclei: the process of beta decay. In 1914 a British scientist, James Chadwick, studying the energies of beta-particles emitted by various radioactive elements, made an astounding discovery. While alpha-particles emitted by a given radioactive element all have exactly the same energy, beta-particles show a wide spread in their energy distribution, ranging from practically zero to the upper limit of a few million electron volts. It almost seemed that the law of conservation of energy was not applicable

to electrons emitted from the nuclear interior, and indeed such great physicists as Niels Bohr considered that possibility seriously. However, the German physicist Wolfgang Pauli was much more conservative in his thinking about the unaccountable energy losses in the process of beta disintegration. Instead of abolishing such a fundamental law of physics as the law of conservation of energy, he preferred to think that the balance of energy is carried away by a mysterious particle which is emitted together with a beta-particle but escapes experimental detection. This particle, which was later named *neutrino*, was supposed to carry no electrical charge and to possess a negligibly small mass, if any. Pauli showed that particles of that nature would have extremely small interaction with matter and would easily slip between the fingers of an experimentalist trying to catch them.

On the basis of this suggestion, the Italian physicist Enrico Fermi published in 1934 a mathematical theory of beta decay as a simultaneous emission of an electron and a neutrino. Fermi's theory stood in perfect agreement with the observed facts and in particular explained in detail the relation between the energies and lifetimes of various beta decays. But, although well recognized on a theoretical basis, neutrinos were still escaping experimental detection, even though in the course of the years there accumulated some indirect evidence of their participation in various nuclear processes. And it was only in 1955 that two American physicists, Fred Reines and Cloyd Cowan, designed a neutrino trap which was effective in catching them. Their trap consisted of a huge chamber filled with hydrogen and surrounded by special Geiger counters, one half of which would count only electrons and another half only neutrons. The chamber was placed outside a thick shield which surrounded one of the atomic piles at the Savannah River project of the Atomic Energy Commission. The shield was thick enough to stop various kinds of nuclear radiation from the uranium-fission processes taking place in the pile, but not to stop the neutrinos. These particles could fly through thick concrete walls as easily as a swarm of mosquitoes flies through chicken-wire. They did not endanger the experimenters because they passed non-stop through their bodies, on through the body of the Earth, and then out into space. But a few of them (very

few!) collided with the protons in the atoms of hydrogen in the chamber, turning protons into neutrons and positive electrons according to the equation:



And each time that one neutron and one electron counter clicked simultaneously, Cowan and Reines knew that a neutrino had made a hit. They found that the ability of neutrinos to penetrate matter is so high that a lead shield several light-years thick would have only a fifty-fifty chance of stopping a neutrino!

We now return to the year 1938 and the discussion of energy-production in stars. While Bethe and von Weizsäcker were developing their theory of the carbon cycle, a young physicist, Charles Critchfield, at that time a student of the author at George Washington University, got an entirely different idea. He realized that in the collision between two fast-moving protons it may happen, though it very rarely does, that one of the protons will lose its positive charge and, thus becoming a neutron, will remain attached to another proton, forming a deuteron, that is, the nucleus of heavy hydrogen. The 'alchemical' formula of this reaction is:



where e^+ stands for the positive electron. Later deuterium nuclei (H^2 , which is also often written D^2) combine with other protons, growing ultimately into a helium nucleus (He^4). Thus H-H fusion leads to the same over-all result as the carbon cycle, although the intermediate steps are quite different (Figure 32). In particular, H-H fusion can proceed in pure hydrogen and does not require catalysts as the carbon cycle does.

Using Fermi's theory of beta decay, Critchfield calculated the rate of energy-production in this process in the conditions existing inside the Sun and found that it is competitive with the carbon cycle. During the next decade or so, the H-H fusion and the C-cycle theories were playing leapfrog. When first proposed in 1938, H-H fusion was an underdog, thought to have a yield some twenty times smaller than the C-cycle. Then, after some nuclear cross-sections were remeasured more carefully, and the central temperature of the Sun was recalculated more exactly, the C-cycle

became subservient, with H-H fusion being top dog. Later the situation changed again several times and finally stabilized on the conclusion that, for the Sun, H-H fusion is the principal

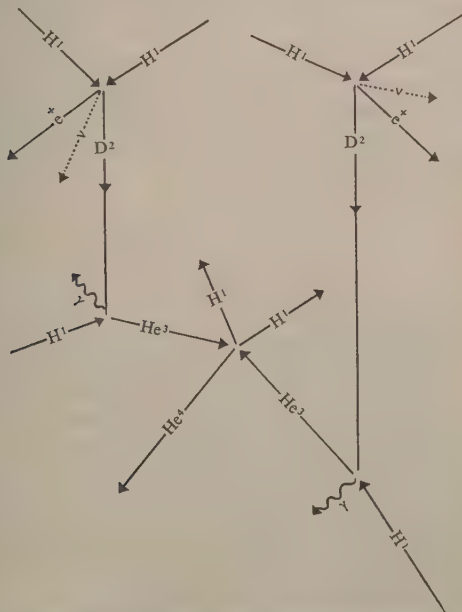
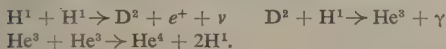


FIGURE 32. The H-H process consists of the transformation of hydrogen into helium by the following reactions:



energy-producing process, with the C-cycle a secondary one. The reason for the long vacillation between the two theories is very simple. The two processes show very different dependence on the temperature T , and whereas the H-H fusion rate increases as $T^{3.5}$, the carbon cycle proceeds as T^{17} . Now it so happens that the central temperature of our Sun is very close to the intersection of

change in the measured nuclear cross-sections or in the calculated central temperature of the Sun throws the balance very strongly one way or the other. But all this is immaterial, because both reactions result in the same product (He^4) and the evolution of a star proceeds in about the same way, independently of which reaction prevails in its interior. The fact is, however, that in all stars brighter than the Sun (and thus somewhat hotter in their interior) the C-cycle prevails, whereas all stars fainter than the Sun (and thus somewhat cooler) live on H-H fusion. Here again we find that our Sun is a very average star.

THE SUN AMONG THE STARS

On a void and misty ocean,
Self-propelled and riding free,
Heavenly lamps, with gentle motion,
Float in choral harmony.

— *Mikhail Lermontov*

Translated by Barbara P. Gamow

HOW MANY STARS ?

CAN you guess how many stars a person with normal vision can see in the sky on a clear night ? A billion ? No. A million ? Wrong again. As a matter of fact there are only about six thousand stars visible to the naked eye in both celestial hemispheres. The number of stars which can be seen in the sky at any one time is hardly more than two thousand because one half of them are hidden under the horizon, and many of those close to the horizon may not be noticed because of poor visibility conditions. The use of ordinary field glasses brings the number of visible stars up to fifty thousand, while an amateur's small two-inch telescope raises that number above three hundred thousand. Modern giant telescopes increase the number of stars which can be seen (or, still better, photographed) almost beyond any limit, and it is estimated that the stellar system of which our Sun is a member contains about one hundred billion stars.

When, in the year 1610, Galileo built the first astronomical telescope and pointed it at the faintly luminous band encircling the sky, he saw 'nothing else but a mass of innumerable stars planted together in clusters'. This observation discredited the ancient Greek legend that the Milky Way was made of milk spilled from the breasts of Hera, the Greek goddess of home happiness, when the hungry baby Hercules pressed them too greedily.

The famous British astronomer Sir William Herschel (1738–1822) was one of the first to formulate the hypothesis that the stellar system to which our Sun belongs occupies a lenticular volume, with the Sun located somewhere inside, near the plane of the lens. Thus, in any direction within this plane, we see a multitude of stars which are located farther and farther away, fusing into the luminous band of the Milky Way. In other directions we see many fewer stars with nothing but a dark background of empty space beyond them.

HOW FAR ARE THE STARS ?

Astronomers measure the distances to various celestial objects by the method of 'parallactic displacement', which is explained earlier in this book. But the distances to the stars are so much larger than the distances within the solar system that even observations from two opposite points on the globe could not reveal any noticeable parallactic displacement with respect to other stars located still farther away. Fortunately it is possible to carry out the observation of parallactic displacement of stars by using a much longer base than the diameter of the Earth. The Earth moves around the Sun along an orbit the diameter of which is 23,000 times larger than its own diameter. Thus, by observing the positions of a star six months apart in time (when the Earth is at the opposite point of its orbit around the Sun), we can increase the precision of parallax measurement by the same factor. The famous Danish astronomer Tycho Brahe (1546–1601), who constructed the most precise astrometrical instruments of his time, tried to observe the semi-annual parallactic displacement of stars but got a negative result. We know now that this was partly because Tycho's instruments were not precise enough to register the displacement and partly because he was just not lucky enough to pick stars that are comparatively close to us. Tycho himself, however, interpreted his negative results as proof that the Earth does *not* move around the Sun, as was claimed by the Copernican system, which was fighting for its life during these early years. And it was only in 1838 that the German astronomer Friedrich Wilhelm Bessel made the first reliable parallax measurement for a

star known as 61 Cygni (constellation of the Swan), obtaining the value of 0.30 angular seconds. Later in the same year the Scottish astronomer Thomas Henderson measured the parallax of a bright star, Alpha Centauri (constellation of the Centaur), obtaining the value of 0.76 angular seconds. From these figures it follows that the distances of 61 Cygni and Alpha Centauri from the Sun are 10.9 and 4.3 light-years respectively.* During later years more stellar parallaxes were measured, and today we know the distances to about three thousand stars.

HOW BRIGHT ARE THE STARS ?

Looking at the starry sky, we notice that some of the stars, such as Sirius, are exceptionally bright, whereas others are barely visible to the naked eye. Two great astronomers of the ancient world, Hipparchus and Ptolemy, arbitrarily graded the visible stars into six 'magnitudes', the first magnitude comprising about twenty brightest stars, and the sixth those which are just barely visible to the naked eye. (The eyes of the ancient astronomers were, of course, always naked.) The division into six groups was made by them so as to have the same *brightness-differences* between the members of the neighbouring magnitude groups. We know now that, according to the general law of the physiology of perception (Fechner's law), the strength of outside stimuli, either the brightness of light or the loudness of sound, appears to us to increase in *arithmetical progression* if the amount of energy which they carry increases in a *geometrical progression*. Thus, if we have three loudspeakers operating at 10, 100, and 1,000 decibels (the unit of sound intensity), our senses will tell us that the difference in loudness between the first and the second is the same as between the second and the third. Similarly, looking at three electric bulbs operating at 10, 100, and 1,000 watts, we will feel that the difference in illumination between the first

* A light-year is a unit used by astronomers for expressing large distances and is defined as a distance covered by light in the course of one year. Since one year contains 3×10^7 seconds, and the velocity of light in a vacuum is 3×10^{10} cm./sec., one light-year equals 9×10^{17} cm., or 9×10^{12} km., which is equivalent to 6×10^{12} miles.

and the second is the same as between the second and the third.*

Thus, whereas for the eyes of Hipparchus and Ptolemy stars of different magnitudes appeared to be spaced *equidistantly*, the intensity of light was actually spaced *equiquotiently*, that is, increasing by the same factor from one magnitude class to another. In standard astronomical practice it is accepted that the stars of the first magnitude are exactly one hundred times brighter than the stars of the sixth magnitude. Thus the actual ratio of the amount of light emitted by two stars differing by one Hipparchus-Ptolemy magnitude is the fifth root of 100, or 2.512. To sum up, the stars of the second magnitude are about two and one half times (2.512) fainter than those of the first magnitude; the stars of the third magnitude are fainter by the same factor than those of the second magnitude, etc. It would of course be more rational to forget about the old-fashioned notion of stellar magnitude and to compare directly the amount of light emitted by different stars, rather than the effect of that light on the human eye. But astronomers *are* old-fashioned, or rather sentimental, about the history of their science, and the notion of stellar magnitudes is retained instead of being converted into more rational intensity-ratios.

Adding to the confusion is the fact that the values of stellar magnitudes are somewhat different, depending on whether the observations are made *visually* (i.e. just by looking into the telescope), *photographically* (i.e. by inspecting the photograph taken by a camera attached to the telescope), or *bolometrically* (i.e. by measuring directly the total energy arriving from any given star with the use of sensitive thermocouples). Thus astronomers talk about 'visual magnitudes', 'photographic magnitudes', 'photo-visual magnitudes', and 'bolometric magnitudes', using special formulas or tables for converting one of them into another. You cannot blame astronomers for retaining all these complicated units, any more than you can blame the British for using pence, shillings, florins, half-crowns, pounds, and guineas. Once you

* Mathematically speaking, Fechner's law states that the intensity of perception increases as the *logarithm* of the actual intensity as measured directly by physical instruments.

learn how to translate quantities like shillings and pence into dollars and cents you do not have any trouble. Today stellar magnitudes are given with one decimal point (one tenth of the magnitude), which corresponds to the smallest difference of stellar brightness that can be noticed by the human eye. For stars too faint to be seen by the human eye, the magnitude scale is extended beyond the sixth, and, using this extension, we find that the 100-inch telescope of Mount Wilson Observatory can photograph stars of the twenty-first magnitude, while the 200-inch telescope at Mount Palomar can photograph those of the twenty-fifth magnitude.

For objects brighter than the brightest stars, negative magnitudes are being used. Thus the planet Venus is of about -4th

TABLE 7 *Number of Stars Brighter than Each Given Magnitude*

Magnitude Limit	Number of Stars	
	Photographic	Visual
4.0	360	530
5.0	1,030	1,620
6.0	2,940	4,850
7.0	8,200	14,300
8.0	22,800	41,000
9.0	62,000	117,000
10.0	166,000	324,000
11.0	431,000	870,000
12.0	1,100,000	2,270,000
13.0	2,720,000	5,700,000
14.0	6,500,000	13,800,000
15.0	15,000,000	32,000,000
16.0	33,000,000	71,000,000
17.0	70,000,000	150,000,000
18.0	143,000,000	296,000,000
19.0	275,000,000	560,000,000
20.0	505,000,000	1,000,000,000
21.0	890,000,000	

magnitude, the Moon -12.5 magnitude, and the Sun -26.7 magnitude.

Table 7 shows the numbers of stars brighter than each given magnitude. The differences between the second and third columns

are due to the difference in sensitivity between the human eye and the photographic plate, which we discussed on page 32.

So far we have been discussing the brightness of stars as they look to us from our observational platform, the Earth. But of course these figures do not represent the actual intrinsic stellar brightnesses, since a very bright star at a very long distance may look to us fainter than a really faint star which is much closer to us. But if we know the distance of a given star, having measured its parallax, we can calculate its *actual* brightness, using the basic law of optics which states that the apparent intensity of any light-source decreases as the inverse square of its distance. The actual brightnesses of stars are expressed in absolute magnitudes, which are defined as the magnitudes that would be observed by terrestrial astronomers if all the stars were brought closer and placed at a distance of 10 parsecs* from the Sun. While the apparent magnitude is usually denoted by the letter *m* (with sub-indices, *vis*, *phot*, *phot-vis*, *bol*, depending on the observational method used), absolute magnitudes are denoted by *M* (with the same variety of indices).

TABLE 8 *Luminosity Function of Stars*

Absolute Magnitude M	Relative Number of Stars	Star's Brightness in Terms of Sun's Brightness
-5	1	10,000
-2.5	90	1,000
0	3,300	100
+2.5	42,000	10
+5	200,000	1
+7.5	350,000	0.1
+10	500,000	0.01
+12.5	600,000	0.001

While the apparent magnitude of a star is a purely subjective notion of its brightness, depending on our position within the Milky Way, absolute magnitudes correspond to cosmic reality. Table 8 gives the relative number of stars having different absolute magnitudes which characterize their actual brightness (this dependence is known as 'luminosity function').

* A parsec is equal to 3.258 light-years, which is 3.084×10^{13} km., or 1.92×10^{13} miles.

The numbers in this table run quite differently from those in Table 7, because of what is known as 'observational selection'. Very bright stars can be seen at much longer distances than fainter ones, so that it appears to us that they are more numerous than they really are. Thus, whereas Table 7 described only the visual impression which we get by looking at the starry sky, Table 8 gives the actual number of stars of different absolute brightnesses. It follows from this that more than 97 per cent of all stars are fainter than the Sun, and only less than 3 per cent are brighter.

THE MOTION OF STARS

Since antiquity the stellar sky has served as a symbol of permanence and invariability, and, in fact, a stellar catalogue compiled in the middle of the second century B.C. in Alexandria by Hipparchus, which shows the positions of 1,080 stars, can be used today with a sufficient degree of precision. Various constellations named by Assyrian, Babylonian, Egyptian, and ancient Greek astronomers stand out on the starry sky just as they stood thousands of years ago. However, observation with very precise astronomical instruments shows that stars do move across the sky, even though that motion is too slow to change to any large extent the general pattern of the stellar sky in the course of just a few thousand years. Astronomers call the slow motion of stars their *proper motion*; it is not actually slow but appears slow to us because of the very large distance. In the same way, an aeroplane flying at a very high altitude seems to move slowly, whereas the same aeroplane roaring close over our heads appears to move very fast. The majority of stars move through space with a velocity of between 5 and 20 miles per second, and there are some stars that move much faster.

In Figure 33a and b, we show the motion of stars forming the constellations of Corona Borealis (constellation of the Crown of the North) and of Ursa Major (constellation of the Great Bear). The arrows indicate the expected displacements in the course of 50,000 years. We notice that in Corona Borealis the motion is quite at random, indicating that the stars forming that

constellation are complete strangers to one another and have come together by sheer chance. On the other hand, all but two stars in Ursa Major move with nearly the same velocity in nearly the same direction, indicating a possible common origin. A very

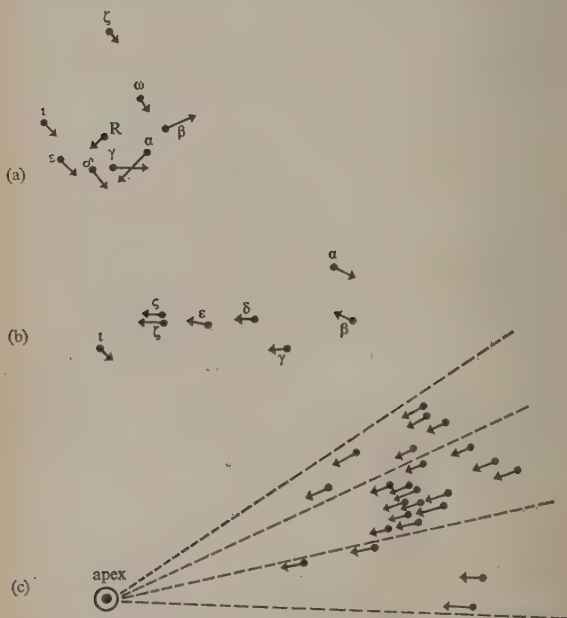


FIGURE 33. Motion of stars in the constellations of the Northern Crown and Great Bear, and in Hyades cluster during the next 50,000 years.

striking picture is presented by the observed proper motion of a large number of closely located stars which form the so-called Hyades cluster of Taurus (constellation of the Bull), shown in Figure 33c. If one continues the arrows showing stellar motion (broken lines), one finds that they converge toward one point.

No, this does not mean that all these stars will collide with one another in some distant future, nor that they started at the same point in some distant past. All these stars actually move along parallel tracks away from or toward us, and what we observe is only an optical illusion, just as parallel railway tracks seem to converge in the far distance.

While observation of the proper motion of stars tells us about the changes of their positions in the direction perpendicular to the line of vision, there exists another method which gives us stellar velocities in the direction of this line. It is based on the Doppler effect, which was mentioned in Chapter 3 in relation to the study of the rotation of the Sun's surface (see page 79). When a star moves at high speed toward us, light waves emitted from it become somewhat shorter, being shifted toward the violet end of the rainbow spectrum. When a star moves away, the light waves become longer and the shift is toward the red end. Even at the great speeds at which stars move, the shift is very small, and it would be impossible to notice it just by slight colour changes. Fortunately the spectra of all stars (including the Sun) are intersected by a multitude of thin dark lines caused by the absorption of light in stellar atmospheres. The positions of these spectral lines can be measured very exactly, and even a small shift toward violet or red can be detected. By combining the methods used on proper motion and observation of the Doppler effect in stellar spectra, the astronomer can get complete information concerning the motion of stars through space. It may be noticed here that, while the shift of spectral lines is independent of the distance of the star from us, the apparent proper motion decreases with that distance and may become unnoticeably small in the case of stars located very far away. Thus, for very distant stars only their radial motion (toward or away from us) can be measured.

EVOLUTIONARY SEQUENCE

As we have seen in Chapter 5, our Sun derives its energy supply from the nuclear reactions which take place in its hot interior. It is only natural to assume that all other stars in the sky obtain their energy in a similar way, although in some cases the particular

nuclear reactions may be different from those taking place inside the Sun. Since the young star has a limited amount of nuclear fuel, this fuel can last only for a limited length of time, so that each star must have a limited life-span from its birth to its ultimate death. Thus, looking at the stellar population of the Milky Way, or of any other galaxy, we may expect to find various stars in different stages of their evolutionary life, just as, studying the population of a city, we find babes-in-arms, schoolchildren, teen-agers, adults, and a scattering of people who have just celebrated their hundredth birthdays. The percentage of people of various age groups gives us reliable information concerning the duration of each particular evolutionary stage of a human being. Similarly, inspecting the stellar population of a galaxy, we can get very definite information concerning various stages of stellar evolution and their relative duration in time.

The study of human populations is simpler than that of stars since, whereas all humans have approximately the same life expectancy, the life expectancy of stars varies as much as from that of a butterfly to that of an elephant. It was found by direct observation, and later confirmed by theoretical considerations, that the fainter stars live much longer than the brighter ones. This conclusion is based on the observed fact that the brightness of a star which is similar to the Sun in its internal structure increases as the cube (third power) of its mass. Thus a star twice as massive as the Sun is eight times brighter; a star three times as massive, twenty-seven times brighter, etc. The brightness of a star is proportional, of course, to the rate at which its nuclear fuel is consumed. On the other hand, the amount of the nuclear fuel allotted to each star at its birth is proportional to its mass (first power). Thus, a star twice as massive as the Sun has twice as much nuclear fuel but consumes it eight times faster. Therefore its life-span is expected to be four times shorter than that of the Sun. Similarly a star ten times more massive than the Sun has ten times more fuel but, consuming it one thousand times faster, will exhaust its supply a hundred times sooner than the Sun.

Therefore, if all the stars within our galaxy were formed at about the same time, we conclude that there must be an upper limit to the brightness of stars that are still in the sky. The stars

that were originally too massive and therefore much too spend-thrift in the use of their nuclear fuel must have all burned out by now. On the other hand, less massive stars, using their fuel more economically, must still be shining today as they were at their birth. The study of stellar luminosities shows indeed that there is an upper limit to the brightness of stars populating our system of the Milky Way. This upper limit corresponds to the stars with an age of about five, and maybe even as high as 10 billion years, a figure which apparently describes the age of our stellar system, the Milky Way.*

To understand this statement, let us suppose that a litter of mice, a litter of puppies, and several babies were born on the same day. According to biological data, mice live about one year, dogs ten to fifteen years, and human beings much longer. Thus, if at some time after their birth we find all the mice dead, some of the dogs still alive, and all the babies in the prime of their youth, we may conclude that ten to fifteen years have passed since their birth.

There is a saying that 'It's the exception that proves the rule', and this applies indeed to the stars of the Milky Way. There are some stars, known as Blue Giants, which shine so brightly that they could not possibly have lasted for more than a hundred million years or even less. There is no way of explaining their existence except by assuming that they are newcomers – *nouveaux riches* – in galactic society, having been formed from the diffuse interstellar material much later than the main bulk of stars.

SPECTRAL CLASSIFICATION

Just as in the case of the Sun, stellar spectra are crossed by dark Fraunhofer lines resulting from the absorption of light in their atmospheres and indicating the presence of various chemical

* There are only a few stars which seem to be older than 5 billion years. They belong to galactic star clusters studied by A. R. Sandage of the Mount Wilson and Palomar observatories. Their very low hydrogen content suggests that they may be twice as old as most of the other stars in the galaxy. However, the possibility is not excluded that the abnormally low hydrogen content of these stars is due not to very advanced age, but rather to an original hydrogen deficiency at the time of their formation.

elements. While the characteristic wave lengths in stellar spectra are identical with those in the spectrum of the Sun, their relative intensities vary a great deal. Some stars, like our Sun, show very strong, sharp lines of various metals, notably calcium, sodium, and iron, while others show only broad lines of hydrogen. Still others exhibit a wealth of thin, narrow-spaced lines fusing into continuous dark bands characteristic of light-absorption by molecules. The presence of molecules in the atmospheres of these stars agrees with the fact that they look red, indicating low surface temperature. At the end of the last century the American astronomer Edward Charles Pickering, with a group of workers at Harvard Observatory, undertook the laborious task of arranging spectra of different stars into a single sequence in such a way that the pattern of Fraunhofer lines would change very little from one class of spectra to another. To get as many stellar spectra as possible, he used a so-called objective prism, that is, a large prism placed in front of the telescope's objective rather than at the eyepiece. The original plan of this classification was to call various spectral classes by the letters of the alphabet, beginning with *A* for stars having strong hydrogen lines and going down to *M*, where hydrogen lines are hardly visible at all. But after the terminology was fixed, some original classes (*C*, *D*, and *H*) were dropped, others were reshuffled, and one new class (*O*) for very hot stars added. Thus, as it is used now, the Harvard spectra classification runs:

Oh, Be, A Fine, Girl, Kiss, Me.

Also three new classes were added to the red stars:

Right, Now, Sweetheart.

The fact that different stars possess different spectra should not, however, be interpreted (as it was in olden times) as the result of their different chemical compositions. As we have seen in Chapter 2, the appearance of Fraunhofer lines of any given element depends to a high degree on the temperature and pressure in the absorbing gas layer. Using the quantum theory of atomic structure, Mehnad Saha was able not only to obtain the correct proportion of elements in the atmosphere of the Sun, but also to

show that various classes of stellar spectra can be explained by assuming the same composition but different temperatures and pressures in the atmospheres of various stars. Thus a sequence of spectral classes became simply a sequence of surface temperatures of stars. Table 9 gives the relation between the spectral class, the

TABLE 9 *Surface Temperature of Stars (after Kuiper)*

Name	Spectral Class	Temperature	Colour
Alpha Crucis	B1	23,000° K.	bluish
Spica	B2	20,400	bluish
Achernar	B5	15,500	bluish
Rigel	B8	12,300	bluish
Sirius	A0	10,700	white
Altair	A5	8,530	white
Procyon	F3	6,800	straw

Dwarf Stars

Name	Spectral Class	Temperature	Colour
Sun	dG0	6,000°	yellow
Sigma Eridani A*	dG5	5,360	yellow
Epsilon Eridani	dK0	4,910	orange
61 Cygni A	dK7	3,900	orange
Lacaille 9352	dM2	3,200	red

Giant Stars

Name	Spectral Class	Temperature	Colour
Capella A	gG0	5,200°	yellow
Kuiper	gG5	4,620	blond
Arcturus	gK0	4,230	orange
Aldebaran	gK5	3,580	orange
Betelgeuse	gM2	3,200	red

* Capital A refers to the brightest component of the binary stars. The letters *d* and *g* before the spectral class indicate dwarf and giant respectively.

surface temperature, and the visual colour of stars, based on the studies of the Dutch-born American astronomer Gerard P. Kuiper.

The reader may be surprised by the fact that a one-stranded sequence, O, B, A, F, splits into two strands for G, K, M. As we shall see later in this chapter, while hot stars can exist only in

one state, a cooler star can exist in two different modifications, possessing the same surface temperature but having entirely different luminosities. Astronomers call such stars Red Giants and Red Dwarfs. The two kinds have very little to do with each other, except for their colour. It may be very surprising too to read the names of stars. Some are clearly personal names given by ancient astronomers, such as Achernar, Sirius, Procyon, Capella, Betelgeuse, etc. Those are the brightest stars in the sky, attracting the attention of both astronomers and astrologists. The stars which have no personal names are called by the constellations to which they belong, using the Greek alphabet to indicate their visual intensity; thus Alpha Crucis is the brightest star in the constellation of the Cross. The Greek alphabet has 24 letters. So when the 24 brightest stars of a given constellation are named by Greek letters, astronomers call the next one by the number 25. Thus, for example, 61 Cygni (for which the first measurement of the parallax was made) is the 37th faintest star after Omega in the constellation of the Swan. But what is Lacaille 9352?! Well, it is just the ordinal number of that star in the stellar catalogue composed by the French astronomer Nicolas de Lacaille. In spite of a Tower of Babel confusion, this system of notation works, and it would lead only to more confusion if a 'more rational' notation system were proposed.

The author cannot help relating here an amusing story – changing, for the purpose, to the first person. Soon after the end of the Second World War I took my family to a ranch near Lake Louise in the Canadian Rockies. We planned a long pack trip, and the question arose which of the ranch horses was strong enough to carry me through a long mountain trip (6 feet 3 inches and 220 pounds – *my* height and weight, not the horse's). The only animal qualified for the task was a tall red stallion, called by the cowboys 'Big Red' for lack of a better name. I decided to rename him Betelgeuse (see Table 9), which is a typical Red Giant star in the constellation of Orion. The star was given its name millennia ago by Arab astronomers and 'Betelgeuse' in Arabic means 'shoulder of a giant' – a most appropriate name for my horse. The pack trip turned out to be very successful, and my Betelgeuse carried me faithfully each day from dawn to dusk.

A number of years later we revisited that ranch. All the cowboys were new, but Betelgeuse was still there. 'What is the name of that horse?' I asked. 'Battle Goose, sir,' answered the stable boy. 'But why Battle Goose?' 'Don't know, sir, but he always was called that.'

HERTZSPRUNG-RUSSELL DIAGRAM

During the First World War two astronomers in different parts of the world, the American Henry Norris Russell and the Dutchman Ejnar Hertzsprung, were carrying out very important studies without knowing of each other's work. Both scientists wanted to know how the surface temperatures (spectral class) of different stars are correlated with their intrinsic brightness (absolute luminosity). As a result came the famous Hertzsprung-Russell diagram reproduced in Figure 34. Here on the ordinate is plotted the spectral class of any given star, while on the abscissa is plotted its luminosity with respect to the Sun in terms of absolute stellar magnitude. The most striking feature of this diagram is a rather narrow band running from the upper left to the lower right corner. It is known as the *main sequence* of stars and includes our Sun sitting roughly in the middle of it. The upper end of the main sequence contains very bright stars with such high surface temperatures that they emit bluish-white light. They are the Blue Giants, mentioned earlier in this chapter as comparatively young stars that spend their nuclear fuel at a high rate. The lower end of the main sequence contains very faint red stars with rather low surface temperature; they are called Red Dwarfs. Our Sun, which is located about half-way between Blue Giants and Red Dwarfs, and which emits mostly yellow light, could be called Yellow Mediocrity.

Apart from the main sequence stars, there are also a few stars which fall in the upper right corner of the Hertzsprung-Russell diagram. They are very luminous, but their surfaces are comparatively cool and emit mostly red light. They are known as Red Giants. The term 'giant' pertains not only to the high luminosity of these stars but also to their large geometrical dimensions. In fact, since their temperature is low, they emit comparatively little

light per unit surface. Thus, in order to explain their high luminosity we must assume that their surfaces, and hence their diameters, are exceedingly large. In Figure 35 the dimensions of some Red Giants are given in comparison with that of our Sun.

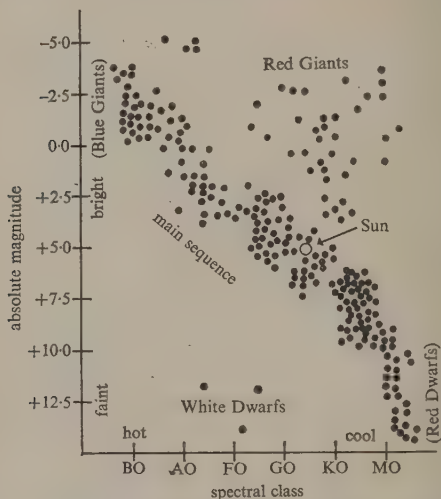


FIGURE 34. Hertzsprung-Russell diagram for 230 brightest stars, showing the break-up into (normal) Main Sequence stars, Red Giants, and White Dwarfs.

In the lower left corner of the Hertzsprung-Russell diagram we see three points corresponding to the stars with very high surface temperature and very low luminosity. These stars are known as White Dwarfs and, reversing the argument about Red Giants, we must conclude that their diameters are exceedingly small. In fact, the first White Dwarf discovered is a faint companion of the brilliant Sirius. Its diameter is comparable with that of the Earth, while its mass is almost equal to (95 per cent of, to be exact) the mass of the Sun. The mean density of the companion of Sirius is

500,000 times that of water, and one cubic centimetre of it must weigh about half a ton.

Very little is known about the birth and the childhood of stars. They apparently condense under Newton's law of universal gravity from the cool and extremely rarefied interstellar material, and become smaller and hotter as the condensation process proceeds. The time necessary for the condensation can be calculated from the Thomson-Helmholtz theory discussed earlier, and is only about one hundred million years, which is very short as cosmic times goes. Having a very low luminosity (if any)

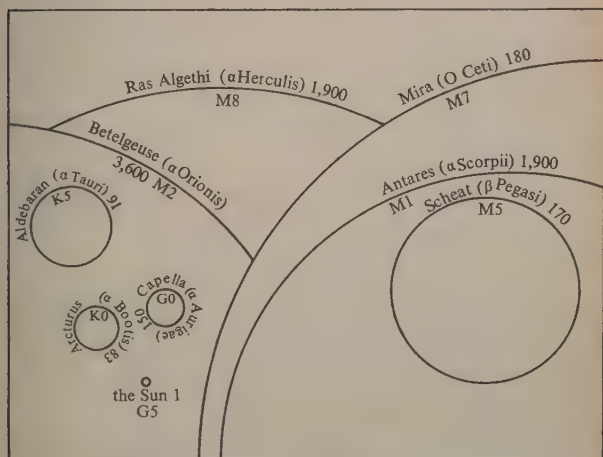


FIGURE 35. The dimensions of various Red Giant stars. The numbers indicate the luminosities of these stars as compared with that of the Sun, while the letters give their spectral class. (Notice that Capella is about half-way between the Sun and the real Red Giants.)

and existing for only a short time, young stars represent a statistical rarity in the stellar world. Similarly, the number of newborn babies in any city represents a very small fraction of the number of schoolchildren or adults.

Once the temperature inside a contracting newborn star hits the mark of about 20 million degrees, which is necessary for starting the thermonuclear reaction, the situation changes completely. Now oodles and oodles of nuclear energy pour from the star's centre, keeping it in status quo as long as the hydrogen fuel lasts. Since hydrogen is plentiful in the Universe, a star will remain in this stabilized state for a much longer period than was necessary for its condensation from interstellar gas. Of course, depending on circumstances, newly born stars may have masses ranging from a fraction of the mass of the Sun to many times that mass. And since, as was mentioned earlier, the rate of nuclear-energy consumption depends on the star's mass, the 'adult' stars will take various positions in the frame of the Hertzsprung-Russell diagram. The more massive ones will be hotter and brighter; the less massive ones will be cooler and fainter. Thus, the main sequence in the Hertzsprung-Russell diagram is simply the spectrum of stars of different masses.

RED GIANTS

While the stars belonging to the main sequence were very well understood on the basis of the hydrogen-helium transformations taking place in their hot interiors, Red Giant stars remained a mystery for some period of time. Why are they so tremendously large, so extremely rarefied, and so comparatively cool on their surfaces? Apparently they could not be built in the same way as the normal stars of the Main Sequence. In the year 1939 the present author, in collaboration with his student Charles Critchfield (the inventor of the H-H reaction), conceived an entirely new idea concerning the internal structure of Red Giant stars. As we have seen, an accepted model of the normal Main Sequence star consists of: (1) central, pointlike nuclear-energy source; (2) a convective zone which supplies fresh hydrogen to the site of thermonuclear reactions; and (3) the outer stagnant envelope through which the radiation diffuses toward the surface. As the transformation of hydrogen into helium proceeds in the centre zone, the convective region surrounding it gradually becomes deprived of hydrogen fuel and enriched by helium pro-

duced in the thermonuclear reaction. What happens when the hydrogen content of the convective zone finally comes to zero? Apparently the reactions in the centre of the stars are extinguished for lack of fuel and no more energy is produced in this region. On the other hand, beyond the limits of the convective zone there is still enough hydrogen to react if the temperature of that region rises high enough to maintain the carbon-cycle or the H-H reaction. Thus, one must expect that the central region of the star will contract until the temperature on the surface of the former convective zone reaches the value of 20 million degrees necessary for maintaining thermonuclear reactions in hydrogen. As a result of this temperature adjustment, the internal structure of the star will slowly change to something rather different from that characteristic for the Main Sequence stars. Thermonuclear reactions in hydrogen will now be taking place on the interface between the dehydrogenized former convective zone and the hydrogen-rich outer envelope. The inner zone will now be formed almost entirely of helium at a constant temperature (isothermal), as is the inside of a kitchen oven. The internal structure of the star will thus change from the original *point-source model* to a *shell-source model* in which nuclear energy is produced not in the centre but on the surface of a sphere, as is shown in Figure 36. As the hydrogen fuel at the boundary is gradually consumed by thermonuclear reaction, 'nuclear fire' will move farther and farther away from the centre toward the surface of the star. The situation is similar to that of a fire started in a dry field of grass by a carelessly dropped match, as the fire ring moves outward, leaving scorched earth behind it. As the high-temperature region in which the nuclear reactions take place moves from the centre outward, the outer body of the star gets heated more and more and expands into the surrounding space. The star swells up to an enormously large volume, and its surface temperature drops because less heat is radiated through each unit area of its surface. In other words, the star becomes a Red Giant and remains in this state until most of the hydrogen of the outer envelope is used up.

Although the qualitative picture of the Red Giant stage of evolution was reasonably clear, exact mathematical calculations of this process were prohibitively difficult because they had to

be done by hand or with the aid of the table computer machines. The best that the present author, collaborating first with Critchfield and later with Geoffrey Keller, could do was to prove that a star which develops a shell source has a 'tendency' to move into the Red Giant region of the Hertzsprung-Russell diagram.

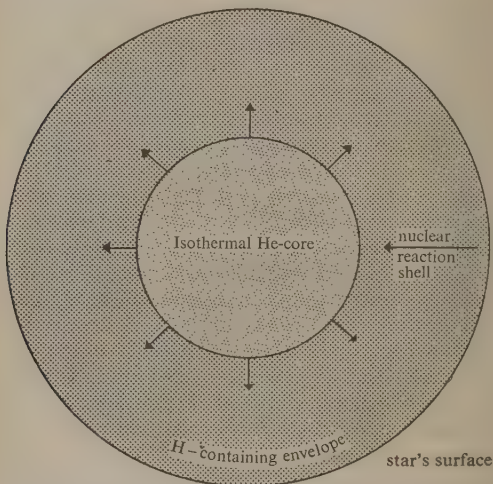


FIGURE 36. The shell-source model of a Red Giant star. (After G. Gamow and C. Critchfield)

After electronic computers, developed during the Second World War mostly for military purposes, became available for purely scientific research, Schwarzschild and A. R. Sandage undertook this calculation on the electronic computer of Princeton University. The results, which completely confirmed the original expectations, are shown graphically, in Figure 37 for five stars with 1, 1.5, 2, 3, and 4 sun-masses respectively. Starting from the Main Sequence, after the hydrogen in the convective zone is exhausted, the star moves toward higher luminosity, lower surface temperature, and a much larger radius. The peculiar kink in the evolu-

tionary line is caused by an entirely new reaction which sets in when shell-source evolution has progressed to some degree. To understand the reason for that new reaction, we must note that the calculation of the shell-source model shows that, while the outer diameter of the star increases, the diameter of the energy-producing shell surrounding the central helium core shrinks. Thus

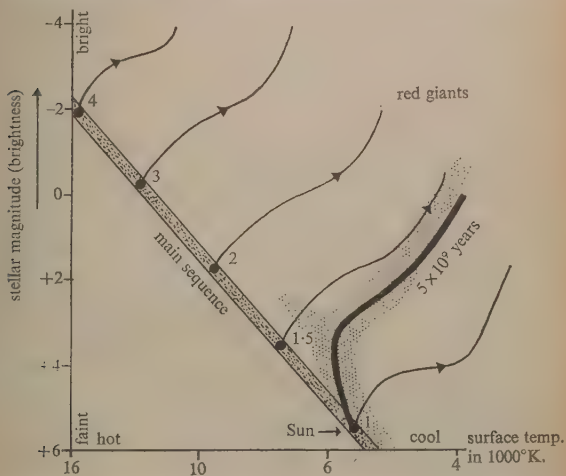


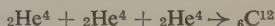
FIGURE 37. Evolutionary tracks in Hertzsprung-Russell diagram of five stars with 1, 1.5, 2, 3, and 4 sun-masses respectively as calculated by Schwarzschild and Sandage on the basis of the shell-source model. Thin lines, starting from the points marked 1, 1.5, 2, 3, and 4, represent individual tracks of the stars. The heavy line represents the expected distribution of stars in the Hertzsprung-Russell diagram, 5 billion years after the stars were formed. The shaded area gives the observed distribution of stars in the stellar population II. Fits very nicely!

the helium core constantly increasing in mass nevertheless decreases in size, which leads of course to a rapid increase in the density of its material. The Australian physicist Edwin Salpeter calculated that at the temperature of about 100 million degrees

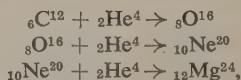
and at very high densities helium nuclei begin to react among themselves, going through a fusion process. The first step of fusion would be:



However, a beryllium isotope with atomic weight 8 does not exist in nature and has never been produced artificially. Natural beryllium is Be^9 , containing one extra neutron, which helps two helium nuclei to stick together. The only way to fuse helium nuclei into the nuclei of heavier elements is to rely on a triple collision:



which results in the formation of ordinary carbon. But triple collisions are very rare, and the rate of this reaction would be expected to be negligibly small even at the very high density of the helium core, if the situation were not saved by one lucky circumstance. It has been shown experimentally that two helium nuclei do hold together for an appreciable amount of time; they do not bounce from each other right away but stay together for a period of time very short from our human point of view but rather long in the nuclear time scale. This brief encounter between two helium nuclei gives a much better chance for the third one to join the company and to form a stable nucleus of carbon. After that is accomplished the rest of the fusion processes are easy, since they require the addition of only one helium nucleus at each step. These reactions are:



etc., producing stable isotopes of well-known elements and liberating large amounts of nuclear energy. When helium begins to 'burn', the locus of stellar energy production moves again to the centre of the star, with the hydrogen-burning shell dwindling on the outside. However, the Salpeter reactions do not go all the way to uranium. It is well known that the largest binding of atomic nuclei occurs in the region of iron-group elements, and that the binding decreases in the groups toward both the beginning and the end of the periods in the periodic system. Thus the fusion

processes can liberate energy only until the nuclei of iron are formed, and, as time goes by, more and more iron is accumulated in the stellar interior. It is possible that, when a star reaches the extreme stage of the Red Giant evolution, its interior consists entirely of iron, chromium, and nickel.

The kinks on the evolutionary curves in the Sandage-Schwarzschild diagram correspond to the onset of the helium-fusion reaction. It must be remembered that, although the evolutionary tracks for stars of different masses look very much alike, they are travelled at quite different speeds. Indeed, as we discussed earlier, the life-span of a star decreases as the square of its mass. Thus the track on the extreme left in the diagram will be covered by an evolving star sixteen times faster than the track on the extreme right. In addition we have seen that most stars of the Milky Way were probably born at about the same time: some 5×10^9 years ago. This statement is particularly true for the stellar groups known as globular clusters. If all stars were born at the same time but have evolved at different rates, depending on their masses, they must be today at different stages of the evolution. The solid curve in Figure 37 shows how the distribution of stars in the frame of the Hertzsprung-Russell diagram should look today if they all had started their evolution 5×10^9 years ago. The shaded area gives schematically Hertzsprung-Russell distribution observed in a typical globular cluster. The agreement between the theory and observation is really magnificent!

According to calculations, our Sun still has enough hydrogen in its interior to last as a normal Main Sequence star for another five billion years. But by the end of that period it is bound to have developed a nuclear shell source and to be evolving in the direction of Red Giant stars. In practically no time (speaking astronomically) the red-hot disk of the Sun will occupy a good part of the sky, as shown in Plate 9. Oceans will boil, and all life will be extinguished on the surface of the Earth. But we still have about five billion years before this disaster!

DOUBLE, TRIPLE, . . . , STARS

To the naked eye, stars look like single bright luminous points scattered over the sky. But telescopic observations show that about half of them consist of two, and sometimes three or more, closely located components rotating around the common centre of gravity under the action of Newtonian gravity. Thus, only about one third of all stars are single, while the other two thirds live in couples, and less often in triplets, quadruplets, etc. Just to make a comparison, let us remember that about one fifth of the men and women in the United States are single. We describe here three examples of stellar families.

The bright star Alpha Centauri consists of two stars revolving about one another with a period of about 80 years. The orbits of the two components are rather eccentric: sometimes they approach as near as 11 astronomical units* (about the distance of Saturn from the Sun) while other times they recede to 35 astronomical units (i.e. about the distance from the Sun to Pluto). The brighter component of the pair is almost the 'splitting image' of our Sun, though maybe slightly more massive, more luminous, and hotter. The fainter component is cooler, less massive, and less luminous. It was discovered recently that there is a third component of the family, much cooler and much fainter. It moves at about 10,000 astronomical units away from the brighter pair and takes about 1 million years to complete its orbit.

Another multiple star is Alpha Ursae Majoris. Its brighter component is composed of two stars which move around each other in 1.83 years. The pair of stars forming the fainter component swing around each other in four days, and the two pairs move around each other with a period of 60 years. The total mass of this quadruplet stellar system is estimated to be 2.27 Sun masses, the two brightest components having 0.93 and 0.35 masses of the Sun respectively.

Another multiple star is the famous Castor (Alpha Geminorum), which was proved to be a double star by the English astronomer James Bradley in 1719. In 1896 the Russian astrono-

* An astronomical unit is defined as the distance from the Earth to the Sun.

mer A. Belopolsky found that the fainter component of Castor is itself a double star with a period of about 3 days. In 1904 R. H. Curtis in England proved that the brighter component of Castor is also a binary with the period of 9 days. Finally Walter S. Adams of Mount Wilson Observatory found another double star not far from the two principal binaries, which is also participating in the general motion of the Castor family. Thus Alpha Geminorum was finally proved to be a sextuple star.

The average distances between the components of multiple stars vary from 0.01 astronomical units ($\frac{1}{40}$ the distance between Mercury and the Sun) to 100,000 units (3,000 times the distance from the Sun to Pluto). It is highly significant that the average distance of planets from the Sun is very close to the average distance between the components of multiple stars.

PLANETARY SYSTEMS

We are living on the third planet (counting from the Sun) of the solar system and should be happy about it. In fact, we would be roasted on Mercury or Venus and frozen into icicles on Jupiter and on the planets beyond. Only Mars may harbour life, but, according to the best available evidence, it has nothing to offer but mosses and lichen.

The problem of the origin of the solar system of planets has occupied the minds of great thinkers for centuries. In the middle of the eighteenth century the French naturalist Georges Louis Leclerc, Comte de Buffon, suggested that planets were formed by the impact of a comet which ran into the Sun on its way through space. A few decades later a quite different hypothesis was formulated independently by the famous German philosopher Immanuel Kant and the no less famous French mathematician Pierre Simon, Marquis de Laplace. According to the Kant-Laplace hypothesis, the Sun gave birth to planets parthenogenetically – that is, without the cooperation of any other body. These men assumed that after its formation from interstellar material the Sun rotated around its own axis, a rotation which increased rapidly as the body of the primeval Sun contracted under the forces of gravity. At a certain stage of the contraction

the Sun's matter started to stream out from its equatorial region, forming a gaseous ring (solar nebula) around it. Subsequently this gaseous ring broke into pieces (Sir Isaac Newton's gravitational forces again) and condensed into the individual planets.

These views were criticized later from the mathematical point of view by the British physicist James Clerk Maxwell, who proved (incorrectly, because of the contemporary lack of information) that the hypothetical solar nebula of Kant and Laplace would not be massive enough to condense into planets under the action of Sir Isaac's forces. Thus, at the turn of the century, cosmogony returned to Buffon's old hypothesis, substituting, however, for a comet another star in a collision with the Sun. The collision theory was rejuvenated by Sir James Jeans in England, and by Thomas C. Chamberlin and Forest Ray Moulton in the United States.

But this theory did not work satisfactorily, and during the last year of the Second World War the German physicist Carl von Weizsäcker rejuvenated the old and forgotten Kant-Laplace hypothesis, using new astronomical information. Weizsäcker's theory was extended by the Dutch-born American astronomer Gerard P. Kuiper, and today there remains hardly any doubt that the Kant-Laplace hypothesis was essentially correct.*

PLANETS V. BINARIES

Earlier in this chapter we noticed an interesting correlation of the mean distances between the components of binary and multiple stars with the average distance of the planets from the Sun. Is this really a coincidence? And why, roughly, are about one half of all stars binaries or multiples, while at least our Sun possesses a system of planets? Of course astronomical observation cannot tell us whether or not some other stars also possess planetary systems, since, even in the case of the nearest stars, planets would be absolutely undetectable even by the largest telescopes. But it would be strange to think of our Sun as a single exception among one hundred billion stars forming the system of the Milky Way.

* For a more detailed discussion on the origin of the planetary system, see the author's book *A Planet Called Earth*.

It seems much more reasonable to assume that the formation of a planetary system is an alternative to the formation of a double or a multiple star. For all we know, the formation of stars from diffuse interstellar material is due to the same good old Sir Isaac and his law of gravity. Sir James Jeans showed, in fact, that a uniform distribution of gas in an infinite (or at least a sufficiently large) space is gravitationally unstable and is bound to break up into lumps of various sizes which will eventually contract into rather dense gaseous spheres (stars).

The large expanses of gas are not only gravitationally unstable but are also subject to turbulent motion. *Turbulence* is a term which until recently was used only by hydrodynamical and aerodynamical engineers, but it is now becoming of paramount importance in many astronomical and cosmogonical problems. This irregularity of fluid motion known as turbulence is a very important factor in all fields of fluid dynamics, from the design of an aeroplane wing to the explanation of the origin of stars and planets. We can see turbulent motion on the surface of a fast-flowing river when we look down from a bridge; similarly we can feel it in the form of discontinuous gusts of wind blowing in our faces. Turbulence in a fluid medium is a very complicated and completely orderless motion, which is not easy to represent by a simple picture or diagram. Probably the nearest representation of what turbulence actually is is given by Figure 38, in which individual turbulent streamers are represented schematically by arrows of different sizes. If we look at this diagram from a distance we first observe large arrows winding around one another. Closer inspection will show, however, that the large arrows consist of a large number of smaller ones, which in turn consist of arrows of still smaller size. Extend this picture in two directions, so that the largest arrows will be almost as large as the entire volume of the fluid and the smallest ones almost as small as individual molecules, and you will have a fairly clear picture of what turbulent motion actually is. So that this picture could be drawn at all, the arrows in each category are shown in Figure 38 as about the same size. In actuality, the hierarchy of turbulent streamers includes all sizes of arrows as well as all directions of motion. Turbulent motion within a fluid includes the rotational as well as the translatory

motions of its different elements, and it is because of the rotational type of motion that turbulent streamers are commonly known as 'eddies'. It would be incorrect to think that turbulent eddies maintain their individuality for a long time, enabling us to represent turbulent motion by giving the positions of different eddies for properly chosen intervals of time. Actually, the lives of individual eddies are very short; they usually disappear after



FIGURE 38. Hierarchy of eddies.

travelling a distance comparable to their own diameter, giving rise to new eddies which may move in an entirely different direction. The presence of turbulent motion in a moving fluid leads to an increase in its internal friction known as turbulent viscosity. If, for example, the propeller of a motor-boat were to produce nothing but a laminar flow behind its stern, the boat would hardly be pushed at all and would move at a snail's pace. The same is true for the propeller of an aeroplane; the wings, on the other hand, should be designed to avoid turbulent motion at their

surface, so that they can slide through the air with the least possible resistance.

At first glance, it would seem impossible to develop a consistent theory for such complex and irregular motion as that presented by a turbulent flow, and until quite recently the study of turbulence was carried out (mostly for engineering purposes) on a purely empirical basis. But during recent years the theory of turbulence has been put on a strictly mathematical basis, mostly by the work of Geoffrey I. Taylor in England, Theodore von Kármán in the United States, A. N. Kolmogoroff in Russia, and Werner Heisenberg in Germany. One of the main results of that research was the derivation of the so-called 'energy spectrum' of turbulent motion. The motion of eddies involves large amounts of kinetic energy, which is being continuously transferred from larger eddies to smaller ones, all the way down from large-scale motion of the fluid to the molecular motion of its constituent particles. We know that in all kinds of friction the kinetic energy of motion changes to heat; in the particular case of turbulent friction this transfer goes down along the hierarchy of eddies of ever-decreasing size. The question we may ask here is: how much energy is stored in eddies of different size (what is their 'energy spectrum'?) or, in other words, what are the velocities of the different irregular streamers that constitute turbulent motion? Theoretical considerations, too involved to be even hinted at here, lead to the conclusion that velocity distribution between the eddies of different sizes is governed by the so-called Kolmogoroff law, which states that the velocity is proportional to the cube root of the size – the smaller the size of an eddy, the smaller the corresponding velocity. Imagine, as an example, a fast turbulent stream of water in a channel 10 metres (1,000 centimetres) in width. The eddies comparable in size with the width of the channel will have velocities comparable to the total velocity of flow. Eddies 1 centimetre in diameter will have about one tenth of the total velocity ($\sqrt[3]{1,000}$), and eddies which are only 10 microns across will have one hundredth of the velocity of the main flow.

We come now to a very important problem, namely that of the conditions under which a regular laminar flow of fluid will break up into turbulent eddies. These conditions were established

purely empirically by the British physicist Osborne Reynolds, who studied the flow of various liquids at different speeds through tubes of different diameters. He found that the smooth laminar flow of a liquid always breaks up into turbulent eddies when the velocity of flow exceeds a certain limit, a limit which is lower for wider tubes and for liquids of lower viscosity. These findings can be summarized by means of a simple empirical formula:

$$R = \frac{\text{density of fluid} \times \text{velocity of stream} \times \text{width of stream}}{\text{viscosity of fluid}}$$

where the dimensionless quantity R is known as the *Reynolds number*. If the density and viscosity of the fluid, and the velocity and width of the flow, are such that the number defined by this equation is less than about 1,000, the flow will remain laminar; if the number is greater than 1,000, turbulent motion will appear. The empirical condition for the appearance of turbulent motion obtained by Reynolds was given a theoretical foundation in the recent work of Werner Heisenberg, but again the theory is much too complicated to be described here.*

Turbulent eddies condensed as a result of gravitational instability form gaseous whirlpools in interstellar space, some of them rotating rather slowly, the others rotating very fast. As they gradually contract (Sir Isaac's forces, again) their rotation speed increases, and various things may happen, depending on how fast the original rotation was. They may split into several pieces, or remain in one piece with a ring of gas extending in the equatorial plane, and later condense into a number of smaller bodies. In the first case we get a multiple star; in the second a planetary system.

What is the chance that a planetary system will be formed? We can answer that question by considering the observed relative masses of the binaries. In some cases they are of comparable mass; in other cases one is much less massive than the other. According to the Weizsäcker-Kuiper theory of the origin of the planetary system, the original solar nebula was about ten times

* The foregoing discussion of eddies was previously published in the author's book *The Creation of the Universe* (New York (Viking), 1961), in which the reader will find more details concerning this topic.

less massive than the Sun itself. The relative number of binaries in which the faint component is at least ten times smaller than its brighter brother is about 10 per cent. Thus, we may say that one in ten stars out of the one hundred billion stars forming the solar system is a candidate for having planets. Applying a safety factor of one tenth, we may conclude that at least one billion stars in the Milky Way possess planetary systems.

WHEN A STAR DIES

Vous, stériles soleils,
Qui n'êtes plus que cendre . . .

— *Charles-Marie Leconte de Lisle*

THE LAST MILE

By the time that a star reaches the final stage of its Red Giant evolution, all nuclear-energy sources in its interior are presumably completely exhausted. Thermonuclear reactions, which for many billions of years counteracted the forces of gravity and near the end blew the body of the star up to giant size, are now extinguished, and the Newtonian attraction between the parts of its body begins to pull the stellar material toward the centre. The star is not yet dead, since the contraction liberates gravitational energy which, being turned into heat, maintains the high temperature in its interior. The star is now described by the Kelvin-Helmholtz model discussed in Chapter 1 of this book. As we have seen there, the evolution of a contractive stellar model proceeds considerably faster than that of the nuclear-source model and, depending on the mass of the star, may take only a few hundred million years or even less.

During this last stage of evolution the radius of the star continuously decreases and its luminosity diminishes. On the other hand, its surface temperature goes up and so does the temperature in its interior. The decrease of luminosity does not contradict the increase of the surface temperature; each unit area of the star emits more and more light, but its surface decreases so quickly that the total energy emission drops down. Thus the star rolls down from the upper right corner of the Hertzsprung-Russell diagram to the lower left corner. Since stars pass through this stage of evolution comparatively fast, the number of stars in this stage seen in the sky is rather small. In the frame of the Hertz-

sprung-Russell diagram there is just a diluted scattering of points indicating where the contractive evolution line runs. But, although the stars passing through this evolutionary stage are rather rare, they are well known to astronomers and are subject to intensive studies because they display great showmanship. While normal stars shine with steady brightness, contracting stars behave very erratically. They puff out their material, change their luminosities within weeks or even days, go through a series of minor explosions, and finally blow up as giant cosmic fireworks.

PULSATING STARS

The best-known misbehaving stars are *Cepheid variables*, or simply cepheids, so called after Delta Cephei, the first star which was observed to have these properties. Figure 39a shows the luminosity curve of a typical cepheid – Delta Cephei itself. In contrast to eclipsing variables, where two stars revolving around the common centre periodically hide behind each other (see Chapter 6), the luminosity curve of cepheids shows smooth variation of brightness, with a rather steep rise and a much slower descent. In 1914 the American astronomer Harlow Shapley suggested that periodic luminosity changes of cepheids are due to the pulsations of their giant gaseous bodies. This suggestion was confirmed by the study of the Doppler shift in the spectral lines of the cepheid variables. If their photospheres periodically rise and fall, thus moving toward us and away from us, spectral lines should show periodic shifts toward the blue and red ends of the spectrum. Such a periodic shift is really observed; the velocity curve of the surface layers of Delta Cephei, shown in Figure 39b, has indeed exactly the same period as the corresponding light curve.

A few years after Shapley's proposal, Eddington developed a mathematical theory of pulsating stars, a theory not much more complicated than that of an oscillating pendulum or a vibrating string. It followed from that theory that the pulsation periods of cepheids must decrease in inverse proportion to the square root of their mean densities. This relation is similar to the familiar acoustical statement pertaining to the vibration of strings in

various musical instruments: the thicker the string the slower will be its vibration period, and the lower will be the pitch of the sound it emits.

The agreement between the theoretical prediction of Eddington and the observation can be seen from Table 10.

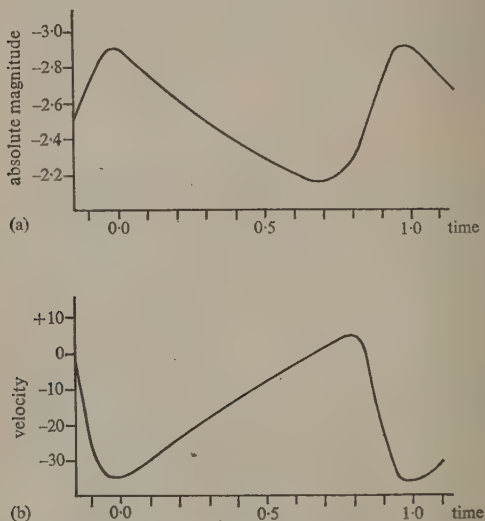


FIGURE 39. Variability curves of Delta Cephei. The upper curve represents the variation of brightness (in terms of absolute magnitude). The lower curve represents the variation of surface velocity (in km./sec). Time is given as a fraction of the pulsation period (which in this case is 5.366 days).

It shows that for the pulsating variables differing in their brightness by a factor of almost a hundred, and in their pulsation period by a factor of sixty, the product of their period by the square root of their mean density remains nearly constant.

But this success of Eddington's pulsation theory is still not spotless. First of all, according to the theory, luminosity and

velocity curves should have equal slopes on both sides, which they do not have. Second, the maxima and minima of luminosity should occur at the time when the pulsating star reaches the smallest and the largest dimension, from which it would follow that at the maxima and minima of the light curve the velocity curve should pass through zero. A glance at Figure 39 shows that this is certainly not the case. Many attempts were made to explain these discrepancies by assuming that the outer layers of cepheids pulsate out of phase in respect to their inner bodies, but the situation still remains rather unclear.

TABLE 10 *Pulsating Stars*

Star	Mass $M \odot = 1$	Spectral Class (Variable)	Mean Luminosity $L \odot = 1$	Period (Days)	Density $\rho \odot = 1$	Period Multiplied by $\sqrt{\text{Density}}$
RR Lyrae	3.7	B9-F2	100	0.57	0.022	0.08
SU Cassiopeia	5	F2-F9	225	1.95	0.0032	0.11
Polaris	8	F7-F8	400	3.97	0.00049	0.09
δ Cephei	9	F4-G6	500	5.34	0.0005	0.12
η Aquilae	11	F2-G9	800	7.18	0.0003	0.13
ζ Geminorum	15	G0-G1	1,400	10.15	0.00009	0.10
χ Cygni	19	F8-K0	2,800	16.39	0.00013	0.19
γ Ophiuchi	23	F8-G7	3,200	17.12	0.00005	0.13
ϵ Carinae	50	F8-K0	9,000	35.52	0.000008	0.10

* The symbol $\odot$ means the Sun. Hence $M \odot$ means the mass of the Sun.

Another important empirical law pertaining to the pulsating stars is the famous *period-luminosity relation*. In 1910 Miss Henrietta S. Leavitt of Harvard was studying the cepheids in two Magellanic Clouds, so named after the well-known sea captain of the sixteenth century who first spotted them in the southern sky. These 'clouds' are actually the aggregates of billions of stars floating in space millions of light-years beyond the limits of our stellar system of the Milky Way. At that time nobody knew how far away they actually were, but it was logical to conclude that all stars belonging to them are at about the same distance from us, just as all Swiss cities are at about the same distance

from New York City. Thus, although Miss Leavitt could not tell what were the absolute brightnesses of these distant pulsating stars, she could be sure that their actual brightnesses are proportional to their apparent brightnesses as observed from the Earth. She was surprised by the fact that there is a definite relation between the observed luminosities of these stars and their pulsation periods: the brighter a star is, the longer it takes to go

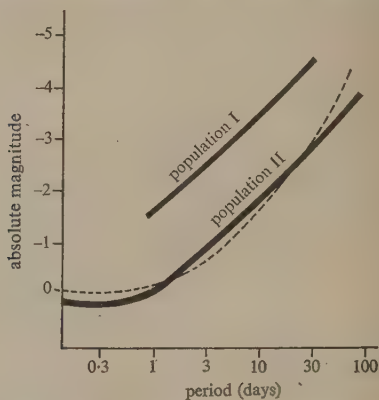


FIGURE 40. Period-luminosity relation for pulsating stars. The two solid curves pertain to two stellar populations, according to Walter Baade. The broken line is the original Harlow Shapley curve.

through the cycle. The relation between the luminosities of various cepheids and their pulsation period is shown graphically in Figure 40. Soon after Miss Leavitt's discovery, Shapley realized the importance of cepheids as a yardstick for the measurement of cosmic distances. Indeed, from the observed pulsation period the absolute luminosity of the star can be found directly from the graph and, comparing it with the visual luminosity, one finds the distance to the star simply by using the inverse-square law for distant light sources.

An important question concerning cepheids is the nature of the energy source which keeps them going. Eddington had calculated

that, because of internal friction of the gases forming a star, its pulsations – whatever causes them – will be rapidly damped and will completely stop within only a few thousand years. This conclusion contradicts the observed fact that the amplitudes of pulsating stars remain practically constant. We have here a situation similar to that of a pendulum clock with the driving mechanism hidden in a box behind the dial. From elementary mechanics we can obtain a simple relation between the length of the pendulum and its oscillation period, but if there were nothing driving the pendulum, friction forces would bring it to rest in five or ten minutes. What is in the box? A spring, an electric motor with a battery, or a squirrel in a wheel? And that is the question which astronomers ask themselves about cepheids. Since cepheids apparently represent the late stage of stellar evolution, it is likely that no nuclear-energy sources are left in their interiors and that their contraction is purely gravitational. But why don't they contract smoothly?

It was suggested by Edward Teller many years ago that such a chronic instability of the stellar body may be caused by purely atomic (not nuclear) phenomena, such as the variation in the degree of ionization deep under their surfaces. This suggestion has been substantiated by more recent theoretical studies carried out by John Cox at the Joint Institute of Laboratory Astrophysics in Boulder, Colorado.

PUFFING STARS

In the year 1867 two astronomers, Charles Wolf and Georges Rayet at the Paris Observatory, spotted a very unusual star. Instead of a continuous spectrum crossed by thin Fraunhofer lines, this star showed broad, bright emission lines superimposed on a somewhat fainter continuous background. During the following years more and more stars possessing that type of spectrum were discovered, and they are now known under the general name of Wolf-Rayet stars. The explanation of these peculiar spectra was proposed later by C. S. Beals of Dominion Astrophysical Observatory, Canada, and, independently, by Donald H. Menzel, then at the Lick Observatory in California. The wide

emission lines could originate only in a rapidly expanding gaseous envelope, and their great width must be due to the Doppler effect in the light emission from the atoms rushing at high speed away from the surface of the star. The gases streaming toward us show a violet shift of spectral lines, those streaming away a red shift, and those streaming at different angles have intermediate shifts which fill the gap between the two extremes. To explain the observed width of emission lines one has to assume that the gases are expelled with tremendous velocities of 3,000 kilometres per second, nearly 2,000 miles per second.

Applying to the expanding envelope the ionization theory, similar to that worked out by Meghnad Saha for the chromosphere of the Sun, one can find the temperature of the expelled gases. Table 11 gives the temperatures obtained from different spectral lines belonging to the highly ionized states of helium, carbon, nitrogen, and oxygen.

TABLE 11 *Temperatures of Highly Ionized Gases in Stars*

Star*	Ion	Ionization Potential (ev)	Temperature (Absolute)
B.D. + 43° 3571	He II	54.14	18,000
	C III	47.6	23,000
	C ⁺ IV	64.2	40,000
	O III	54.6	28,000
	O IV	77.0	70,000
	O V	113.0	80,000
B.D. + 35° 4001	He II	54.14	44,000
	N III	47.2	40,000
	N IV	77.0	75,000

* Here the name of star refers to *Berliner Durchmusterung*, a stellar catalogue prepared in Potsdam Observatory.

The reader may be surprised that for the same star the temperatures estimated from different spectral lines are so different. But this is quite natural, because the light emission originates from different depths of the extended stellar envelope. Thus the radiation of highly ionized atoms, such as O V, originates in hot, deep layers of the expanding envelope ($T = 80,000^\circ \text{K.}$), whereas

the radiation of C III originates in the cooler outer layers ($T = 23,000^{\circ} \text{K.}$).

Hydrogen emission lines are conspicuously absent in Wolf-Rayet spectra, probably because the original hydrogen supply has completely turned into helium in this late stage of stellar evolution.

What causes the star to blow its material away at this stage? Most probably it is the pressure of the radiation emitted from its hot surface. It is estimated that the surface temperature of the central Wolf-Rayet star is $100,000^{\circ} \text{K.}$ (as compared with $6,000^{\circ} \text{K.}$ for the Sun), and at such temperature radiation pressure on the surface may well exceed the forces of gravity which tend to hold the star in one piece. Thus it proceeds to blow itself out of existence.

An interesting example of a Wolf-Rayet star was discovered by the Harvard astronomer Sergei Gaposchkin, who started his career as a sailor on board a life-saving ship on the Black Sea shore and saved thousands of lives before he ever looked at a star through the telescope. The star is called B.D. + $38^{\circ} 4010$ and is an eclipsing binary, one of the components being a Wolf-Rayet star. From his observations Gaposchkin was able to calculate that the Wolf-Rayet component is 10 times more massive than the Sun, 6 times larger in diameter, and 230 times as bright. So far, this is the only Wolf-Rayet star for which all these data could be obtained.

Similar to Wolf-Rayet stars are the so-called P Cygni stars and the B emission stars, but it is impossible to go into details about these in the framework of this book.

Closely related to Wolf-Rayet stars are the so-called *planetary nebulae*, which were badly misnamed because they have actually nothing to do with what we call planets. Scanning the stellar sky, astronomers of olden times found a number of luminous objects which looked like disks, instead of looking like mere points as do ordinary stars. One of these objects, a planetary nebula in Lyra, is shown in Plate 10. We know now that planetary nebulae are the expanding gaseous envelope surrounding the ageing stars and fluorescing under the action of ultra-violet radiation from very hot stellar bodies in their centres. But the gas masses ejected

from the central bodies of planetary nebulae are moving out considerably more slowly than in the case of Wolf-Rayet stars (at only about 20 kilometres per second, or 12 miles per second) and are consequently considerably denser. Thus we can see and photograph them as large spheroidal bodies. Table 12 gives information on some of the planetaries.

Just as in the case of the solar corona, when astronomers suspected the existence of a new element, 'coronium', which later turned out to be simply highly ionized iron, the study of planetary

TABLE 12 *Planetary Nebulae*

Name*	Radius in Terms of the Diameter of the Earth Orbit	Tempera- ture (° K.)	Density Atoms/cm. ³	Mass in Terms of Sun Mass
NGC-7027	10,600	9,500	5×10^3	0.07
NGC-6572	8,600	9,200	1×10^4	0.07
NGC-6826	13,600	7,200	4×10^3	0.12
NGC-6547	10,800	6,000	1×10^4	0.07

* NGC refers to the ordinal number in Dreyer's *New General Catalogue*.

nebulae led to the hypothesis of another new element, 'nebulium', responsible for the two green lines in the observed spectrum. But a closer study has led to the conclusion that these lines are simply due to double- and triple-ionized oxygen, which were at the earlier time unknown to the terrestrial spectroscopists.

THE NOVAE

In contradiction to an ancient belief that the starry sky is eternal and unchangeable, things happen up there all the time. Even before the invention of the telescope, astronomers noticed radical changes in the stellar pattern, such as the sudden appearance and subsequent disappearance of bright new stars in the framework of the familiar constellations. They were called *novae* because they were believed to be entirely new stars and not, as we know today, temporary flare-ups of old ones. The first new star recorded in the Western World was Nova Cassiopeia, 1572,

studied by Tycho Brahe, who concluded that it belonged to the stellar world outside the solar system.*

It is estimated that at least twenty-five novae visible through a telescope with a one-inch aperture flare up in the sky every year, but only about ten of them since the beginning of the century could have been seen by the naked eye. Novae appear most often within the belt of the Milky Way and are strongly concentrated in the constellation of Sagittarius. The discovery of a nova is more or less a matter of chance, since one cannot look all over the sky every night, but it certainly requires a good knowledge of the constellations. However, since there are so many stargazers, amateurs as well as professional astronomers, novae are usually sighted on the first night of their appearance. A good example of a nova which was easily visible to the naked eye is Nova Aquilae, which flared up on 8 June 1918. That evening the noted American astronomer Edward E. Barnard was driving with some colleagues in an open car, leisurely staring at the starry sky above his head. Suddenly, as his glance passed the constellation of Aquila, he exclaimed, 'That star should not be there!' Indeed, a new star of the first magnitude was in the sky, outshining all other stars except Sirius and Canopus. Considering the exceptional brightness of Nova Aquilae, it is strange that, apart from Barnard, it was noticed that night by only ten men on Earth.

The study of photographs of that region of the sky, taken by chance only two days earlier and also many years earlier, indicated that the star had been there but had been about a hundred times fainter and thus invisible to the naked eye. After reaching the maximum the next day after its discovery, Nova Aquilae began to fade, rapidly at first and then more slowly, and in less than a year it disappeared from the naked-eye view, returning to the brightness which it had had before. Faint as it looks from the Earth, it still is about five times brighter than the Sun and at its outburst was five hundred times as bright.

About six months after the appearance of Nova Aquilae, a faint greenish aureole was observed around the star. It was

* As we will see later, Chinese astronomers beat Tycho Brahe by more than five centuries by discovering a new star in Taurus. Damned clever, these Chinese!

expanding at the rate of about two seconds of arc per year. This observation, combined with the Doppler shift in the spectrum of the expanding aureole, permits astronomers to determine the distance of the exploded star. In fact, the Doppler shift gives us the velocity of ejected gases in kilometres per second, which, being combined with the observed angular expansion rate, give the distance of the luminous object from us. In the case of Nova Aquilae the distance was 1,200 light-years.

Why do novae explode? We cannot understand this any better than we can fathom why cepheids pulsate. Probably because of some instability in the body of an aged star deprived of its nuclear-energy sources and contracting toward its ultimate doom. We know that the energy liberated in a nova explosion is negligible compared with the total thermal energy contained in its body, so that there is no reason why the same star cannot become a nova again and again at intervals of maybe a few thousand years. A few thousand years is a long interval for observational astronomy, but some stars blow off more often than that.

In the year 1898 a new star appeared in the constellation of the Serpent Bearer, was named RS Ophiuchi, and faded out within a year. Thirty-five years later E. Loreta of Bologna noticed that the same star flared up again. The luminosity curve of the 1933 flare-up was very similar to that of 1898, and it is quite likely that this star flares up periodically every thirty or forty years. A similar case is presented by U Scorpii, which flared up in 1863, 1906, and 1936, and by T Pyxidis, which did so in 1890, 1902, and 1960. These stars, known as recurrent novae, differ, however, from the regular novae, having a much smaller increase of brightness during the flare-ups.

Then there are the U Geminorum type of stars, some of which blow up every fortnight, while others do so every two or more months. But in these cases the increase of luminosity at the maximum is much smaller than for recurrent novae. Two Russian astronomers, B. W. Kukarkin and P. P. Parenago, established a very interesting relationship between the periods of explosions and the brightness at the maximum. The longer the time lapse, the more vigorous is the explosion. In fact, the product of the time lapse between the consecutive explosions and the amount of

energy liberated in each explosion is the same for all U Geminorum stars. The recurrent novae fall neatly on the same line, and if we use that graph to find what should be the period between the explosions of the regular novae, we get the value of about ten thousand years. This may explain why the regular novae have never been observed twice.

THE SUPERNOVAE

An entry in the *Records* of the Royal Observatory of Peking reads:

In the first year of the period Chihha, the fifth moon, the day Chi-chou, a great star appeared approximately several inches south-east of T'ien-Kuan (i.e. Zeta Tauri). After more than a year it gradually became invisible.*

If one looks at the spot in the sky indicated in this quotation, one finds a rather peculiar object known as the Crab Nebula (Plate 11). It does not look exactly like a crab, but it certainly resembles a crab much more closely than the constellation Virgo resembles a virgin or Canis Major a big dog. The Crab Nebula has been known to astronomers for a long time, but it was only about three decades ago that Walter Baade and Fritz Zwicky, both from Mount Wilson Observatory, associated it with the new stars observed by Chinese astronomers nine centuries ago. Comparing older photographs of the Crab Nebula with more recent ones, one notices that it is gradually growing larger and, dividing its present visual diameter by the observed rate of expansion, one finds that the luminous gases must have begun to expand from a central point about nine hundred years ago. Thus there is no doubt that the Crab Nebula is a remnant of the new star observed by the Chinese.

The Doppler shift in the spectrum emitted by the Crab Nebula shows that it expands with the velocity of 1,111 kilometres per second. Combining this information with the observed visual rate of expansion (0.18 angular seconds per year), we find that it is about 5,000 light-years away from us. From the Chinese

* 4 July 1054, in our calendar.

statement that this far-distant star was visually of the first magnitude, it becomes apparent that at the maximum its brightness was comparable with the combined brightnesses of all the stars forming the Milky Way. Whereas ordinary novae are about a hundred thousand times brighter than the Sun, the Chinese new star in Taurus was almost a billion times brighter. Such giant stellar explosions are called *supernovae* and are much rarer than the ordinary novae. Since the Chinese supernova of 1054, there have been recorded only two similar giant explosions in the Milky Way. One took place in 1572 in the constellation of Cassiopeia and was described in detail by Tycho Brahe in his book *De Nova Stella*. Another took place only thirty-two years later and was recorded by Brahe's former assistant, Johannes Kepler. No supernovae have since been observed in the Milky Way and, since they seem to appear at an average rate in a particular galaxy of about one in three hundred years, the time is ripe for the next one. When it comes it will give the astronomers a wonderful time, as they will be able to use telescopes, spectrographs, and other astronomical hardware which was not available in the days of Brahe and Kepler.

But, even though supernovae appear rarely in our stellar system of the Milky Way, there are plenty of other stellar systems, or galaxies, floating far away in space but not so far away that a supernova cannot be seen in them. As recently as 1885 a bright new star appeared in our nearest cosmic neighbour, the Great Nebula of Andromeda. It did not cause much excitement, since at that time astronomers did not know that the so-called spiral nebulae are giant stellar groups comparable to our Milky Way, floating in space millions of light-years away. On the contrary, they were assumed to be similar to the gaseous nebulae (like that in Orion) located between the stars of the Milky Way. Thus the new star in Andromeda was classed as an ordinary nova. Only after Edwin P. Hubble had proved that the Great Spiral in Andromeda is really much farther away than was originally believed did astronomers realize that the new star which appeared in it was exceptionally bright and thus quite different from the common novae. Taking this hint, Baade and Zwicky organized systematic observations of a few hundred neighbouring stellar

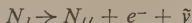
galaxies and, within a few years, bagged a dozen or more supernovae. Plate 12 shows photographs of a supernova which flared up in a galaxy known as IC-4182 in August of 1937 and had faded out of sight by 1942.

As we have stated, the total amount of energy liberated in a nova explosion, tremendous as it is, is but a small fraction of the total heat and gravitation energy contained in the body of a star. Thus ordinary novae can keep exploding again and again over a long period of time. In the case of a supernova the situation is quite different, since the total energy emitted in a single explosion (10^{41} calories) requires the total energy content of the star. Thus a supernova explosion can be considered *rigor mortis*, as the physicians would say. What causes these suicidal explosions at the end of the long-lasting and glorious life of a star? Just as in the case of the erratic behaviour of ageing stars during the last few hundred million years of their lives, we probably do not deal here with some spontaneous liberation of the latent nuclear energy. It must be some kind of instability of the star's body which permits the tremendous amounts of heat energy accumulated in its interior to be released into the surrounding space.

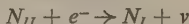
Several theories have been proposed to account for the catastrophic events leading to supernova flare-ups. In 1941 the author of this book, working with a Brazilian student, Mario Schoenberg, suggested that a collapse of a supernova might be the result of a sudden drop of pressure in its interior, caused by the emission of neutrinos, those highly penetrating elementary particles which were discussed in Chapter 5. This particular neutrino-producing nuclear reaction received the name of 'Urca Process' after the Casino da Urca in Rio de Janeiro, where the two authors of this theory first met. A closer interpretation of that name compares the rapid disappearance of energy in the interior of an old star with an equally fast disappearance of money from the pockets of the players crowded around the roulette tables in the Casino da Urca. The official explanation given to the editor of the *Physical Review*, where the article presenting the theory was published, was that the name is just a shortening of '*unrecordable cooling agent*'.

What is the Urca Process? As we have seen in Chapter 5, the emission of a beta-particle (negative electron) by an unstable

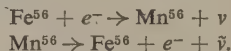
nucleus (N_I) is always accompanied by the emission of a neutrino (or antineutrino, in modern terminology) which carries the balance of energy:



where e^- stands for a negative electron and $\bar{\nu}$ for the antineutrino. According to the basic physical principle of reciprocity, there must also exist a reverse reaction:

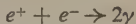


in which a nucleus emits a neutrino after absorbing an electron. A concrete example of this reversible reaction is given by the 'alchemical' equations:

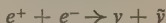


In these two equations, Fe^{56} stands for the most abundant isotope of iron, and Mn^{56} for an isotope of manganese. Since the neutrinos and antineutrinos produced in these two reactions pass without any difficulty through the body of the star and escape into interstellar space, the reaction goes: Tick-tock-tick-tock- Fe^{56} - Mn^{56} - Fe^{56} - Mn^{56} - Fe^{56} -tick-tock... , and the neutrinos, along with the antineutrinos, pour out of the star, carrying with them large amounts of energy. Thus the interior of the star cools, the pressure drops, and the old-man star collapses in a burst of light and heat.

More recently a much more vigorous process for neutrino production was proposed by Philip Morrison from Cornell University. He pointed out that in the hot interior of ageing stars there must be a large number of positive and negative electrons. Normally the encounter of a positive and a negative electron results in their mutual annihilation with the emission of two gamma-ray quanta:



But one can calculate that once in 100,000,000,000,000,000,000 cases a neutrino and an antineutrino are formed instead:



Meagre as it seems, the rate of neutrino production in hot interiors of stars coming to the end of their contractive evolution

would become of paramount importance, and the entire stellar body would collapse even faster than it would as a result of the Urca Process. Ladies and gentlemen, this is the supernova, a star that dies glamorously!

DEAD STARS

In the centre of the Crab Nebula, Baade discovered a very faint star, so faint that even though it can be seen on the original negatives it does not show on the photograph of the Crab Nebula reproduced (Plate 11). This must be the remnants of the star the explosion of which fascinated Chinese astronomers nine centuries ago. The mass of that star is estimated to be comparable to that of the Sun, whereas the mass of expanding luminosity is only about one tenth of the Sun's mass. The puzzling point is that, in spite of the apparent faintness of that central star, it must be held responsible for the illumination of the entire nebula. It is like a little candle that makes a large white screen shine with brilliant light. No, this is not a paradox; but for those who know the laws of light this fact carries a very important message. It simply means that the surface of the central star of the Crab Nebula is so hot that it emits almost all its energy in the form of short-wave ultra-violet rays, with only a very small fraction falling into the visible part of the spectrum. While passing through the gaseous shell surrounding the central star, the ultra-violet radiation is absorbed by various ionized atoms in the nebula and re-emitted in the form of visible light. The situation is similar to that of a dancer on a darkened stage whose clothes shine brightly under the action of ultra-violet light invisible to the public. R. Minkowski from Mount Wilson and Palomar Observatories calculated that in order to produce the Crab Nebula effect the central star must have a surface temperature of about $500,000^{\circ}\text{K}.$, that its radius must be 50 times smaller than that of the Sun, and that its mean density must be about 25,000 times higher than that of water. This represents a highly collapsed state of matter, deprived of nuclear sources and pulled together by the forces of Newtonian gravity.

How can matter be compressed to such tremendously high

density? As we have seen in Chapter 4, matter in the state of plasma behaves as a perfect gas and is highly compressible until its density approaches that of nuclear fluid. In the case of the Sun and other normal stars the plasma forming their giant bodies is prevented from contraction under the action of gravity forces by the nuclear energy which is continuously liberated in their interior. But what happens when these nuclear energy sources are not available, as in the case of old stars which have completely exhausted their nuclear fuel? To answer this question we have to investigate the behaviour of plasma under conditions of very high densities and comparatively low temperatures.

In the normal state of matter, when all electrons are confined to their nuclei they move along various quantum orbits, forming individual atoms of various elements. When the atoms are crushed, by either very high pressures or very high temperatures, the electrons are free to move all around, which does not mean, however, that they can completely throw off the yoke of the quantum rules. Instead of being quantized within the individual atomic volumes, they become quantized within the entire volume occupied by plasma. The de Broglie waves associated with the motion of electrons no longer run around the nuclei along closed orbits, but rather to and fro between the walls of the container. Just as in the case of a single atom, we have here discrete energy levels, each with two electrons, but now as a 'quantized commonwealth' of free electrons, which are confined not by the electric attraction of the nuclei but by the walls of the container. It does not matter here how large the container is; it may be one cubic centimetre or it may be the entire volume of a star. The larger the volume of the container, the larger is the number of free electrons within it, and one factor compensates for the other.

We do not have to fly to stars to find material containing free electrons. A copper wire, a silver spoon, or an iron bar would do. Metals differ from all other substances in being good conductors of electricity, and the electric conductivity of metals is due to the presence of free electrons which are torn from the outer shells of metallic atoms by strong interactive forces. Properly speaking, the 'free' electrons responsible for the conductivity of metals are not exactly free, and have to shoulder their way through the

tightly packed atoms which form a crystal lattice of a solid. When the famous Dutch physicist Paul Ehrenfest, shortly after his visit to Russia in 1931, delivered in Leyden a lecture on the electron theory of metals, he compared the motion of electrons in a metal with the crawling of lice through the hair of Russian peasants. As a rule, free electrons move in a completely disorganized fashion in all possible directions. But if an electric tension is applied to a metallic wire they begin to move preferentially in the direction of electric force, thus forming an electric current.

Whereas in metals only a small fraction of electrons are stripped from the atoms, which still maintain their sovereignty over the rest of the electrons, in the case of high-temperature plasma all electrons become free and form what is known as an electron gas. How this gas behaves in respect to the temperature and the density of gas depends on the numerical relation between these two quantities. If the temperature is high and the density comparatively low (as in the Sun and other normal stars), the electron gas behaves as the perfect gas discussed earlier in this chapter. But, if the density becomes very high while the temperature remains about the same, all lower quantum levels become crowded with electrons, and the resistance to further compression is highly increased. Such cases are referred to as 'degenerated Fermi electron gas'. This stops further contraction of the stellar body, but not until the density of the material shoots sky high. The central star of the Crab Nebula is probably on the way to that collapsed state but has not yet quite reached it, since the catastrophe took place quite recently on an astronomical scale.

But there are stars in the sky which have reached that collapsed stage, being presumably remnants of supernovae which flared up hundreds of millions of years ago. Those are the so-called White Dwarfs which were mentioned in Chapter 6. A notable example of such dead stars is the companion of Sirius (Sirius B) which, although about as massive as the Sun, has a diameter only slightly larger than that of the Earth. The mean density of that star is a million times larger than that of water, and its central regions must be even denser. The surface temperature of the star is about $10,000^{\circ}$ K., but the total amount of light it emits is very small because of its extremely small surface. Only

about a hundred White Dwarfs are known today, but their frequency in the universe is much greater than that figure would imply. In fact, because of their very low intrinsic luminosity, they can be seen even through the largest telescopes only when they are comparatively close to the Sun, and this drastically



FIGURE 41. The relationship between the radii and the masses of cold stellar bodies, according to the calculations of the Indian astrophysicist S. Chandrasekhar. The symbols $\ominus$, $\oplus$, h , and J respectively represent the Moon, the Earth, Saturn, and Jupiter. Note that for masses greater than 460,000 times the mass of the Earth, the radius becomes zero! The words for mass and radius are in Dr Chandrasekhar's original Tamil.

limits the number that can be observed. It has been estimated that White Dwarfs may form as much as 5 per cent of all the stars of the Milky Way.

Being deprived of nuclear energy, and not being able to contract any more, they are hot and luminous only because they have not yet had enough time to cool. It has been estimated that a White

Dwarf may last billions of years because of the tremendous heat-capacity of its dense body and the very small surface through which the heat may escape into the surrounding space. To them apply the words of Leconte de Lisle, quoted at the beginning of this chapter: 'You, sterile suns, which are nothing more than cinders . . .'

Detailed mathematical analysis of the inner structure of such collapsed stellar configurations was carried out by the noted Indian theoretical astronomer S. Chandrasekhar, who is now at Yerkes Observatory, University of Chicago. Using the quantum formula describing the behaviour of the degenerated electron gas, he was able to calculate the radii of collapsed stars of different masses. The results of these calculations are shown by the descending part of the curve in Figure 41. Chandrasekhar was also able to show that the pressure of the degenerated electron gas is able to support the collapsed stellar bodies only if their mass does not exceed a certain critical value equal to 1.5 Sun masses. For the stars of larger masses Chandrasekhar's curve intersects the ordinate axis, indicating that, being deprived of sources of nuclear energy, these stars are expected to contract to much higher densities, equal to that of nuclear fluid (10^{14} grams per cubic centimetre). Collapsed stellar configurations with densities about a million times higher than that of water are thoroughly familiar to astronomers.

COLLAPSED PLANETS ?

As was mentioned earlier, the complete break-up of electron shells of atoms can be caused not only by very high temperatures (as in the case of our Sun and other normal stars) but also by very high pressures, even though the temperature may be quite low. The electron-shell structure cannot resist the overwhelming pressure caused by the weight of the outer envelope of the celestial body, and the atoms are crushed, just as are eggs if a heavy weight is placed on top of an egg basket. Once the electron shells are crushed and the assembly of atoms is turned into a plasmoidal mixture of bare nuclei and free electrons, matter regains high compressibility and its density begins to increase in direct proportion to the pressure acting on it.

The Indian astronomer D. S. Kothari used the quantum theory of atomic structure in order to calculate the pressure which is necessary to break up the electron envelopes of various atoms, and thus to turn cold material bodies (rocks) into a highly compressible state. He found that this pressure is of the order of 100,000,000 Earth atmospheres, varying only slightly for atoms of different chemical elements. The pressure in the centre of the Earth is only 4,000,000 atmospheres, which is nowhere near enough to break up the electron shells of any atom. Thus we can rest assured that the atoms of the molten iron which forms the inner core of our planet are just about as unbroken as the atoms of the molten iron that is poured into casting moulds in our foundries. But for planets more massive than the Earth the internal pressure becomes considerably stronger and, in the case of Jupiter (the largest planet of the solar system), approaches the value necessary for breaking up the electron shells of iron atoms. Thus in a planet more massive than Jupiter (and it is quite likely that such planets may exist in the planetary systems of other stars), the material of the inner core is expected to be pressure-ionized and to become highly compressible. In this case a further increase of the mass of a planet will result in a decrease of its radius and a rapid increase of its mean density. The results of Kothari's calculations are shown in Figure 42, where the radii of very large cold bodies are shown as the function of their mass (with respect to the Sun). The two continuous curves pertain to the theoretical results calculated under the assumption of pure hydrogen and pure iron cores. The dots on the left side of the diagram represent the observed value for the celestial bodies from the Moon to Jupiter. We notice that the radii, which increase as the cubic roots of the mass (constant mean density) for the lighter planets, reach the maximum for about one-thousandth of the Sun's mass (Jupiter) and then begin to decrease for still larger masses. The three dots on the right side of the diagram correspond to the three White Dwarf stars: Sirius B, σ_2 Eridani B, and Procyon B. Even though these three stars are about a thousand times more massive than Jupiter, their radii are considerably smaller and their mean densities are much higher. The cross in the diagram corresponds to our Sun. It is much

larger than any of the White Dwarfs of comparable mass. But when, within several billion years, the nuclear reactions in the solar interior stop because of the exhaustion of hydrogen fuel, the Sun will shrink to a size comparable to that of Sirius B. And

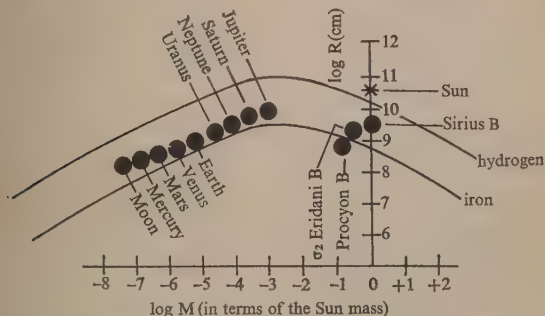


FIGURE 42. Kothari's curves for the radius and mass relation of collapsed celestial bodies compared with the data for planets and White Dwarfs.

then, after the heat still contained in the solar body is finally dissipated by radiation, the Sun will become a dark, dense body of about the size of Jupiter.

Since this book has been adorned with famous poetical quotations at the beginning of each chapter, it is only proper to finish it by quoting a verse, probably the first ever written by A. S. Pushkin. When Pushkin was a schoolboy, his teacher asked the class to write a composition on the theme: 'What would happen if the Sun were to rise in the West?' Wrote Pushkin, somewhat awkwardly:

*The sacred King of Heaven rose one morning in the West.
The people were distressed.
'What do we do,' they said,
'Jump in or out of bed?'* *

(No political implications, please!)

* Translated by Barbara P. Gamow.

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The cover shows eruptive and quiescent prominences of the sun (photo V-Dia-Verlag, Heidelberg)

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